Unit-1

IC: Integrated circuit

→ It is a combination of transistors connected in a single chip (substrate).

→ IC's are two types they are

IC
- Monolithic
- Hybrid

→ again the IC's are happ classified into 3 they are.

IC's
- Analog
- Digital
- Mixed

Two laws for transistor:

Moore 1st law:

This law states that the no. of transistors per chip will be double for every 2 years of period. (every 18 months).

Integration levels :-

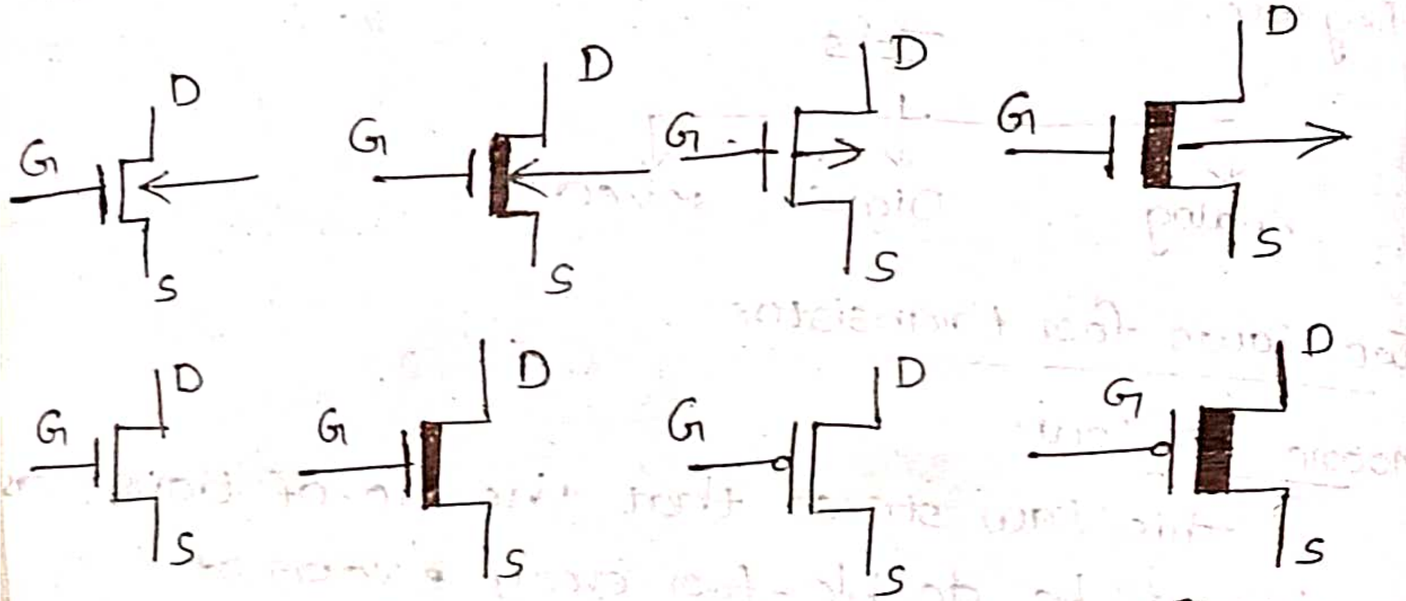
Integration levels	gates/chip	No. of transistors
SSI	→ 3 to 30	→ 10 - 100
MSI	→ 30 to 300	→ 100-1000
LSI	→ 300 to 3,000	→ 1000-20,000 (1L)
VLSI	→ 3,000 to 30,000	→ 20,000 - 1,00,000 (1L 10L)
ULSI	→ >30,000	→ 1,00,000 - 10,000,000 (>10L)

Basic MOS and VLSI related technology:-

MOS → Metal oxide semiconductor

There are having different MOS are shown in below

MOS
- Nmos
  - enhancement mode
  - depletion mode
- pMOS
  - enhancement mode
  - depletion mode
- CMOS
- BiCMOS



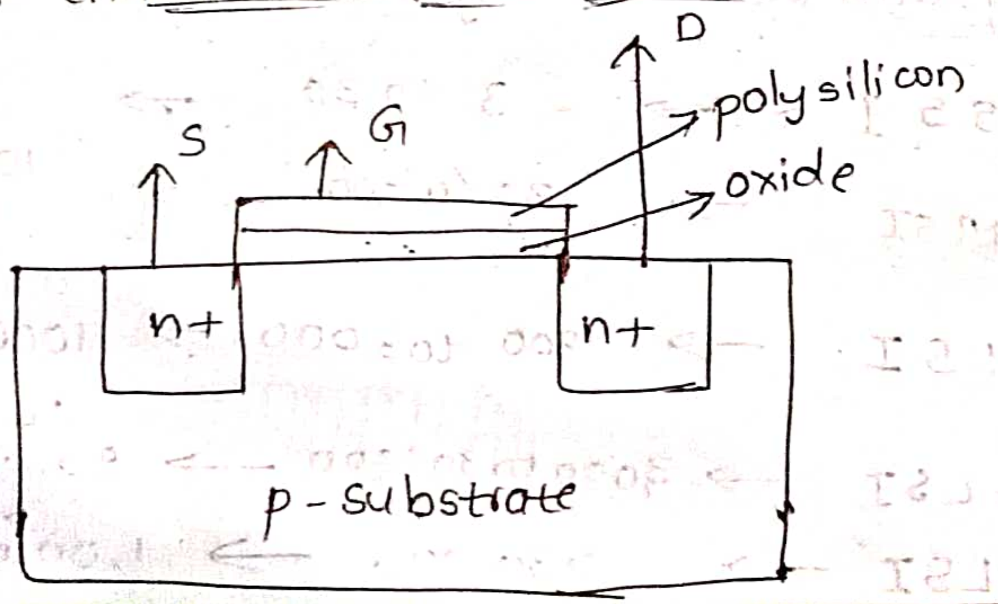
enhancement mode | depletion mode | pmos enhancement | pmos depletion

Structure of enhancement mode n-mos transistor:

case-1

$V_{gs} > V_{th}$

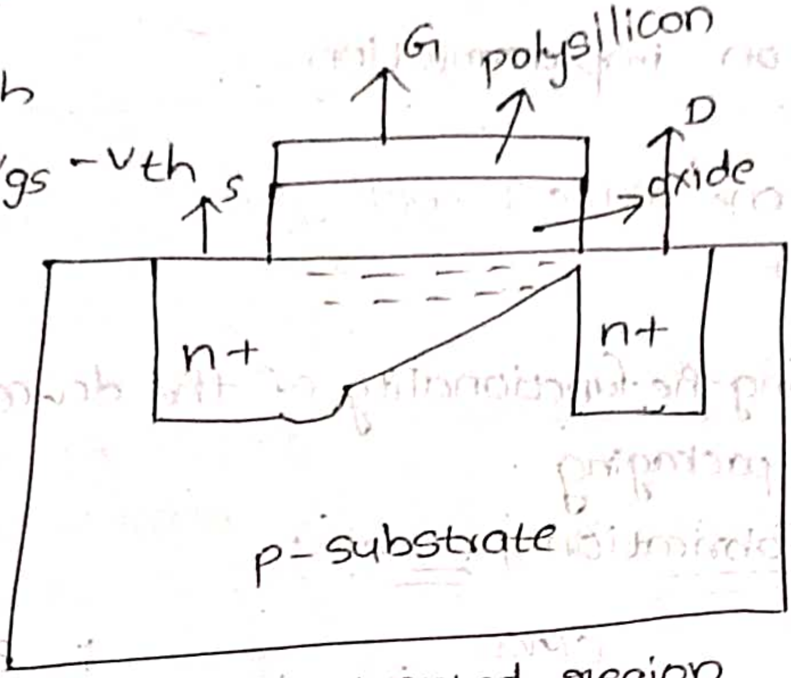
$V_{ds} = 0V$



Case-2 :-

$V_{gs} > V_{th}$

$V_{ds} < V_{gs} - V_{th}$

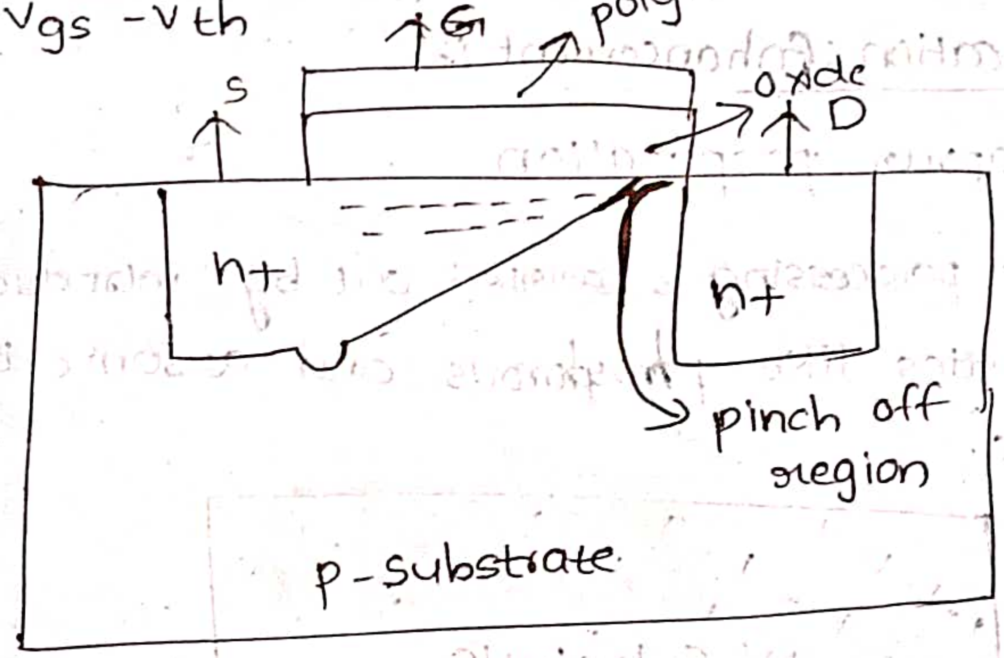


Non-saturated region

case-3 :-

$V_{gs} > V_{th}$

$V_{ds} > V_{gs} - V_{th}$



Saturated region

IC production (or) fabrication process:-

→ Silicon manufacturing

* The raw material of Sand

→ water processing

→ lithography

* It means making of patels.

→ Oxide growth and removal
→ Diffusion and ion implantation
→ Annealing
* It means heat it one make it cool.
→ silicon deposition
→ Metalization
→ Testing → checking the functionality of the device
→ Assembling and packaging

MOS and CMOS fabrication process:-

i, MOS
- nmos
- pmos

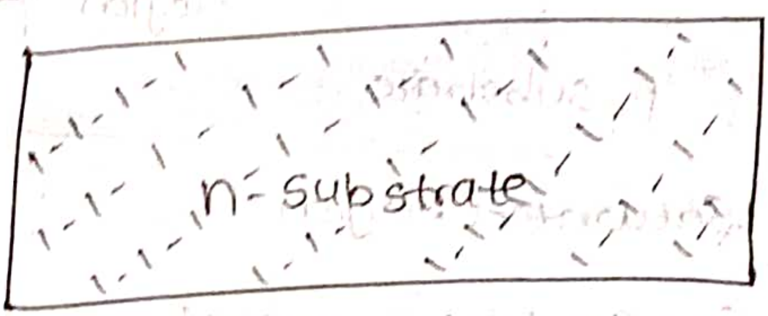
ii, CMOS
- p-well
- n-well
- Twin tuby well
- SOL → silicon on insulator

iii Bicmos

Pmos fabrication : Enhancement :-

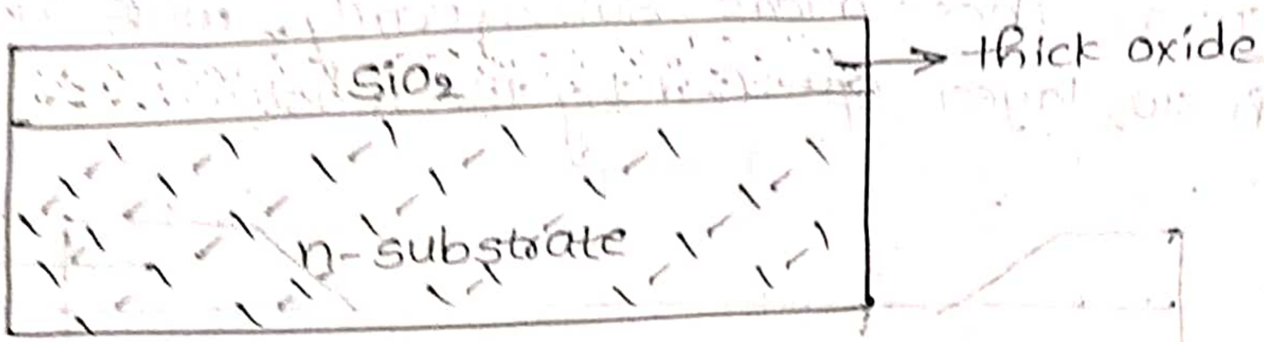
Step:- 1 → vapour preparation

The vapour processing is carried out by introducing n-type impurities like phosphorous and arsonic to form n- substrate.

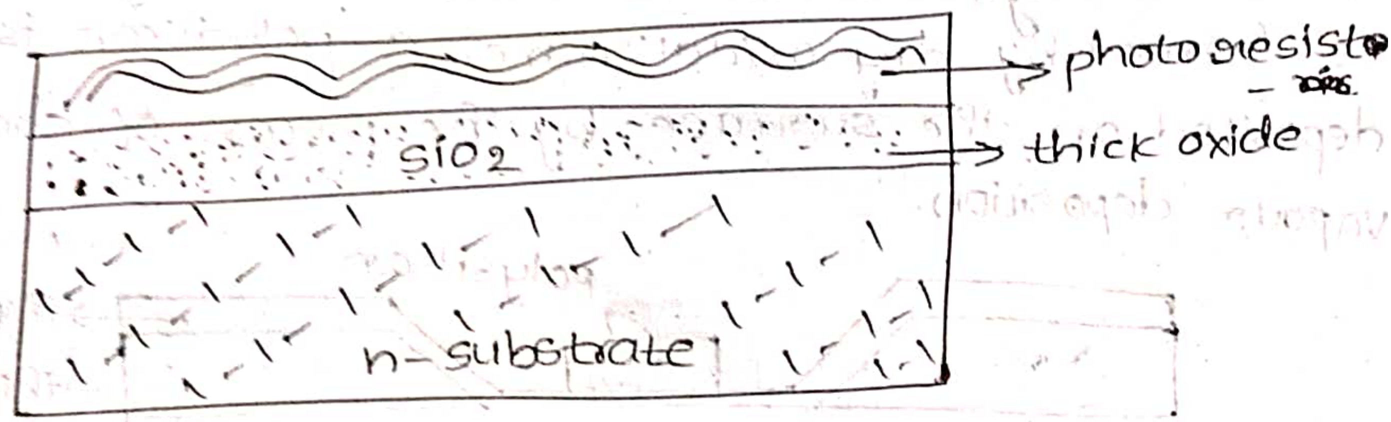


Step 2: Formation of oxide layer

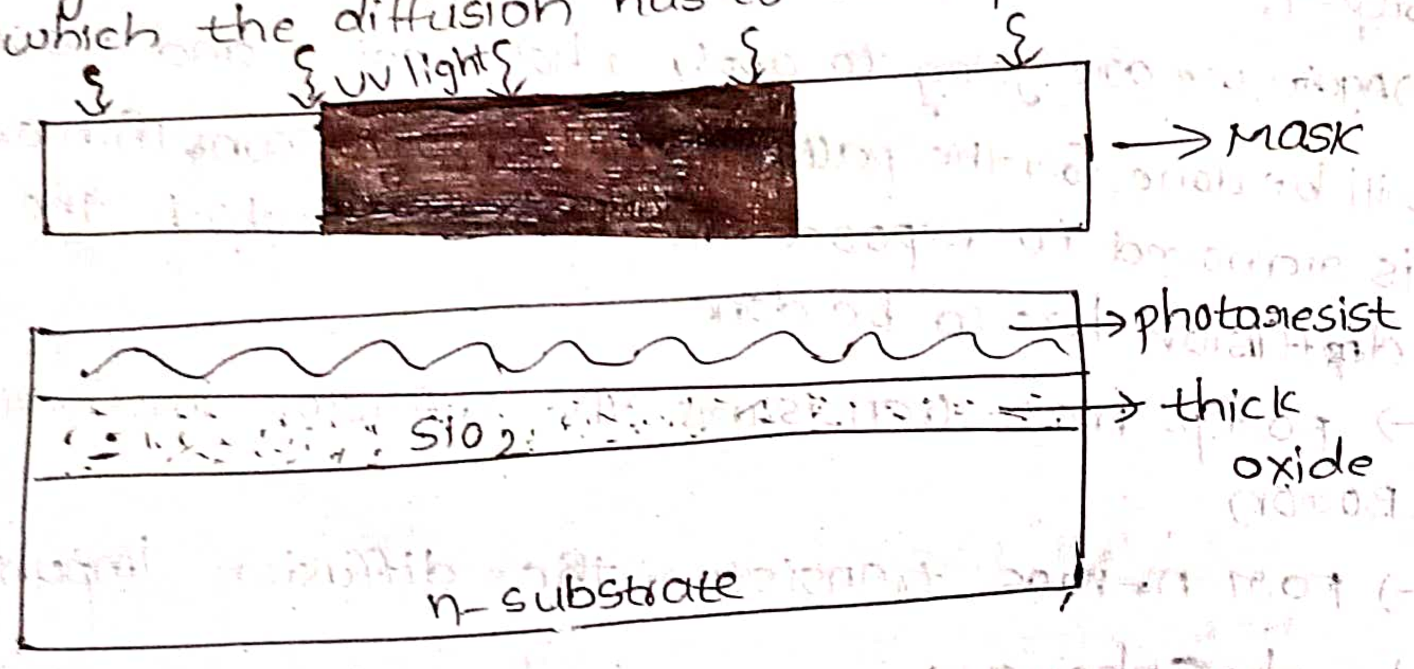
the thick oxide layer of silicon dioxide is formed all over the surface of the vapour to protect the surface. It acts as an insulating material



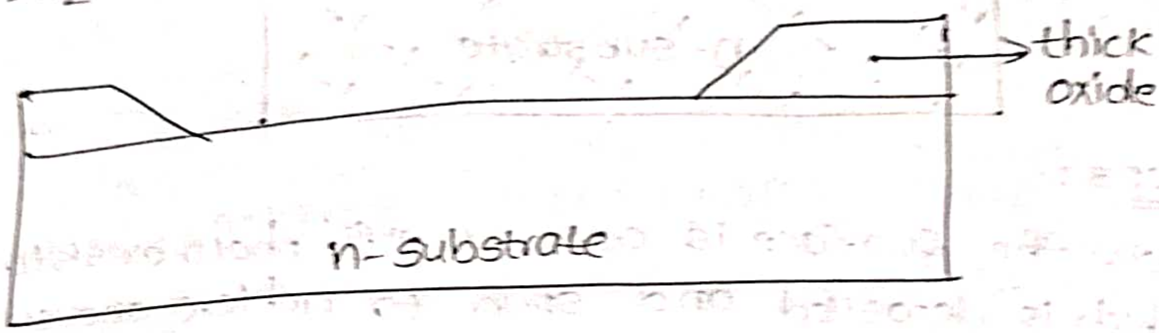
Step 3 :-
Now the surface is covered with photo resistor which is deposited and spun to achieve an even distribution of required thickness.



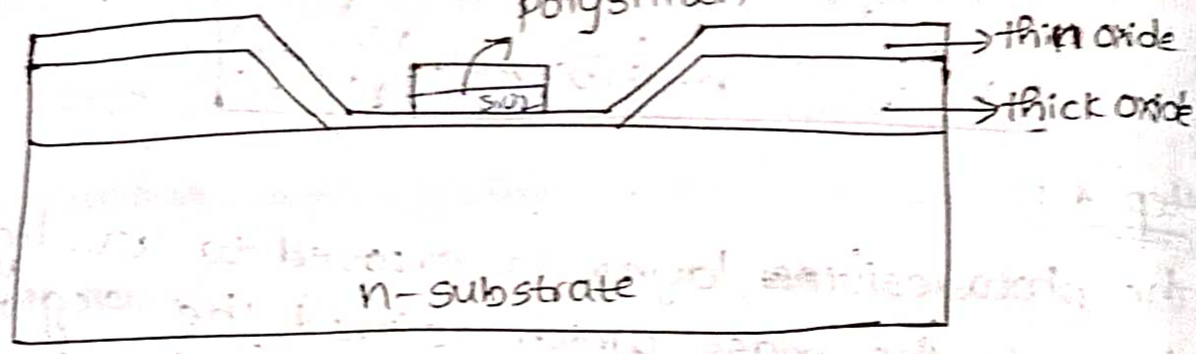
Step 4 :-
The photoresistor layer is exposed to UV light through the mass which defined the regions into which the diffusion has to take place.



step:-5: These areas subsequently readly ~~edge~~ etching with underlying with SiO_2 layer specified by the mask



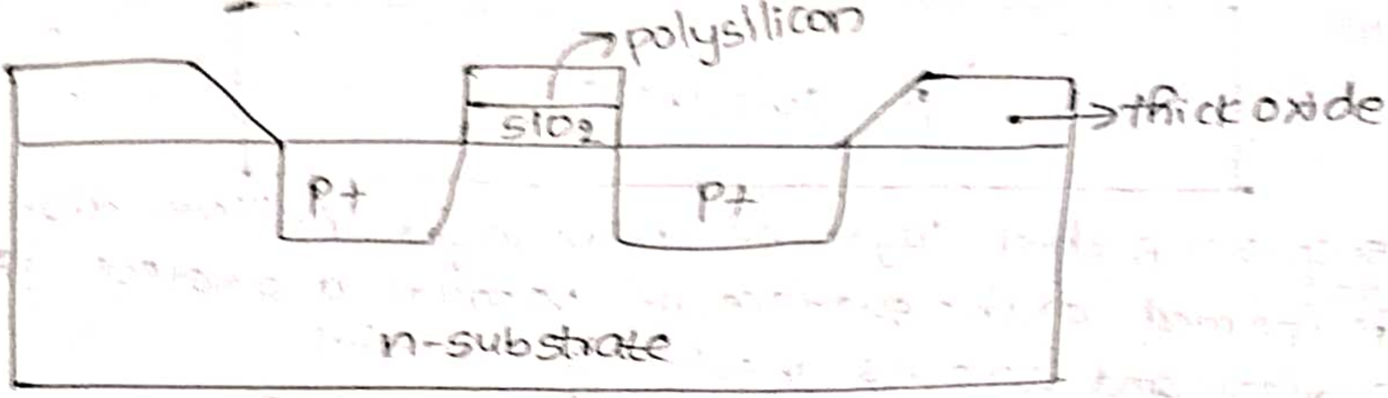
step-6:-
The remaining photoresist is remove a thin layer of SiO_2 is formed by dry oxidation method. a polysilicon is deposited on the surface by the process of chemical vapour deposition.



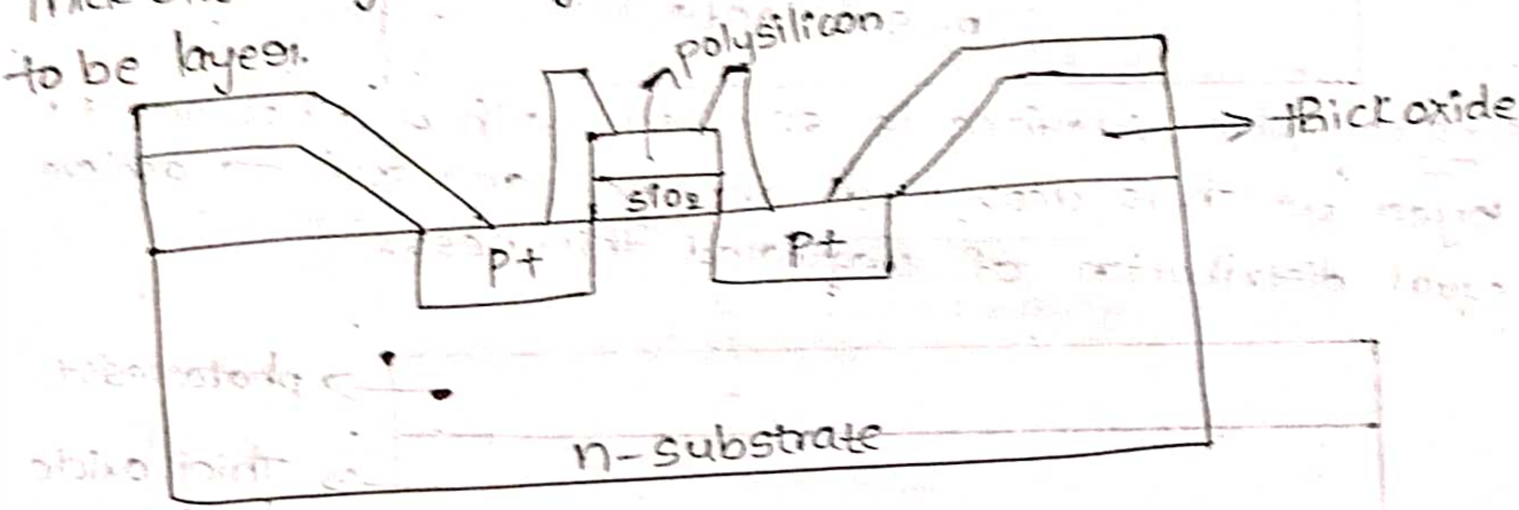
Step-7:-
→ Again we are going to apply photoresist and masking will be done for the patterning the polysilicon & thin oxide is removed to exposed ~~the areas~~ in which the diffusion has to be done

→ For p-mos transistors the diffusion impurities is Boron

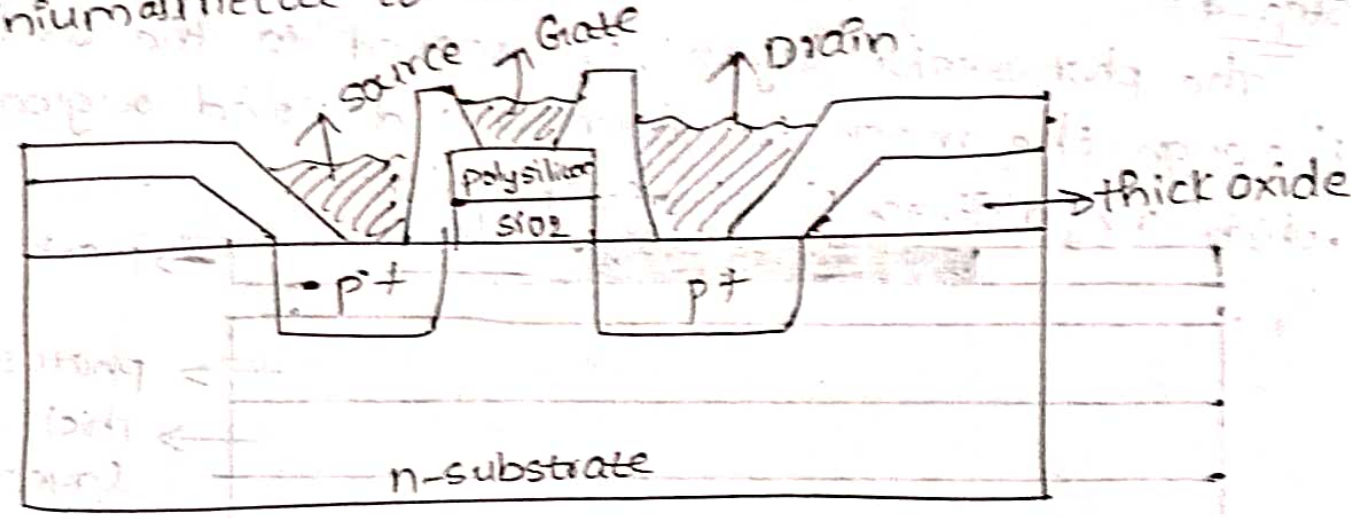
→ For n-mos transistor the diffusion impurities is phosphorous.



Step-8:-
Thick oxide layer is grown and mask for the connected to be layer.

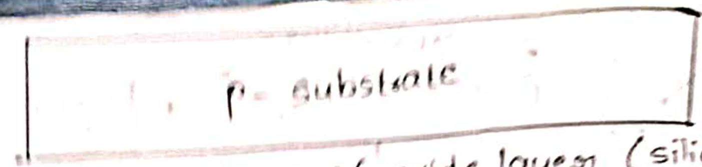


Step-9:- metalization
The metalization process is carried out by considering allumminium metal to construct the pins

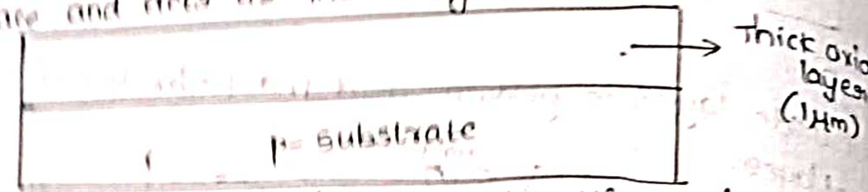


Enhancement NMOS fabrication:-

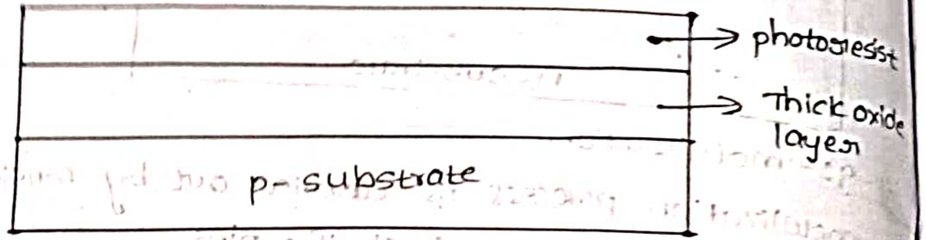
step 1:- The vapour processing is carried out by introducing p-type impurities like boron and galium for the formation of p-type substrate



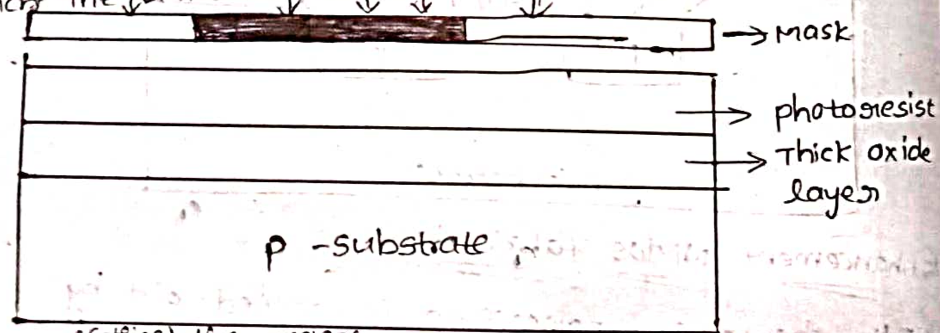
Step 2: A thin layer of oxide layer (silicon dioxide) is formed on the surface of vapour to protect the surface and acts as insulating material



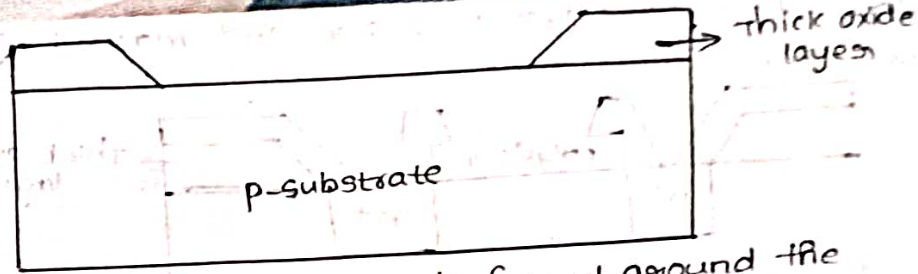
Step 3: The surface is covered with a photoresist layer which is deposit at centre, and spun to achieve equal distribution of required thickness.



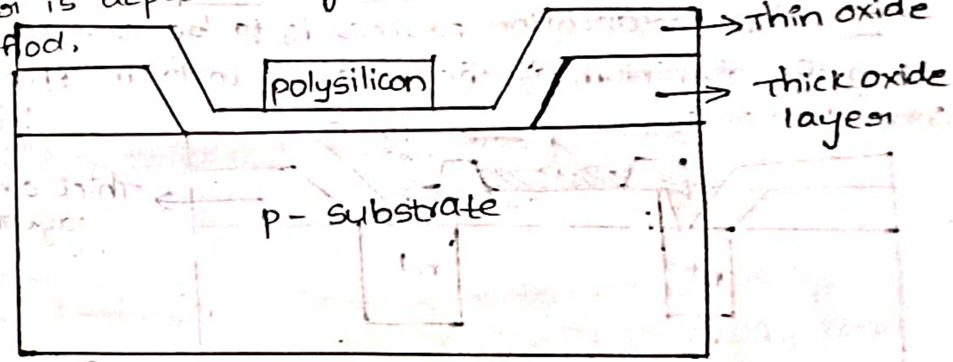
Step-4:
The photoresist layer is exposed to the UV light through the mask to form the desired regions in which the diffusion has to take place



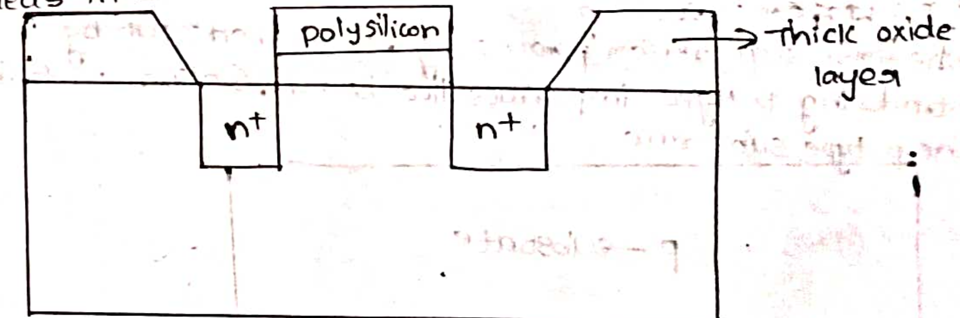
Step 5 (Etching) the regions: The desired areas are subsequently readily etched with underlying SiO_2 layer has specified by the mask.



Step 6:- The thin oxide layer is formed around the surface by dry oxidisation method and a polysilicon layer is deposited by the chemical vapour deposition method.

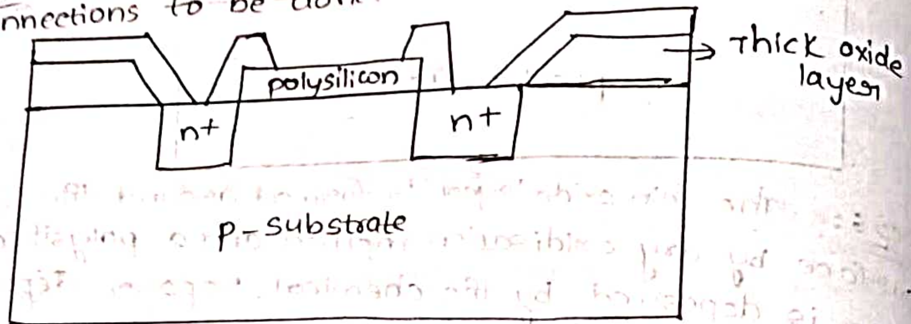


Step 7:- Again we are going to apply the photoresist and making will be done for patterning the polysilicon and thin oxide is removed for expose the areas in to which the diffusion is to be done.

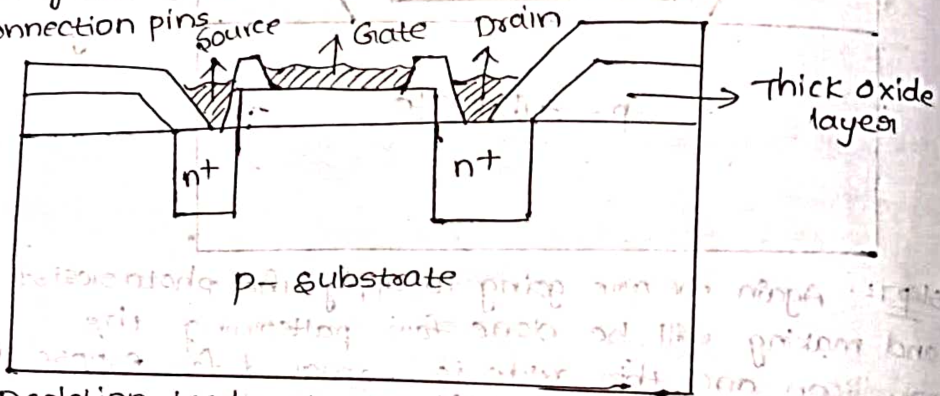


* And for the diffusion process for n mos transistor we use phosphorous
* For the diffusion process for p mos transistor, we use boron

Step 8: A thick oxide layer is grown and masked for connections to be done.



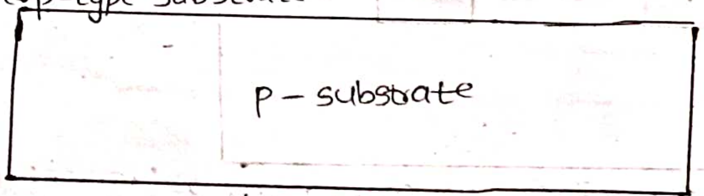
Step 9: The metalization process is to be carried out by using the aluminium as the material to form the connection pins.



* Depletion Mode N-MOS fabrication:-

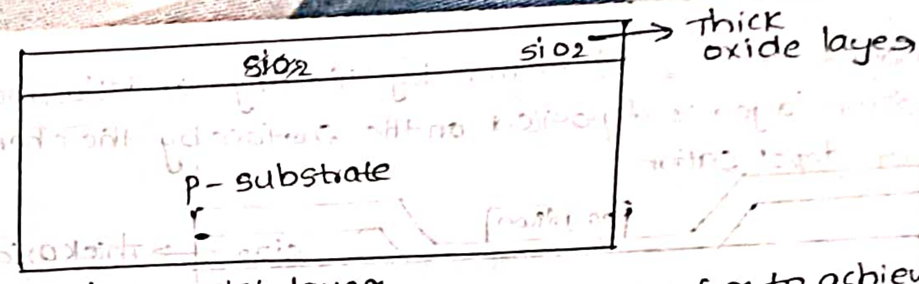
Step 1: ~~wafer~~ vapour processing

The first step wafer processing is carried out by introducing p-type impurities like boron, Galium to form the p-type substrate



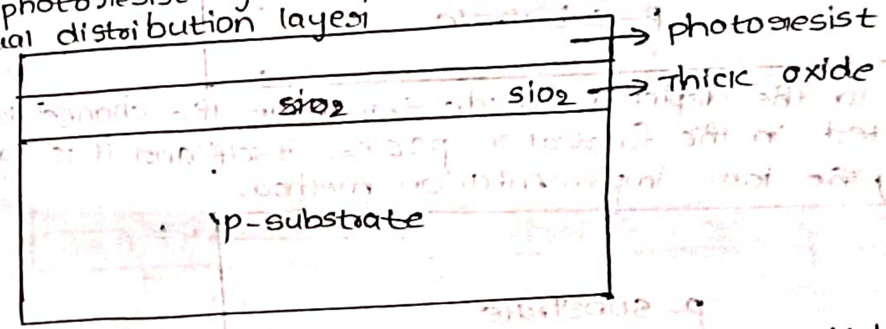
Step-2:- Formation of thick SiO_2 layer

A layer of thick oxide layer (SiO_2) is formed all over the surface of the substrate of wafer to protect it from the contaminations and it acts as protecting layer.



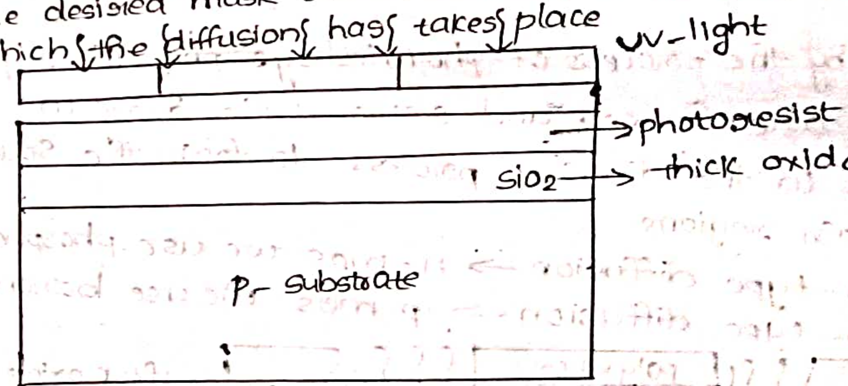
Step 3: photoresist layer

A photoresist layer is applied and spun wafer to achieve equal distribution layer



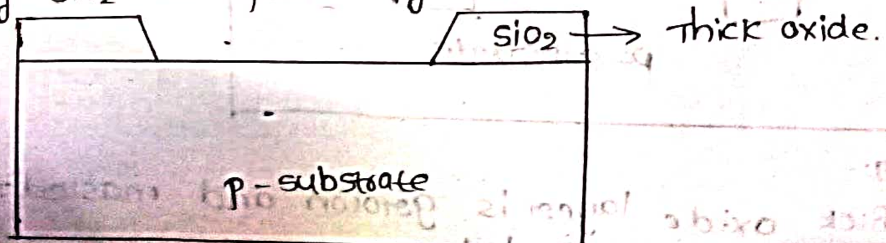
Step 4:

The photoresist layer is when exposed to UV light to the desired mask which defines the resources into which the diffusions has takes place



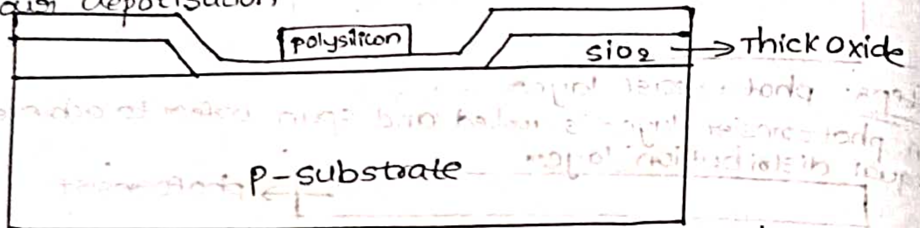
Step 5: Etched

These areas are subsequently readily etched with underlying SiO_2 as specified by the mask.

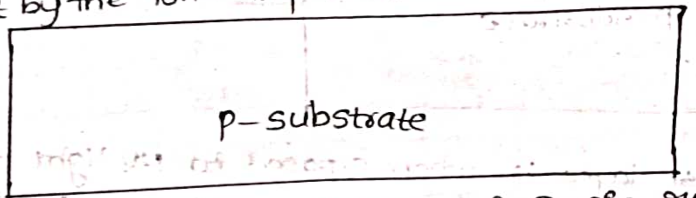


Step 6: Formation of gate

A thin layer of SiO_2 is formed by the dry oxidation method.
A polysilicon layer is deposited on the surface by the chemical vapour depotisation



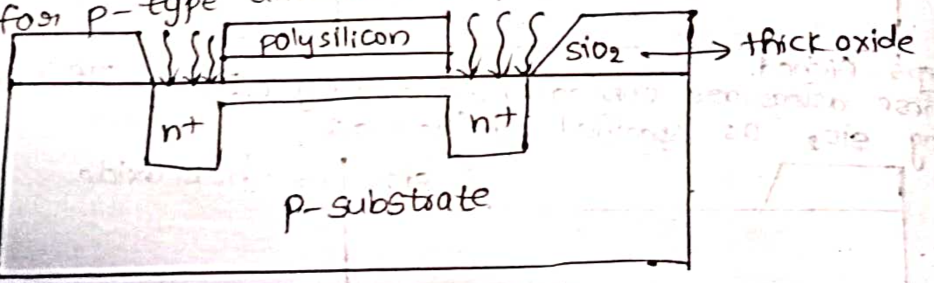
Step 7: To the depletion mode transistor the channel is fabricated in the fabrication process itself and it is carried out by the ion- implementation method.



Step 8: Formation of Source & Drain regions.

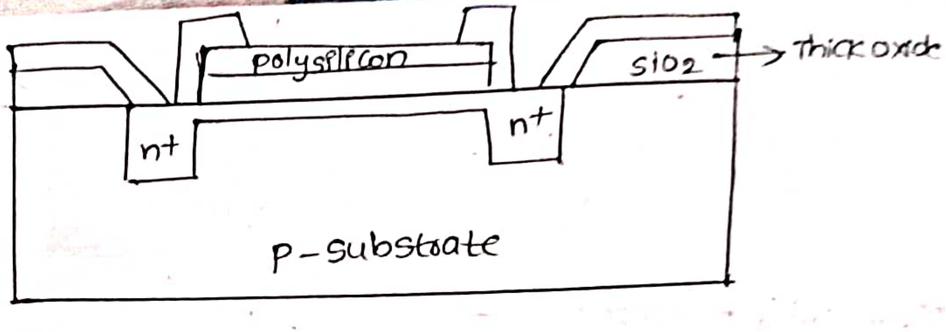
Again by the process of photolithography the required pattern are formed and removed the areas to expose it to the diffusion process to form the Source and drain regions.

* for n-type diffusion → n-mos we use phosphorus
* for p-type diffusion → p-mos we use boron



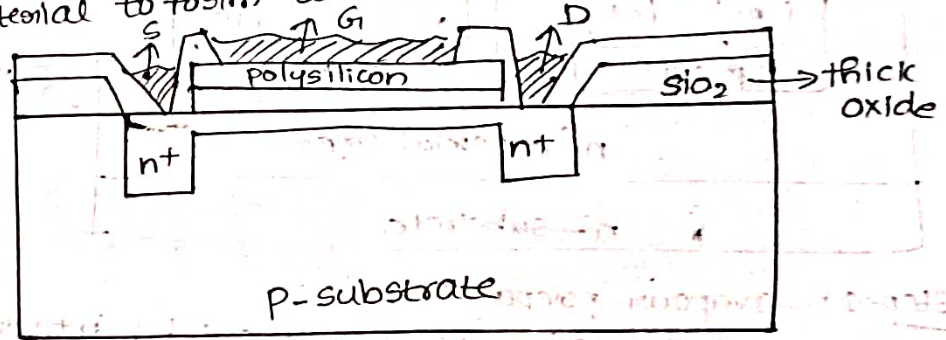
Step 9:-

A thick oxide layer is grown and masked for the connections to be ~~made~~ layer

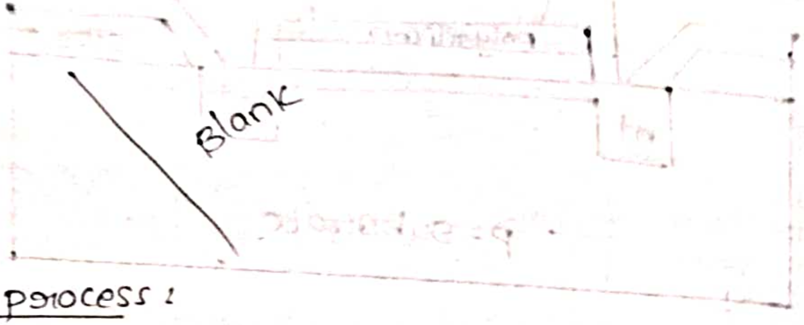


Step 10: metalization

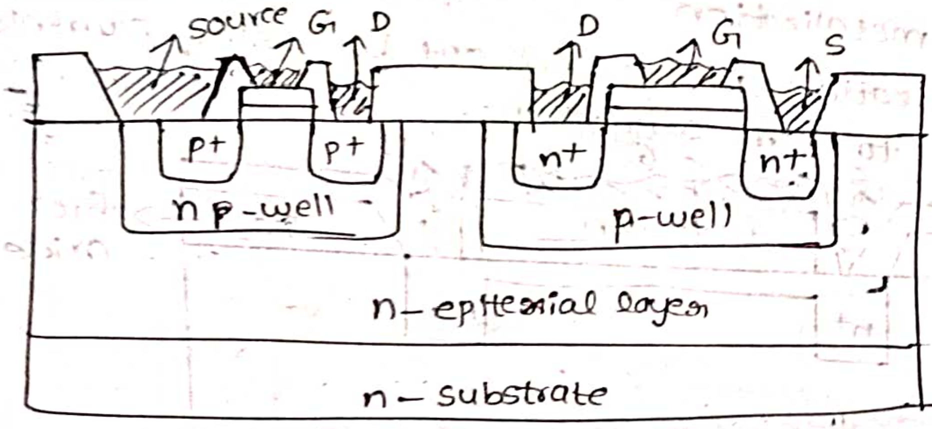
metalization is carried out by using Aluminic material to form connections.



Blank



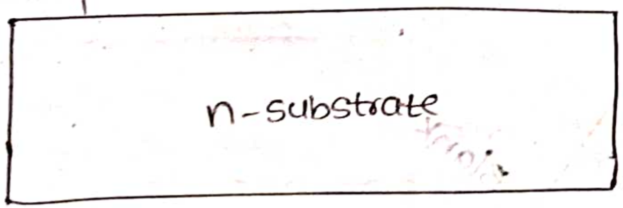
Twin tuby process :



→ final diagram

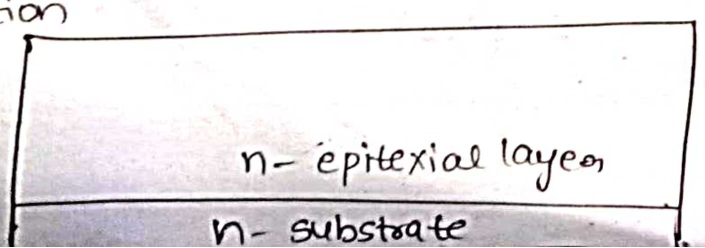
Step-1 : → vapour preparation

The vapour processing is carried out by introducing n-type impurities like phosphorous and arsonic

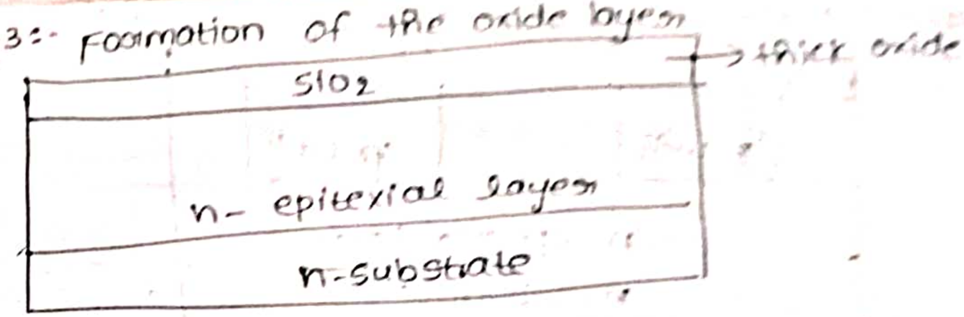


to form n-substrate.

Step-2 : Formation of epitexial process for the wells creation

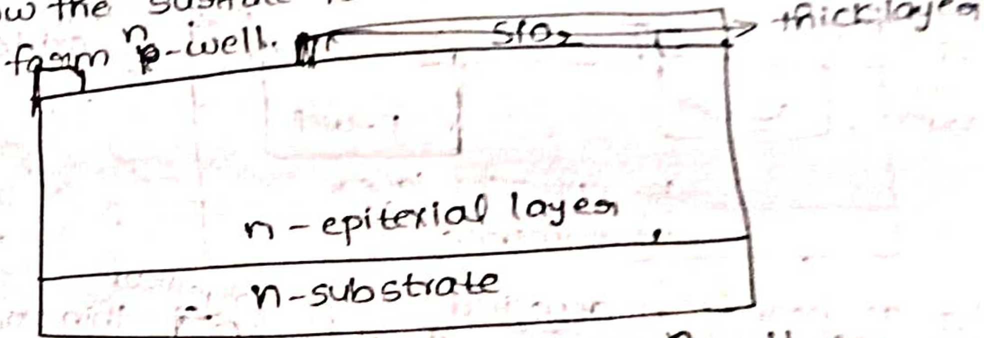


Step-3 :- Formation of the oxide layer

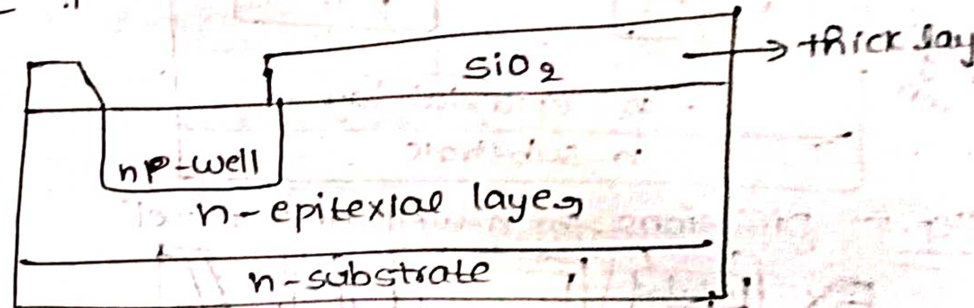


Step-4 :

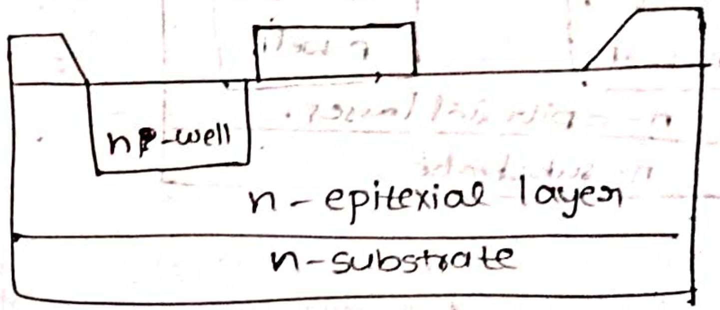
Now the surface is covered with photoresist and form n-well.



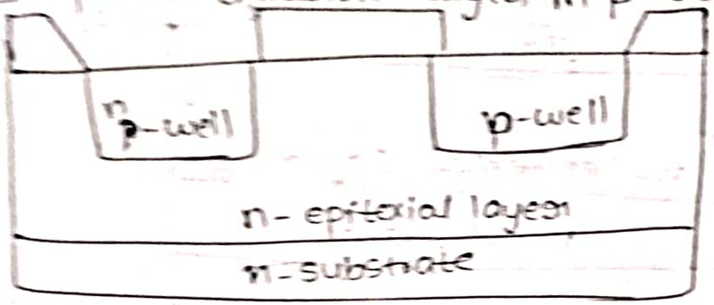
Step 5 :- Form diffusion layer in n-well



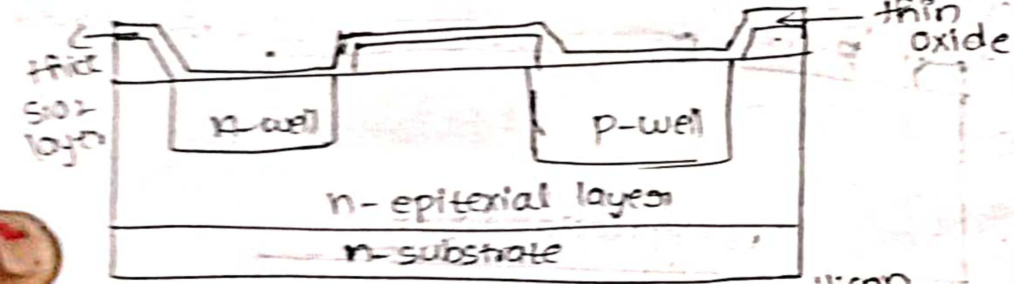
Step 6 : Now form the p-well



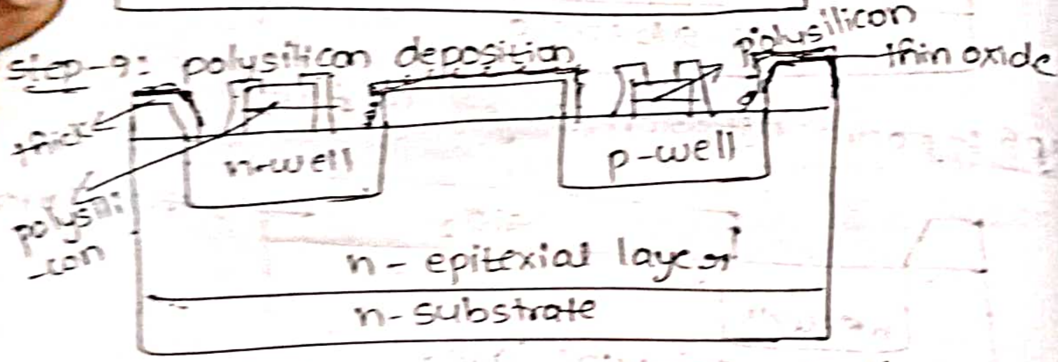
Step-7:- Form diffusion layer in p-well



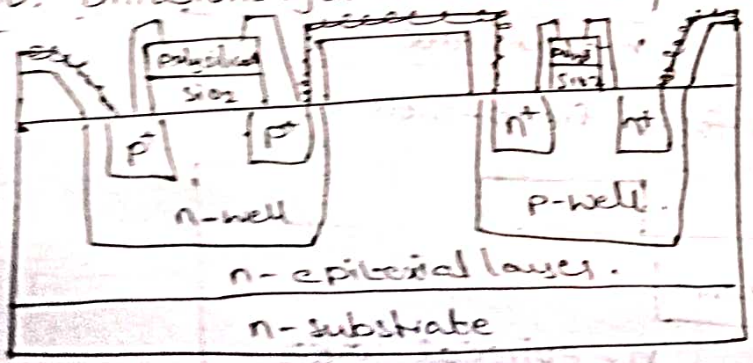
Step-8:- Formation of data oxide



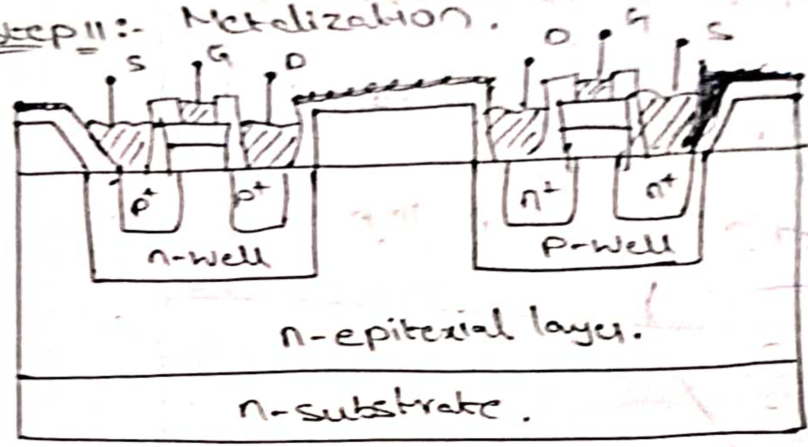
Step-9: polysilicon deposition



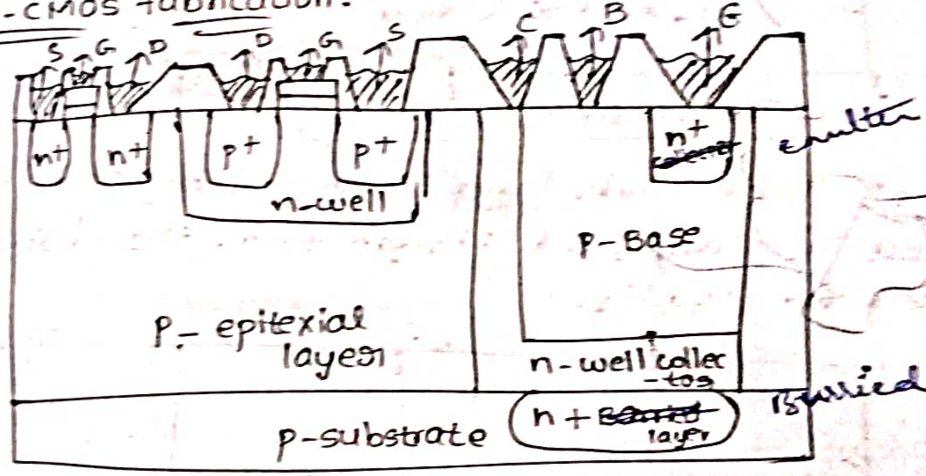
Step-10:- Diffusions for n-well and p-well



Step 11:- Metalization.

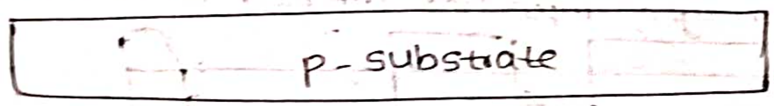


Bi-CMOS fabrication:-

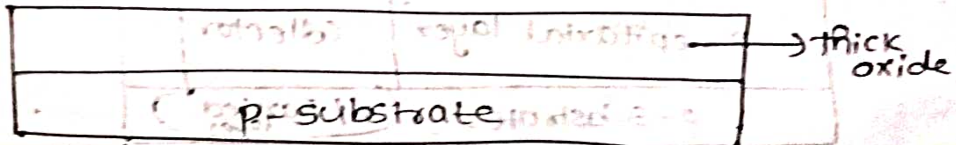


Step:-1: Vapour preparation

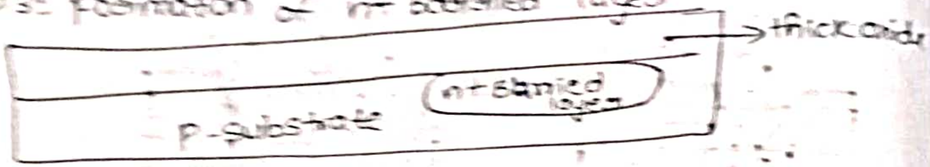
The vapour processing is carried out by introdu-cing p-type impurities like phosphorous and arsonic



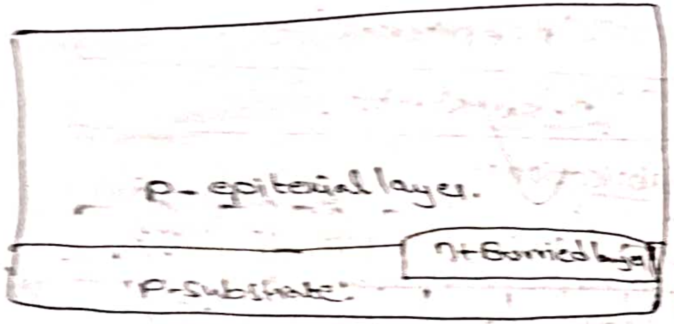
Step 2: formation of thick oxide



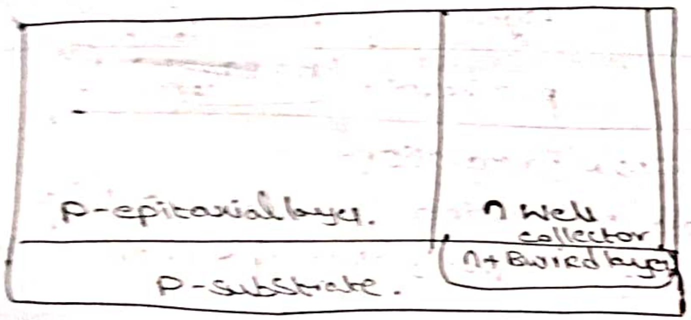
Step 3:- Formation of n+ Barried layer



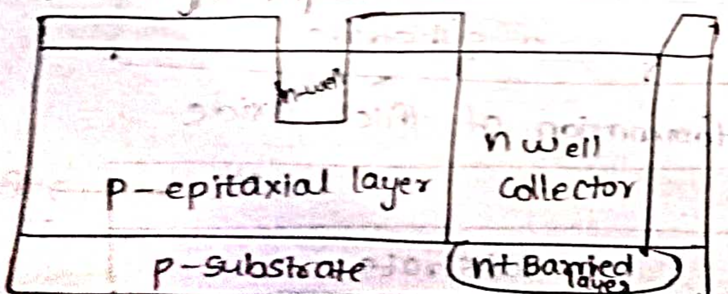
Step 4:- formation of epitexial layer



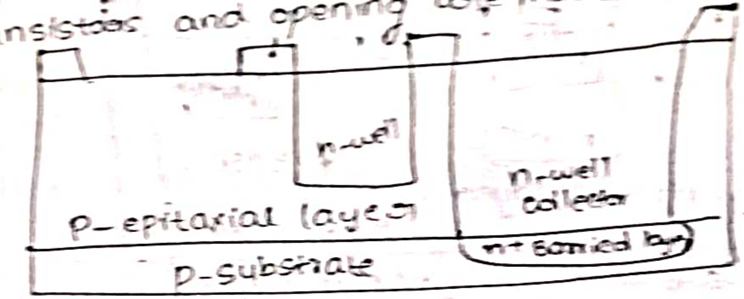
Step 5:- Thick oxide layer is formed by photo resistic process is carried out through form the desired regions for n wells.



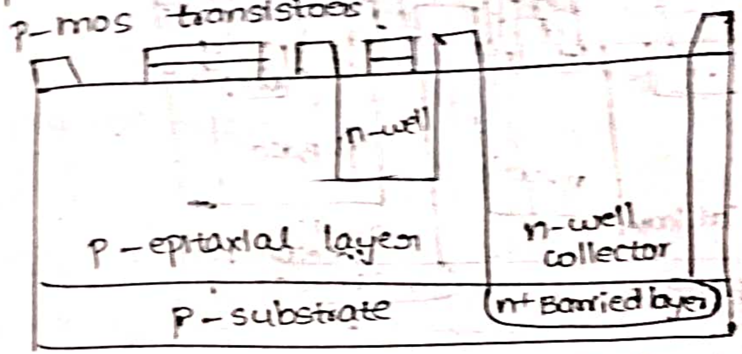
Step 6: The n type impurities diffused to form n-well



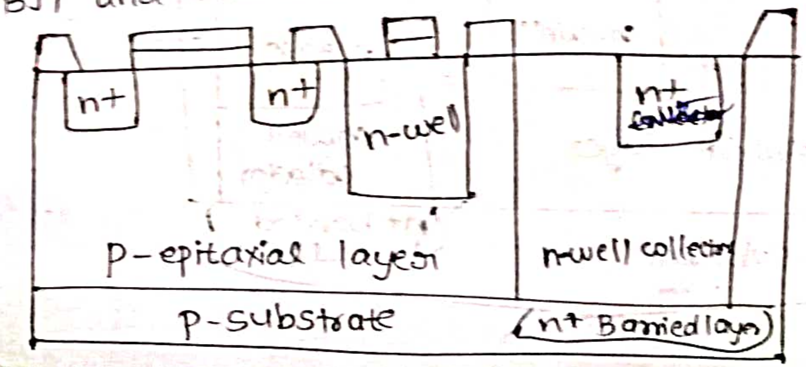
Step 7: again through photolithography process the regions are marked for the formation of n-mos transistors and opening are made in SiO_2 layer



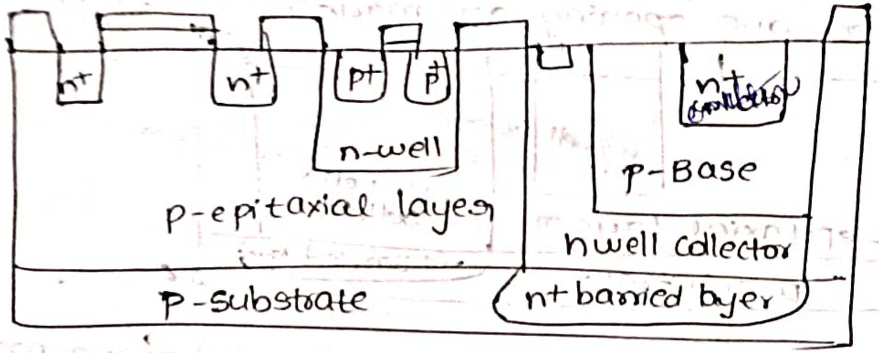
Step-8:
Formation of gate gate terminals and for n-mos & p-mos transistors.



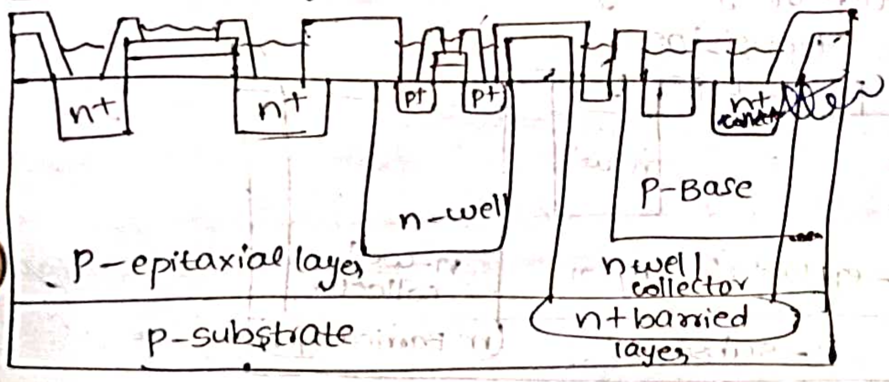
Step-9: p-type impurities are added to form the base terminal of BJT and enter impurities are heavly doped for the formation of emitter of BJT and source and drain regions of n-mos transistor



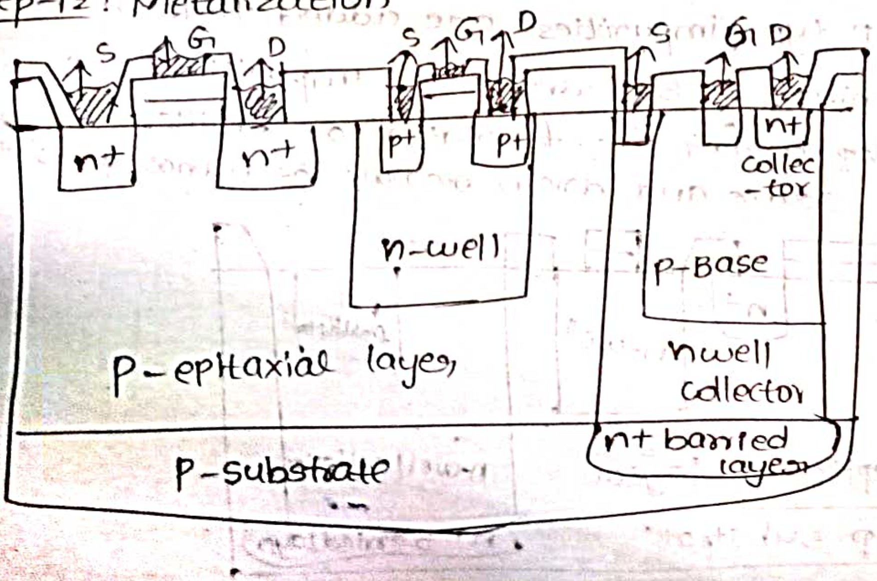
Step 10:- The p-type impurities are heavily diffused to form p-mos transistor for the base



Step-11: oxide layer formation & connections.



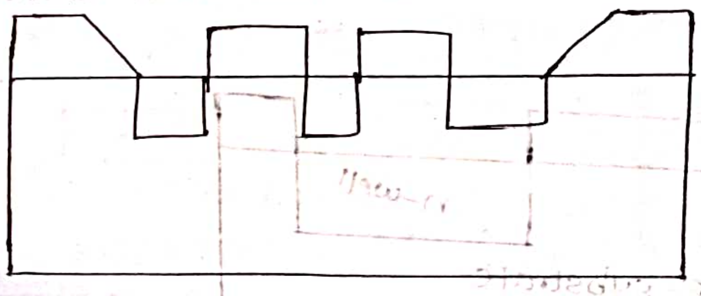
Step-12: Metalization



*Comparision b/w CMOS, and bipolar Bipolar logic

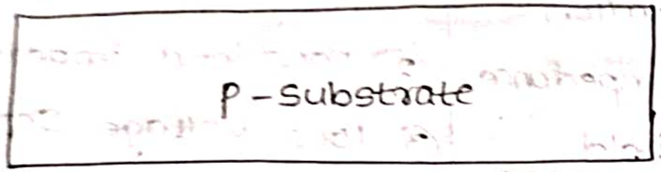
CMOS	Bipolar logic
① low static power desipation	① High power desipation
② High input impedance.	② Low input impedance
③ scalable threshold voltage	③ low voltage Series
④ High noise region	④ low package density
④ High package density	④ No noise region
⑤ High delay sensitivity to load	⑤ low delay sensitivity to load
⑥ low output drive current	⑥ High output drive current
⑦ low gm (gm ∝ vm)	⑦ High gm (gm ∝ e^{vm})
⑧ Bidirectional capability	⑧ High f_t at low current
⑨ ideal sensitivity (FOM) device	⑨ essential unidirectional
⑩ Speed is low	⑩ Speed is more FOM

Envelop fabrication process:-
→ We have to form both the transistors

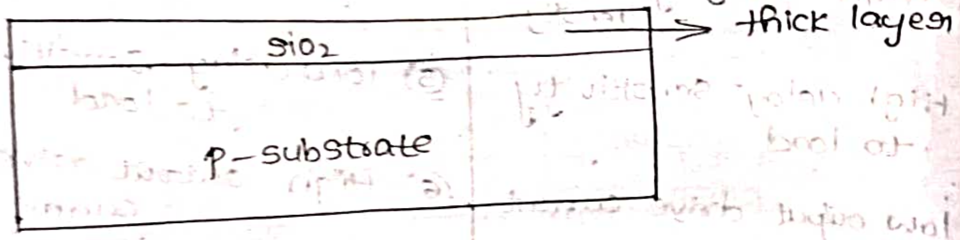


Step 1:- vapour for processing

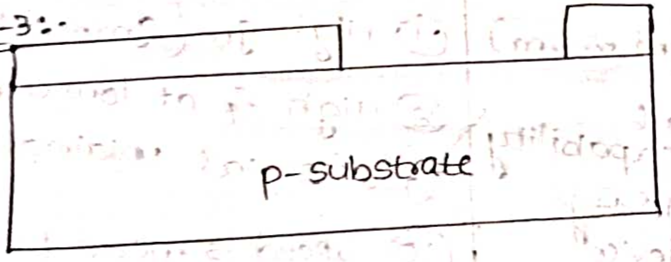
The vapour for processing cascaded at by introducing p-type impurities by the phosphorous and asonic to form p-substrate



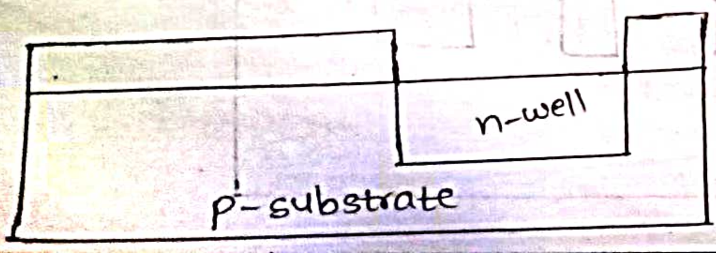
Step-2:- Formation of the envelop 'a thick oxide → layer of silicon & dioxide is formed along the surface of the way for two product



Step-3:-

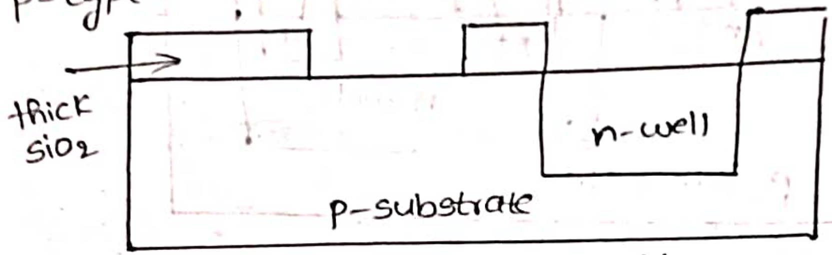


Step-4:-

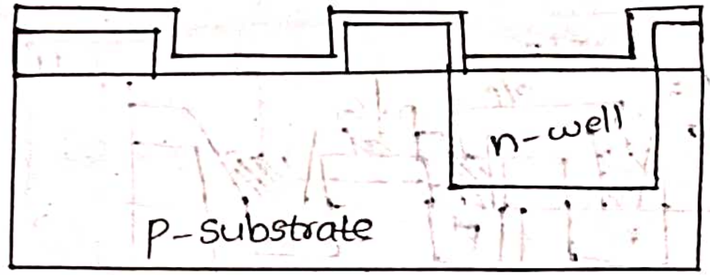


Step-5:-

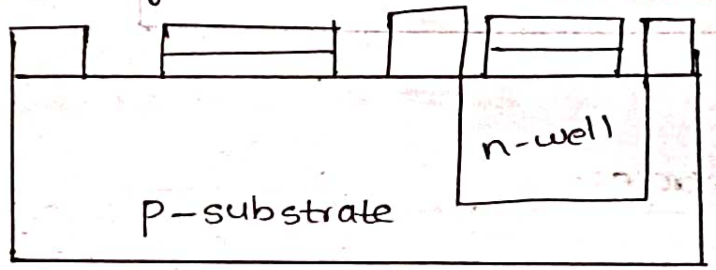
U make the pattern of formation of transistors p-type.



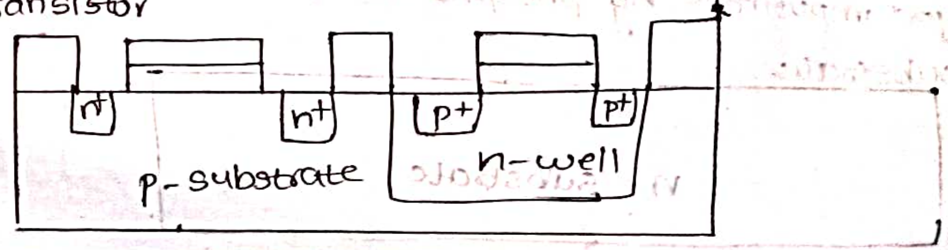
Step-6:- Formation of data oxide



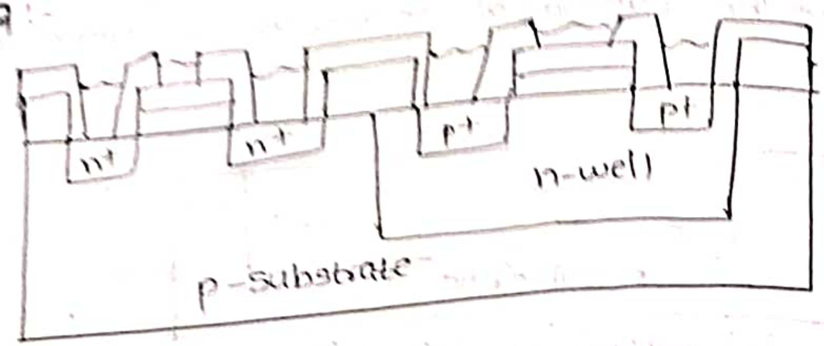
Step-7:- polysilicon deposition



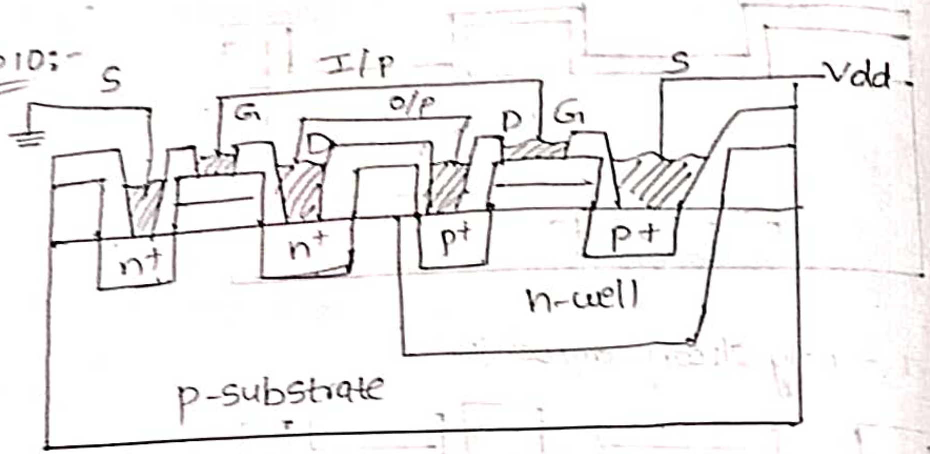
Step 8:- Formation of nmos transistor by the position transistor



Step 9 :-



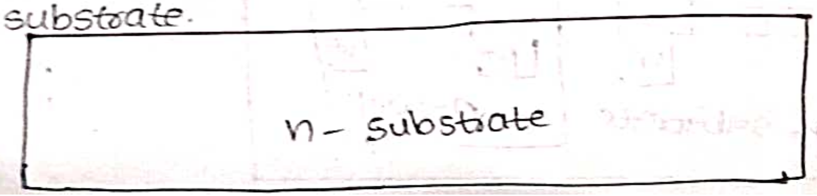
step 10:-



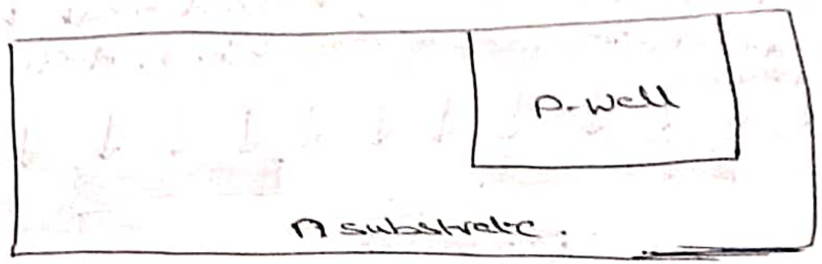
P-well, n-substrate:-

step 1:-

vapour for processing cascaded out by introducing n-type impurities by phosphorous & asonic to form n- substrate.

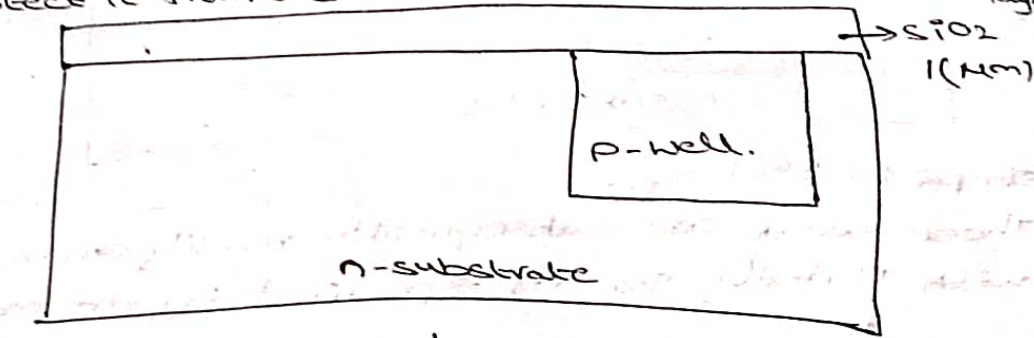


Step 2:- formation of well



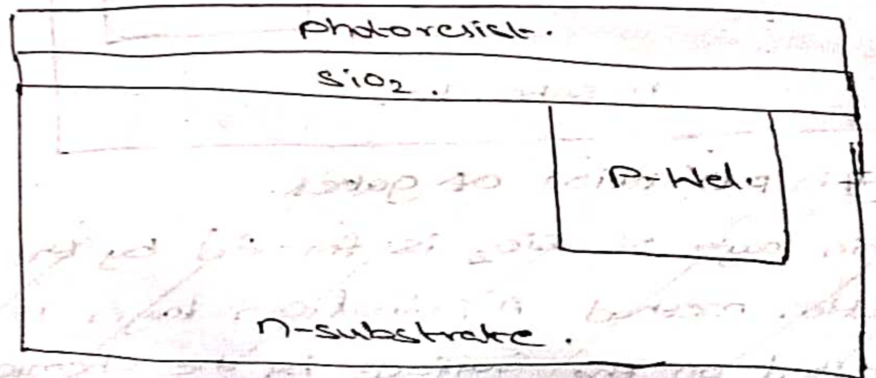
Step 3:- Formation of thick SiO_2 layer.

A layer of thick oxide layer (SiO_2) is formed all over the surface of the substrate of vapour to protect it from the contaminations and it acts as protect layer

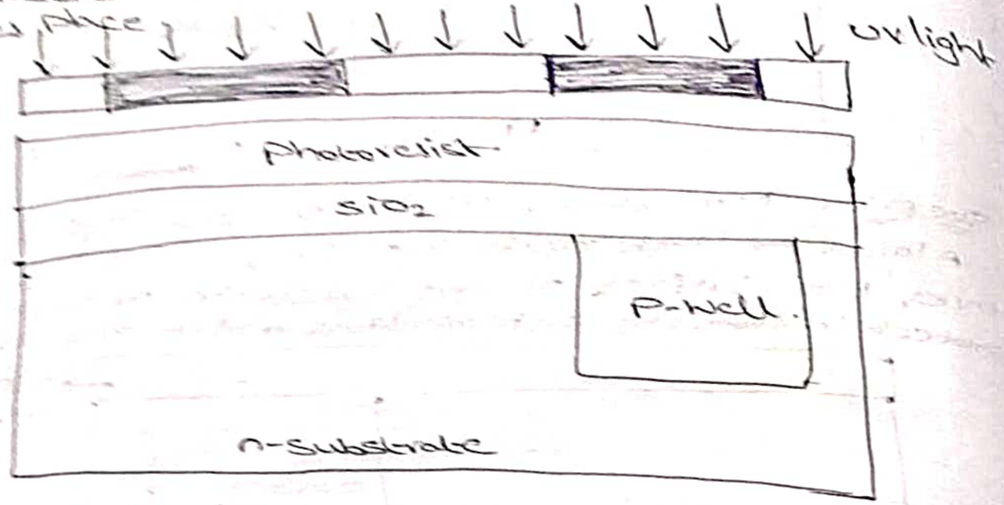


Step 4:- Photoresist layer.

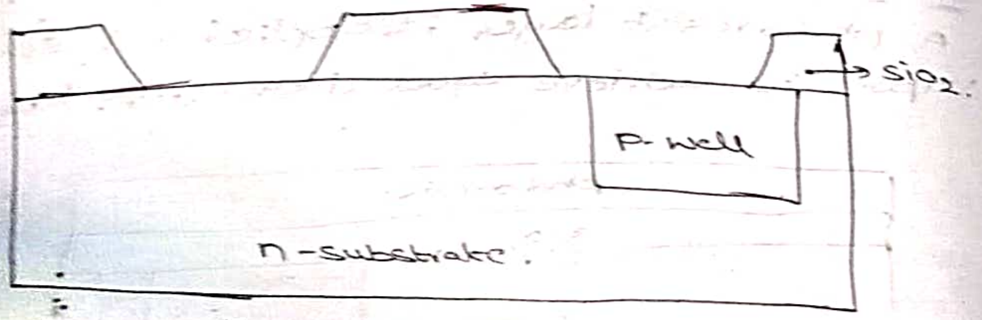
A photoresist layer is applied and spun vapour to achieve equal distribution layer.



Step 5:- The photoresist layer is when exposed to UV light to the desired mask which defineds the sources into which diffusion has takes place.

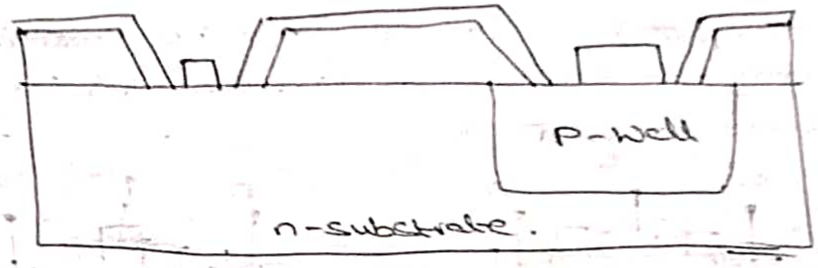


Step 6:- Etching.
These areas are subsequently readily etched with underily SiO_2 as specified by the mask.

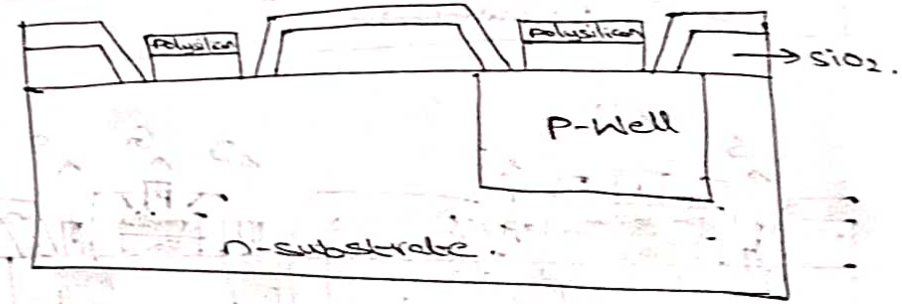


Step 7:- Formation of gates.
~~A thin layer of SiO_2 is formed by the dry oxidation method. A polisilicon layer is deposited on the surface by the chemical vapour deposition.~~

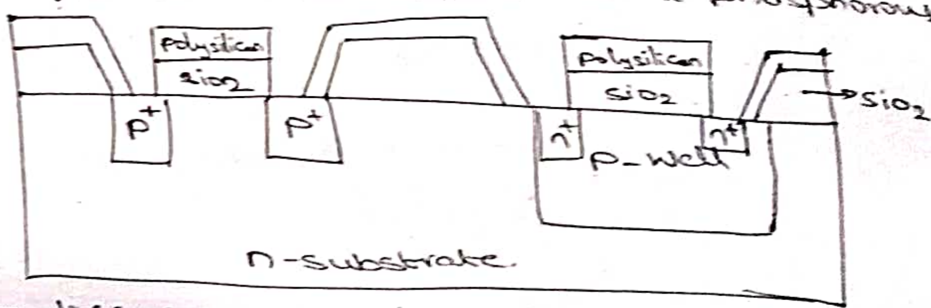
Formation of thick oxide layer.



Step 8:- For the formation of gate.
A polysilicon layer is deposited on the surface by the chemical vapour deposition to form the gate terminal



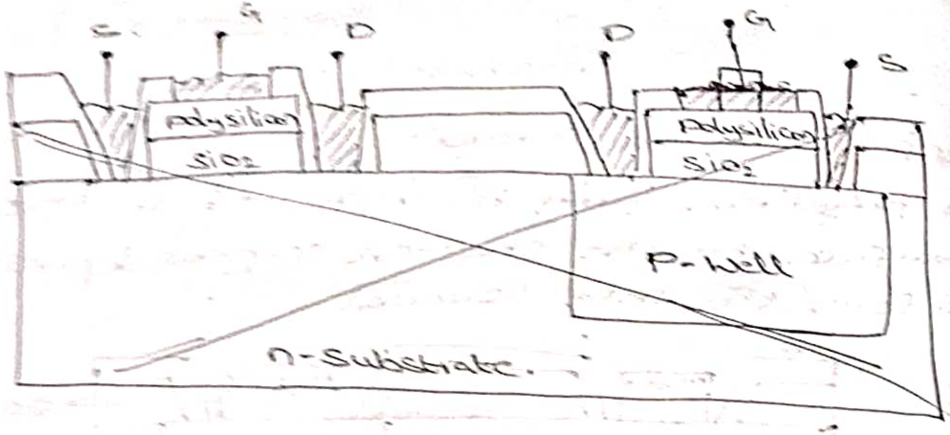
Step 9:- Again we are going to apply the photoresist and making will be done for patterning the polisilicon and the thin oxide is removed for expose the area in to which the diffusion is to be done.
* For diffusion n-mos transistor we use phosphorous

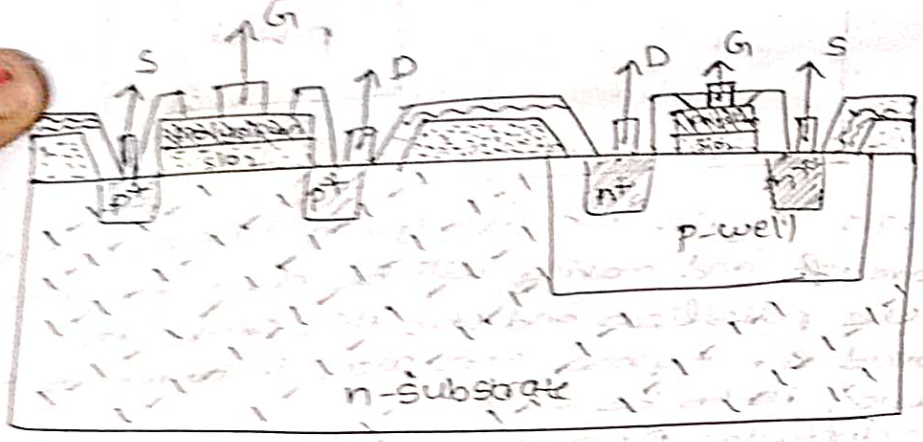


* For diffusion p-mos transistor we use boron.

Step 10:- Metallization.

The metallization process is to be carried out by using the aluminium the material to form the connection pins.



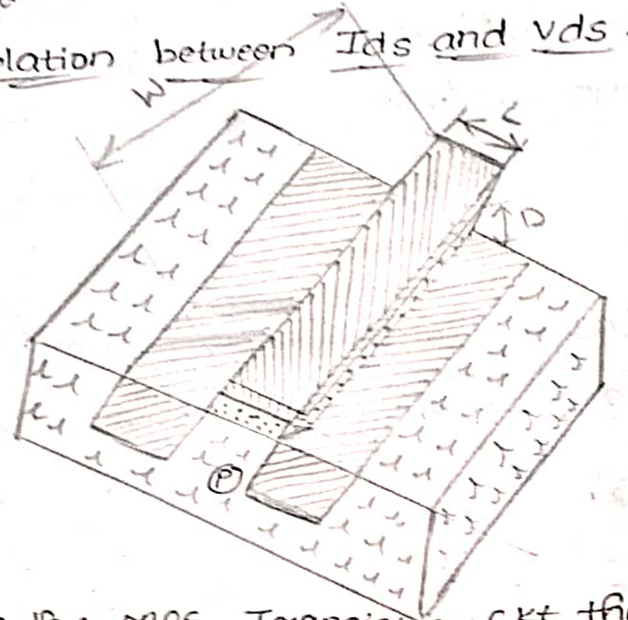


Unit-2:-

BASIC ELECTRICAL PROPERTIES OF MOS TRANSISTOR

conform

Relation between Ids and Vds :-



* In the MOS Transistor ckt the whole concept is depends on the gate voltage and the charge induced in the channel in between drain to source where the charge induced in the channel is depends on gate voltage (Vgs) and drain to source voltage (Vds).
* The drain to source current (Ids) = -Isd

$$I_{ds} = -I_{sd} = \frac{\text{charge in channel } (Q_c)}{\text{Transit time } (\tau_{ds})} \longrightarrow eq ①$$

$$\tau_{ds} = \frac{\text{length of channel } (L)}{\text{velocity } (v)} \longrightarrow ②$$

$$v = \mu E_{ds} \longrightarrow ③$$

μ = Mobility of electrons/ holes

E_{ds} = electric field b/w drain to source

Where

mobility of electrons $\mu_n \rightarrow$ 650 cm^2/v sec

mobility of holes $\mu_p \Rightarrow$ 240 cm^2/v sec

From eq'n ③

$$E_{ds} = \frac{V_{ds}}{L}$$

Substitute E_{ds} in eq'n ③

$$v = \mu \frac{V_{ds}}{L} \rightarrow ④$$

eq'n ④ sub in eq'n ②

$$\tau_{ds} = \frac{L}{\mu \cdot \frac{V_{ds}}{L}}$$

$$\boxed{\tau_{ds} = \frac{L^2}{\mu V_{ds}}} \rightarrow ⑤$$

Where the transistor has two regions

1. Saturation region
2. Non-saturation region

<u>For Non-saturation region:-</u>

In Non-saturation region the current I_{ds} is depends on the gate voltage (V_{gs}) and drain to source voltage (V_{ds}).

→ The current in the channel is varies w.r.t to the distance.

→ Then the average voltage in the channel is $\frac{V_{ds}}{2}$ and the effective gate voltage $V_g = V_{gs} - V_t$

Where $V_g \Rightarrow$ effective gate voltage

V_{gs} = gate to substrate voltage

V_t = threshold voltage

→ the charge for unit area can be written as

$Q_c = E_g \varepsilon_0 . \varepsilon_{ins}$

→ The effective charge $Q_c = E_g \varepsilon_0 \varepsilon_{ins} WL$

ε_0 = permittivity of free space = 8.85 x 10^{-14} F/cm

ε_{ins} = permitivity of insulating material = 4.0 ⇒ $\frac{\varepsilon_{ins}}{\varepsilon_0}$

E_g = electric field ~~from~~ at gate terminal

$$\therefore E_g = \frac{V_g - \frac{V_{ds}}{2}}{D}$$

$$E_g = \frac{(V_{gs} - V_t) - \frac{V_{ds}}{2}}{D} \rightarrow ⑥$$

Substitute eq'n ⑥ in effective charge Q_c then

$$Q_c = \frac{(V_{gs} - V_t) - \frac{V_{ds}}{2}}{D} \cdot \varepsilon_0 \varepsilon_{ins} WL \rightarrow ⑦$$

From eq'n ①, ⑤, ⑦

$$I_{ds} = \frac{\frac{(V_{gs} - V_t) - \frac{V_{ds}}{2}}{D} \cdot \varepsilon_0 \varepsilon_{ins} WL}{\frac{L^2}{\ldots}}$$

$$I_{ds} = \frac{Q_c}{T_{ds}} = \frac{\varepsilon_0 \varepsilon_{ins} \cdot \mu}{D} \cdot \frac{W}{L}\left[(V_{gs}-V_t) - \frac{V_{ds}}{2}\right] V_{ds}$$

$$\boxed{I_{ds} = \frac{\varepsilon_0 \varepsilon_{ins} \cdot \mu}{D} \cdot \frac{W}{L}\left[(V_{gs}-V_t) V_{ds} - \frac{V_{ds}^2}{2}\right]} \rightarrow (8)$$

Sub $\frac{\varepsilon_0 \varepsilon_{ins}\, \mu}{D} = K$

$$I_{ds} = K \cdot \frac{W}{L}\left[(V_{gs}-V_t) V_{ds} - \frac{V_{ds}^2}{2}\right] \rightarrow (9)$$

$$K \cdot \frac{W}{L} = \beta$$

Sub in eq'n (9)

$$I_{ds} = \beta\left[(V_{gs}-V_t) V_{ds} - \frac{V_{ds}^2}{2}\right] \rightarrow (10)$$

→ the Ids and Vds relation can be defined in terms of the gate capacitance and gate capacitance per unit area can be defined as $C_g = \frac{\varepsilon_0 \varepsilon_{ins} WL}{D}$

Substitute C_g in eq'n (8) then, we get

$$I_{ds} = C_g \cdot \frac{\mu}{L^2}\left[(V_{gs}-V_t) V_{ds} - \frac{V_{ds}^2}{2}\right] \rightarrow (11)$$

→ In terms of gate capacitance per unit area

$C_g = C_0 WL$

C_g is sub in eq'n (11)

$$\boxed{I_{ds} = \frac{C_0 W \mu}{L}\left[(V_{gs}-V_t) V_{ds} - \frac{V_{ds}^2}{2}\right]} \rightarrow (12)$$

For Saturation region:-

⇒ If the transistor is in saturation we consider

$V_{ds} = V_{gs} - V_t$

From eq'n (8)

$$I_{ds} = \frac{\varepsilon_0 \varepsilon_{ins} \mu}{D} \cdot \frac{W}{L}\left[(V_{gs}-V_t)(V_{gs}-V_t) - \frac{(V_{gs}-V_t)^2}{2}\right]$$

$$= \frac{\varepsilon_0 \varepsilon_{ins} \mu}{D} \cdot \frac{W}{L}\left[(V_{gs}-V_t)^2 - \frac{(V_{gs}-V_t)^2}{2}\right]$$

$$I_{ds} = \frac{\varepsilon_0 \varepsilon_{ins} \mu}{D} \cdot \frac{W}{L}\left[\frac{(V_{gs}-V_t)^2}{2}\right] \rightarrow (13)$$

Sub $K = \frac{\varepsilon_0 \varepsilon_{ins} \mu}{D}$

$$I_{ds} = \cancel{\varepsilon_0}\, K \cdot \frac{W}{L}\left[\frac{(V_{gs}-V_t)^2}{2}\right] \rightarrow (14)$$

Sub in eq'n (9)

$$K \cdot \frac{W}{L} = \beta$$

$$I_{ds} = \beta\left[\frac{(V_{gs}-V_t)^2}{2}\right] \rightarrow (15)$$

In terms of C_g

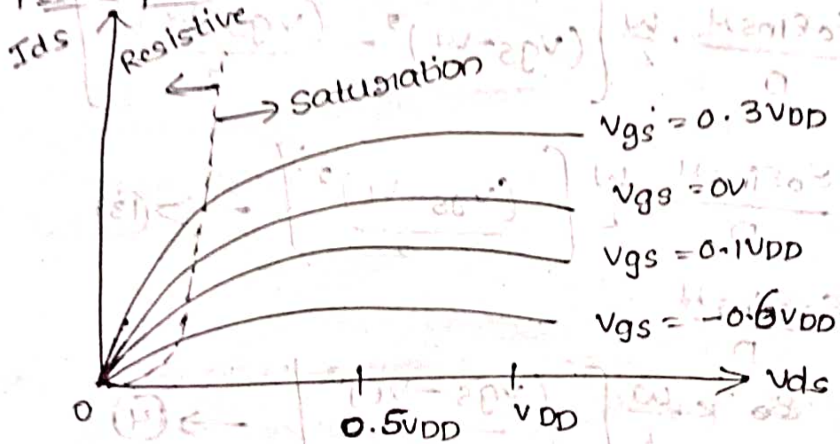
$$I_{ds} = C_g \cdot \frac{\mu}{L^2}\left[\frac{(V_{gs}-V_t)^2}{2}\right] \rightarrow (16)$$

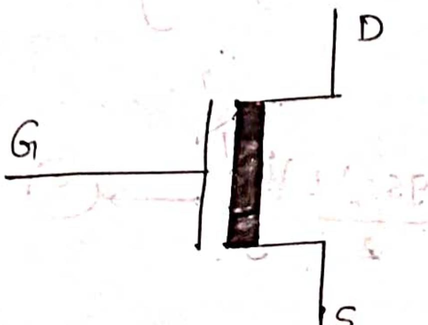
In terms of C_0

$$I_{ds} = C_0 \mu \frac{W}{L}\left[\frac{(V_{gs}-V_t)^2}{2}\right] \rightarrow (17)$$

Where we have two types of transistors such as enhancement and depletion mode. let's consider N-mos transistor the threshold voltage for the depletion transistor will be negative. For enhancement mode transistor the threshold voltage is positive. mode transistor the threshold voltage is positive. let consider the response of the transistor as shown in bellow figure

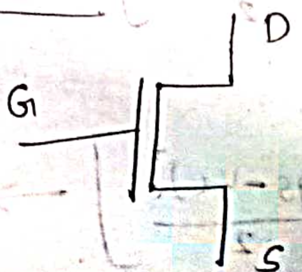
For Depletion:-



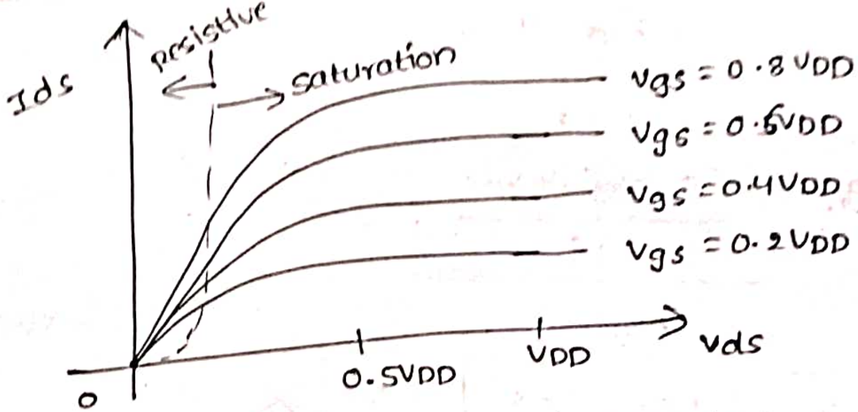


Depletion mode N-mos

For enhancement:-



Enhancement mode N-mos



Transconductance (gm) and output Conductance (gds):-

The transconductance can be defined as the ratio of output current and the input voltage (Vgs) at constant (Vds).

$$g_m = \frac{\delta I_{ds}}{\delta V_{gs}} \Big| \text{constant } V_{ds}$$

Where ~~Ids~~ $I_{ds} = \frac{Q_c}{\tau_{ds}}$

Therefore, $\delta I_{ds} = \frac{\delta Q_c}{\tau_{ds}}$

Where $\tau_{ds} = \frac{L^2}{\mu V_{ds}}$

$$\delta I_{ds} = \frac{\delta Q_c}{\frac{L^2}{\mu V_{ds}}}$$

$$\delta I_{ds} = \frac{\delta Q_c \, \mu . V_{ds}}{L^2}$$

$$\delta Q_c = C_g . \delta V_{gs}$$

$$\delta I_{ds} = \frac{C_g \cdot \delta V_{gs}\, \mu \cdot V_{ds}}{L^2}$$

$$g_m = \frac{\dfrac{C_g \cdot \cancel{\delta V_{gs}}\, \mu \cdot V_{ds}}{L^2}}{\cancel{\delta V_{gs}}}$$

$$\boxed{g_m = \frac{C_g \cdot \mu \cdot V_{ds}}{L^2}} \longrightarrow (1)$$

Initially $C_g = \frac{\varepsilon_0 \varepsilon_{ins} WL}{D}$

$$g_m = \frac{\varepsilon_0 \varepsilon_{ins} W\cancel{L}}{D} \cdot \frac{\mu V_{ds}}{L^{\cancel{2}}}$$

$$\boxed{g_m = \frac{\varepsilon_0 \varepsilon_{ins} \mu}{D} \cdot \frac{W}{L} \cdot V_{ds}} \longrightarrow (2)$$

Where $K = \frac{\varepsilon_0 \varepsilon_{ins} \mu}{D}$

$$\boxed{g_m = K \frac{W}{L} V_{ds}} \Rightarrow K\frac{W}{L} = \beta \longrightarrow (3)$$

$$\boxed{g_m = \beta V_{ds}} \longrightarrow (4)$$

In the transistor is in saturation region

$$\boxed{V_{ds} = V_{gs} - V_t} \longrightarrow (5)$$

above eq'n sub in eq'n (1)

Therefore $\boxed{g_m = \frac{C_g \cdot \mu \cdot V_{gs} - V_t}{L^2}}$

In terms C_g in eq'n (2)

$$\boxed{g_m = \frac{\varepsilon_0 \varepsilon_{ins} \mu}{D} \cdot \frac{W}{L}(V_{gs} - V_t)}$$

Sub eq'n (5) in eq'n (3) then

$$\boxed{g_m = K \frac{W}{L} (V_{gs} - V_t)}$$

Sub eq'n (5) in eq'n (4) then

$$\boxed{g_m = \beta(V_{gs} - V_t)}$$

Generally we required more transconductance that can be possible by increase the width of the transistor. Because of these the size of the transistor will increases and the input capacitance is also increases. The another possibility to improve the transconductance is by decreasing the length of transistor. Because of these the gain will decreases aswell as the o/p conductance is also decreases.

where the o/p conductance (g_{ds}) can be defined as the ratio of change in o/p current to the change in i/p voltage.

$$g_{ds} = \frac{\delta I_{ds}}{\delta V_{gs}}$$

$\lambda \cdot I_{ds} \propto \frac{1}{L^2}$

$\lambda \propto \frac{1}{L}$ & $I_{ds} \propto \frac{1}{L}$

$\longrightarrow$ o/p Trans-conductance

Transistor figure of Merit (W_0) :-

$$W_0 = \frac{g_m}{C_g}$$

$$W_0 = \frac{\cancel{C_g} \cdot \mu \cdot V_{ds} / L^2}{\cancel{C_g}}$$

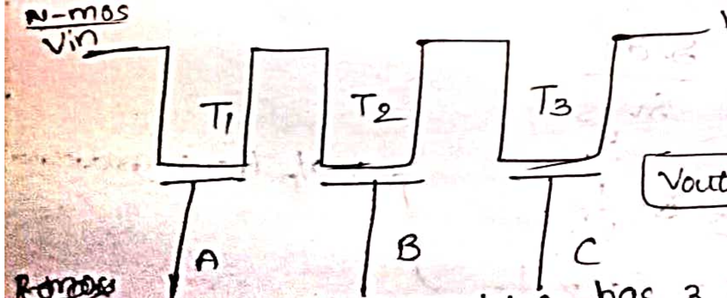
$$\boxed{W_0 = \frac{\mu \cdot V_{ds}}{L^2}}$$

Therefore, FOM (or) speed of MOS transistor is depends on the mobility of the charge carriers, drain to source voltage and inversely proportional square of the length of the channel.

Pass Transistors:-

The MOS transistors can be used as switch. let us consider a multiple transistors which are connected in series format which can carriers a different logic levels. The arrangement of these transistors are known as pass Transistors.

N-mos

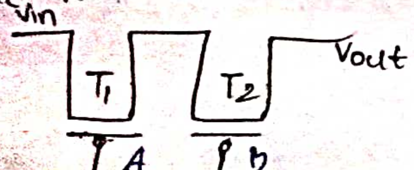


$\boxed{V_{out} = ABC}$ → AND

A	B	C	Vout
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

~~Rmos~~ Where N-mos transistor has 3 input and gate

P-mos



Truth Table

A	B	Vout
0	0	1
0	1	0
1	0	0
1	1	0

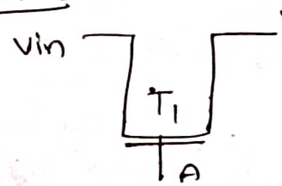
$Y = \overline{A}\,\overline{B}$

$Y = \overline{A+B} \rightarrow$ NOR

$\boxed{V_{out} = \overline{A+B}}$

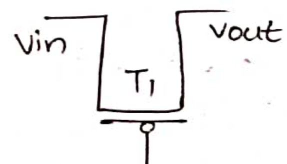
Where p-mos transistor has 2 input NOR gate.

For N-MOS :-



When the i/p '0' then the o/p produces '0' as a o/p in N-mos. That is known as buffer.

For P-mos :-



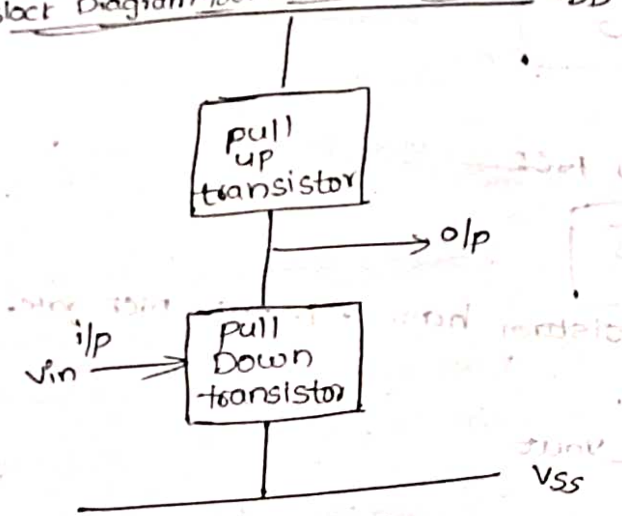
When the i/p is '0' then the o/p produces '1'. Then it is known as NOT gate.

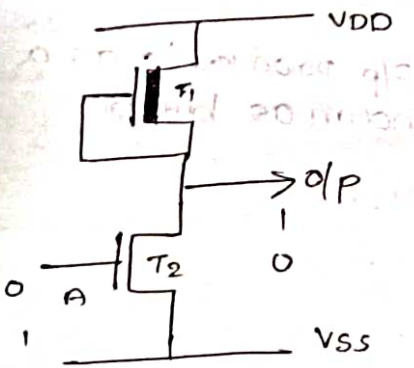
N mos Inverter:-

The inverter very much important in the designing of any logic circuits specially in combinational and special logic circuits. We use a different inverters.

→ To design N-mos inverter we require '1' enhancement mode N-mos transistor and '1' depletion mode N-mos transistor.

Block Diagram for N-mos Inverter:





∴ The o/p is inverter

Truth Table :-

I/p (A)	o/p
0	1
1	0

Steps:-

① The output draws no current when both the transistors are in ON state

② If vin = 0 the Transistor T_2 OFF state can it acts as open switch then the o/p becomes '1'.

③ If vin = 1 in the Transistor T_2 ON state can it acts as closed switch then the o/p becomes '0'.

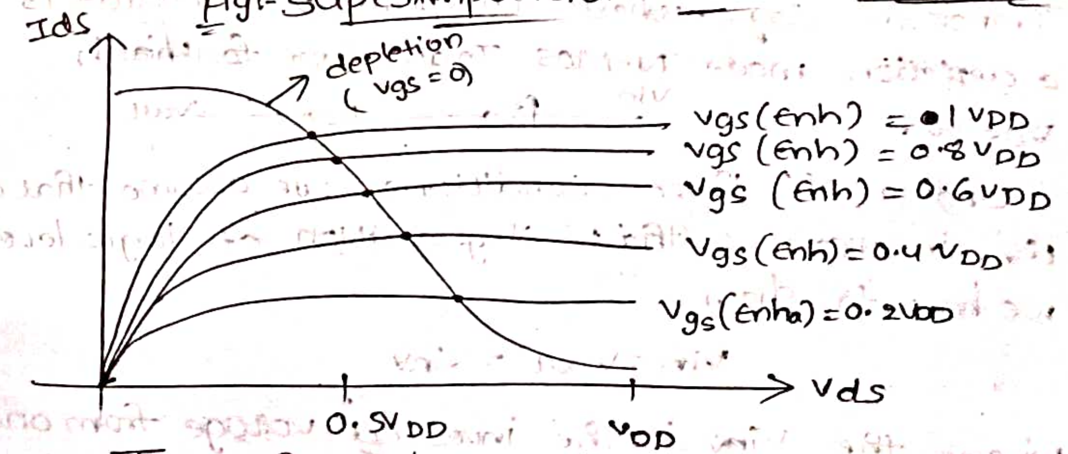
④ From the circuit the depletion mode N-mos transistor is known as pull up transistor.

⑤ The enhancement mode N-mos transistor is pull down transistor.

⑥ Where for the pullup transistor the gate to source voltage is $\boxed{V_{gs} = 0}$

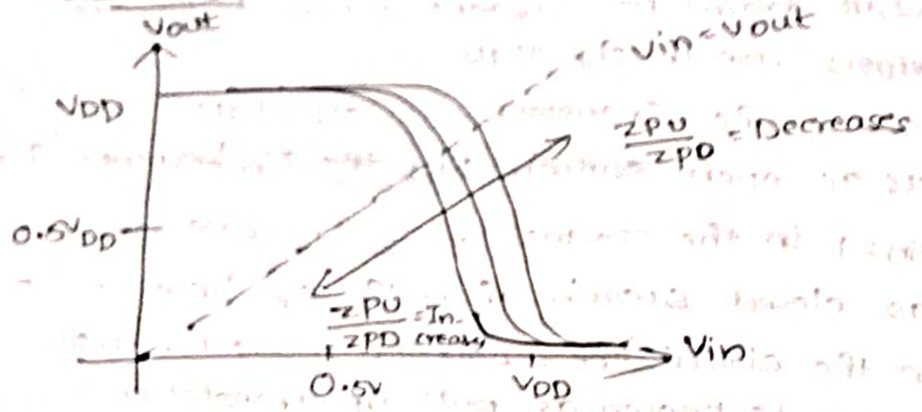
⑦ The transfer characteristics of N-mos inverter can be obtain by the superimpose of the transfer characteristics of pull up transistor on to the transfer characteristics of pull down transistor.

Fig:- Superimposition of Enh in deple mode



Transfer characteristics

Inverter graph:



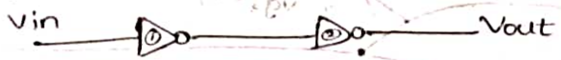
→ From the transfer characteristics of the N-mos inverter we can observe the response is changes based on $\frac{ZPU}{ZPD}$ ratio.

→ The gain of the transfercharacteristics can be obtained as gain = $\frac{\delta V_{ds}}{\delta V_{gs}}$

100% V.V.Imp pull-up to pull-down ratio:-

→ pull-up to pull-Down ratio of a N-mos inverter driven by another N-mos inverter:-

→ Consider the following arrangement using N-Mos Transistor logic. where the pull-up transistor is a depletion mode N-MOS Transistor for which $V_{gs} = 0$.



→ Under all these conditions we assume that all the inverters without degradation of logic levels we have to show

$\therefore V_{in} = V_{out} = V_{inv}$

where the V_{inv} is the inverted voltage from one level to another level.

→ For equal margins around threshold voltage of the inverter we will consider $\boxed{V_{inv} = 0.5V_{DD}}$

→ If the transistors are in saturation then

$$I_{ds} = k\frac{W}{L}\frac{(V_{gs}-V_t)^2}{2} \longrightarrow ①$$

→ If the pull-up transistor is depletion mode then the

$$I_{ds} = k.\frac{W_{P.U}}{L_{P.U}}\frac{(-V_{td})^2}{2} \longrightarrow ②$$

→ If the pull-down transistor in enhancement mode then the

$$I_{ds} = k\frac{W_{P.D}}{L_{P.D}}.\frac{(V_{inv}-V_t)^2}{2} \longrightarrow ③$$

From eq'n ② & ③

$$\cancel{k}.\frac{W_{P.U}}{L_{P.U}}\frac{(-V_{td})^2}{\cancel{2}} = \cancel{k}\frac{W_{P.D}}{L_{P.D}}\frac{(V_{inv}-V_t)^2}{\cancel{2}}$$

$$Z = \frac{L}{W}$$

$$\frac{1}{Z_{P.U}}(-V_{td})^2 = \frac{1}{Z_{P.D}}(V_{inv}-V_t)^2$$

Apply square root on b.s

$$\frac{1}{\sqrt{Z_{P.U}}}(-V_{td}) = \frac{1}{\sqrt{Z_{P.D}}}(V_{inv}-V_t)$$

$$V_{inv} - V_t = \sqrt{\frac{Z_{P.D}}{Z_{P.U}}}(-V_{td})$$

$$V_{inv} - V_t = \frac{1}{\sqrt{\frac{Z_{P.U}}{Z_{P.D}}}}(-V_{td})$$

$$V_{inv} = V_t - \frac{V_{td}}{\sqrt{\frac{Z_{P.U}}{Z_{P.D}}}}$$

lets us consider some typical values such as

$V_{inv} = 0.5 V_{DD}$

$V_t = 0.2 V_{DD}$

$V_{td} = -0.6 V_{DD}$

$$0.5V_{DD} = 0.2V_{DD} + \frac{0.6V_{DD}}{\sqrt{\frac{Z_{P.U}}{Z_{P.D}}}}$$

$$0.3V_{DD} = \frac{0.6V_{DD}}{\sqrt{\frac{Z_{P.U}}{Z_{P.D}}}}$$

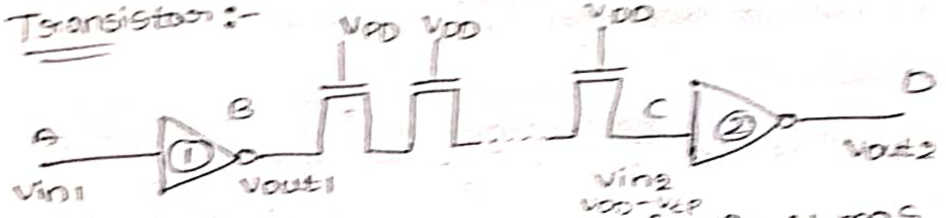
$$0.3\sqrt{\frac{Z_{P.U}}{Z_{P.D}}} = \frac{0.6\cancel{V_{DD}}}{0.3\cancel{V_{DD}}}$$

$$\sqrt{\frac{Z_{P.U}}{Z_{P.D}}} = 2$$

Square on b.s

$$\boxed{\frac{Z_{P.U}}{Z_{P.U}} = \frac{4}{1}}$$

pullup to pull-Down ratio of a N-mos inverter driven by another N-mos inverter through pass Transistor :-



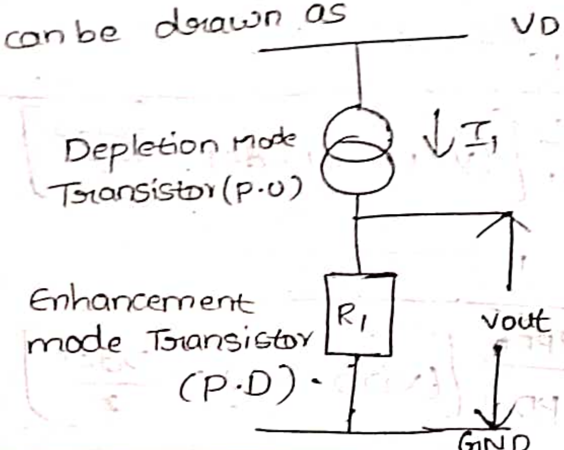
lets us consider the arrangement for the N-mos transistor logic is shown in above

From the diagram we assign 4 nodes ABCD. where A is the i/p of the 1st inverter, B is o/p of the 1st inverter, C is the i/p of the 2nd inverter, D is the o/p of the 2nd inverter.

For the ideal case at Node A the i/p voltage is at 'O' and Node B the o/p we got the value of VDD(1), at node c we have an i/p of $V_{DD} - V_{tp}$ where V_{tp} → threshold voltage of the pass transistors

i.e; $\boxed{V_{in2} = V_{DD} - V_{tp}}$

→ let consider the 1st inverter the equivalent circuit can be drawn as



From the ckt the o/p voltage V_{out1} is

$$\boxed{V_{out1} = I_1 R_1}$$

From the pull-up transistor the current I_1 can be obtained as

$$I_1 = I_{ds} = k\frac{\omega}{L}\left[\frac{(V_{gs}-V_t)^2}{2}\right]$$

where the pull-up transistor is a depletion mode N-mos then $\boxed{V_{gs}=0}$

$$\therefore\ I_1 = I_{ds} = k\frac{W_{PU1}}{L_{PU1}}\frac{(-V_{td})^2}{2} \longrightarrow (1)$$

From the pull-down transistor the I_{ds1}

$$I_{ds1} = k\cdot\frac{\omega}{L}\left[(V_{gs}-V_t)V_{ds} - \frac{V_{ds}^2}{2}\right]$$

$$I_{ds1} = k\frac{W_{PD1}}{L_{PD1}}\left[(V_{DD}-V_t)V_{ds1} - \frac{V_{ds1}^2}{2}\right]$$

according to ohm's law

$$\therefore R_1 = \frac{V_{ds1}}{I_{ds1}} \longrightarrow (2)$$

$$R_1 = \frac{V_{ds1}}{k\cdot\frac{W_{PD1}}{L_{PD1}}\left[(V_{DD}-V_t)V_{ds1} - \frac{V_{ds1}^2}{2}\right]}$$

$$R_1 = \frac{1}{k\cdot\frac{W_{PD1}}{L_{PD1}}\left[(V_{DD}-V_t) - \frac{V_{ds1}}{2}\right]}$$

$$V_{out1} = \left[\cancel{k}\frac{W_{PU1}}{L_{PU1}}\frac{(-V_{td})^2}{2}\right]\cdot\frac{1}{\cancel{k}\cdot\frac{W_{PD1}}{L_{PD1}}\left[(V_{DD}-V_t)-\frac{V_{ds1}}{2}\right]}$$

$$V_{out1} = \frac{1}{Z_{PU1}}\frac{(-V_{td})^2}{2}\times Z_{PD1}\times\frac{1}{\left[(V_{DD}-V_t)-\frac{V_{ds1}}{2}\right]}$$

where V_{ds} is $< V_{DD}$ then $\frac{V_{ds}}{2} << V_{DD}$

So, neglect the $\frac{V_{ds1}}{2}$

$$V_{out1} = \frac{1}{Z_{PU1}}\frac{(-V_{td})^2}{2}\times Z_{PD1}\frac{1}{(V_{DD}-V_t)}$$

$$V_{out1} = \frac{Z_{PD1}}{Z_{PU1}}\frac{(-V_{td})^2}{2[V_{DD}-V_t]} \longrightarrow (3)$$

Similarly, for the 2nd inverter

$$I_2 = I_{ds} = k\frac{W_{PU2}}{L_{PU2}}\frac{(-V_{td})^2}{2}$$

$$R_2 = \frac{1}{k\cdot\frac{W_{PD2}}{L_{PD2}}\left[(V_{DD}-V_{tp}-V_t)\right]}$$

$$V_{out2} = I_2 R_2$$

$$V_{out2} = \frac{Z_{PD2}}{Z_{PU2}}\frac{(-V_{td})^2}{2[V_{DD}-V_{tp}-V_t]} \longrightarrow (4)$$

Now, we can equate the V_{out1} & V_{out2}

$V_{out_1} = V_{out_2}$

$$\frac{Z_{PD1}}{Z_{PU1}} \cdot \frac{(-V_{td})^2}{2(V_{DD}-V_t)} = \frac{Z_{PD2}}{Z_{PU2}} \cdot \frac{(-V_{td})^2}{2[V_{DD}-V_{tp}-V_t]}$$

$$\frac{Z_{PU2}}{Z_{PD2}} = \frac{Z_{PU1}}{Z_{PD1}}\left[\frac{V_{DD}-V_t}{V_{DD}-V_{tp}-V_t}\right]$$

let us consider some typical values such as

$V_t = 0.2V_{DD}$

$V_{tp} = 0.3V_{DD}$

$$\frac{Z_{PU2}}{Z_{PD2}} = \frac{Z_{PU1}}{Z_{PD1}}\left[\frac{V_{DD}-0.2V_{DD}}{V_{DD}-0.3V_{DD}-0.2V_{DD}}\right]$$

$$\frac{Z_{PU2}}{Z_{PD2}} = \frac{Z_{PU1}}{Z_{PD1}}\left[\frac{0.8V_{DD}}{0.5V_{DD}}\right]$$

$$\frac{Z_{PU2}}{Z_{PD2}} = \frac{Z_{PU1}}{Z_{PD1}}[1.6]$$

$$\frac{Z_{PU2}}{Z_{PD2}} = 2 \cdot \frac{Z_{PU1}}{Z_{PD1}}$$

$$\frac{Z_{PU2}}{Z_{PD2}} = 2\left(\frac{4}{1}\right) \quad \boxed{\because \frac{Z_{PU1}}{Z_{PD1}} = \frac{4}{1}}$$

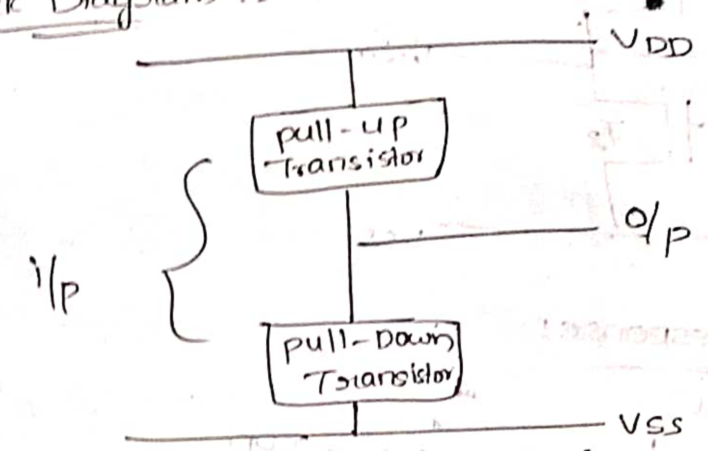
$$\boxed{\frac{Z_{PU2}}{Z_{PD2}} = \frac{8}{1}}$$

Alternate forms of pull-up:-

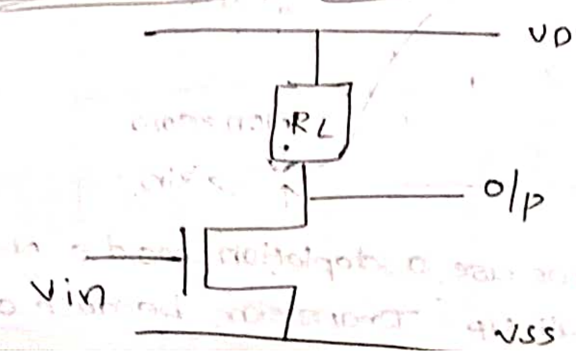
there are different types of pull-up transistors such as

1. load resistor as a pull-up
2. Depletion mode N-MOS Transistor as a pull-up Transistor
3. Enhancement mode N-MOS Transistor as a pull-up Transistor.
4. Enhancement mode P-MOS Transistor as a pull-up Transistor

Block Diagram :-



load resistor as a pull-up Transistor :-



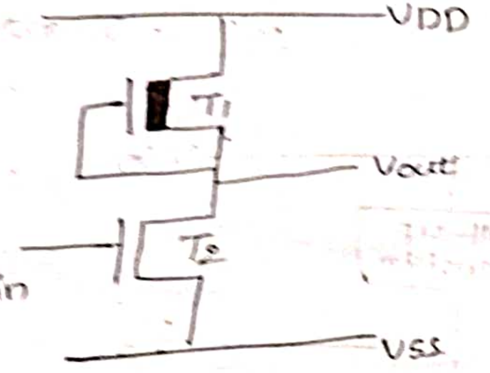
Transistor

... we use a load resistor as a pull-up [illegible] because of these load resistor we have the following drawbacks such as

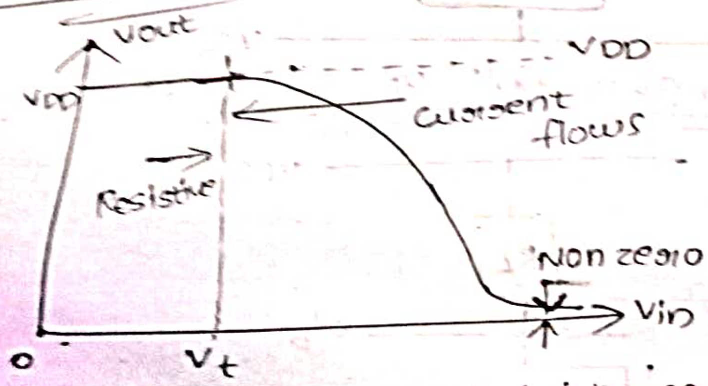
① It has more voltage drop
② The area of occupance is more.
③ It is very difficult to fabricate.

Depletion mode N-MOS Transistor as a pull-up:-

Circuit Diagram:-



i/p and o/p response:-

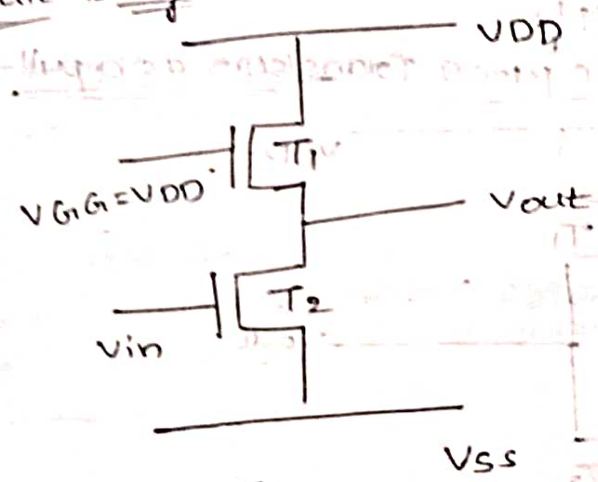


In this logic we use a depletion mode N-MOS Transistor as a pull-up Transistor because of these following drawbacks such as

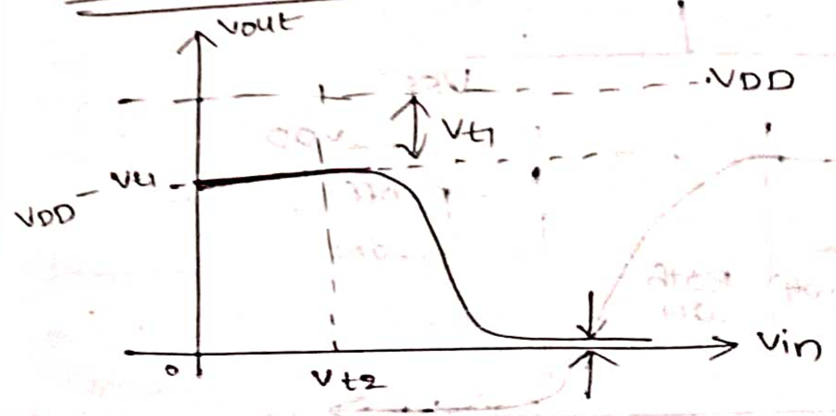
① The power desipation is more in b/w rail to rail (or) VDD to VSS when the Vin = 1
② The o/p logic level will changes from 1 to 0 when the there i/p voltage is greater than the threshold voltage (Vt).
③ In this case the output voltage will never reach to logic '0'.

Enhancement mode N-MOS Transistor as a pull-up:

Circuit Diagram:-



i/p & o/p response:-



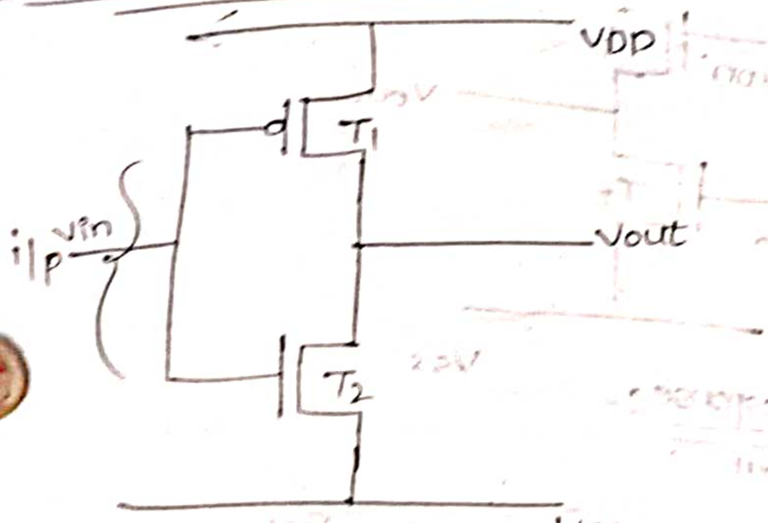
In this logic we use enhancement mode N-mos Transistor as a pull-up Transistor because of the following drawbacks such as.

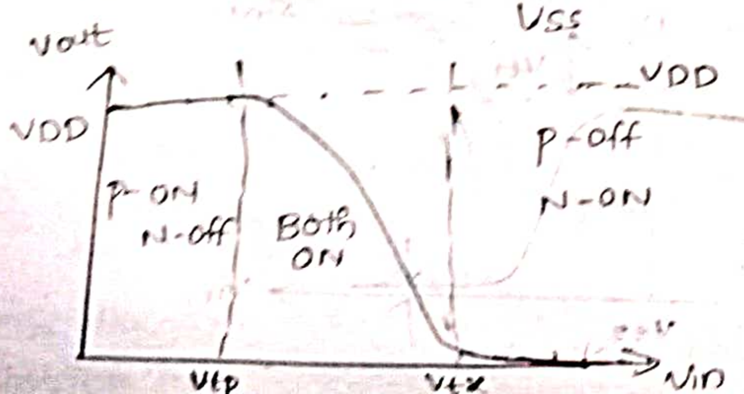
① power desipation is more in b/w rail to rail when the i/p $V_{in} = 1$.

② the o/p is never reach to the maximum V_{DD} we will get maximum of $V_{DD} - V_{t1}$. Where the V_{t1} is the threshold voltage of the 1st pull-up Transistor.

③ If V_{gg} → the i/p of the pull-up transistor. The V_{gg} is greather than VDD the circuit require more power supply.

Enhancement mode PMOS Transistor as a pull-up:





→ In this logic we use enhancement P-mos Transistor as a pull-up and these arrangement is called as CMOS Logic.

→ In this logic we have following advantages such as

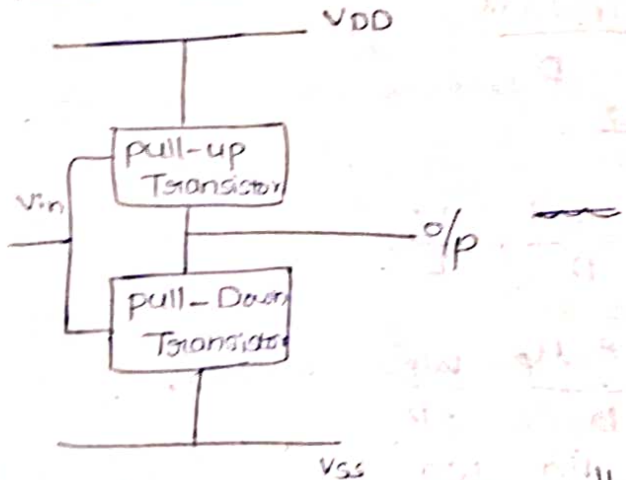
① The power desipation is less because of no direct path is establish in b/w VDD and Vss

② we will get max o/p of VDD and min o/p of zero.

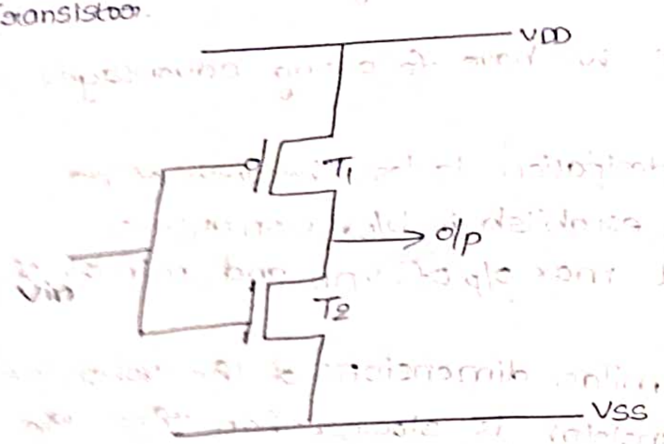
③ for the similar dimensions of the transistors the p-mos transistor is slower than then the N-mos Transistor.

CMOS Inverter:-

The CMOS logic circuit can be design by the combination of enhancement P-Mos and enhancement mode N-mos transistor.



The CMOS inverter will have 1 pull-up Transistor and 1 pull down Transistor where pull-up transistor is enhancement mode P-MOS and the pull down Transistor has enhancement Mode N-mos Transistor.



If the transistors are in Non-Saturation then, the current $I_{ds} = K \frac{W}{L}\left((V_{gs}-V_t)V_{ds} - \frac{V_{ds}^2}{2}\right)$

If the transistors are in Saturation then, the current

$$I_{ds} = K \cdot \frac{W}{L} \frac{(V_{gs}-V_t)^2}{2}$$

where $K = \frac{\varepsilon_{ins}\varepsilon_0 \mu}{D}$

$$\beta = K \frac{W}{L}$$

$$\beta = \frac{\varepsilon_{ins}\varepsilon_0 \mu}{D} \cdot \frac{W}{L}$$

For p-MOS, $\beta_p = \frac{\varepsilon_{ins}\varepsilon_0 \mu_p}{D} \cdot \frac{W_p}{L_p}$

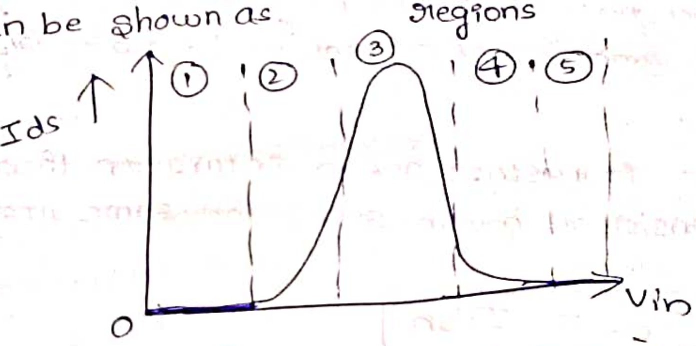
For N-MOS, $\beta_n = \frac{\varepsilon_{ins}\varepsilon_0 \mu_n}{D} \cdot \frac{W_n}{L_n}$

where, μ_p and μ_n are Mobility's of p-MOS & N-MOS Transistors respectively

W_p & L_p are the width and length of the p-MOS Transistor respectively.

W_n & L_n are the width and length of the N-MOS Transistor respectively.

→ The current response for a C-MOS inverter can be shown as



→ In region ① the i/p voltage is exact zero then the p-MOS transistor is ON & N-MOS transistor is off then there is no current flow from VDD to VSS and the o/p voltage has exactly VDD.

→ In region ⑤ the i/p voltage is at one (logic 1) then the N-MOS transistor is ON & p-MOS transistor is off - there is no current flow from VDD to Vss and the o/p has directly connected to ground ($V_{out}=0$)

→ In region ② we are increase the i/p voltage is greater than V_{tn} the p-mos transistor is completely in saturation and the N-mos transistor is slightly in saturation then low amount of current will flow in b/w VDD to Vss.

→ In region ④ we are increase the i/p the voltage is greater than V_{tp} the N-MOS transistor is completely saturation and P-MOS transistor is slightly in saturation then low amount current flows from VDD to VSS

→ In region ⑤ both the transistors are in saturation then a direct path is establishing b/w VDD to VSS and a large amount of current will flow in b/w VDD to VSS.

→ If the Two Transistors are in saturation then both the transistors are in series the same amount of current will flow

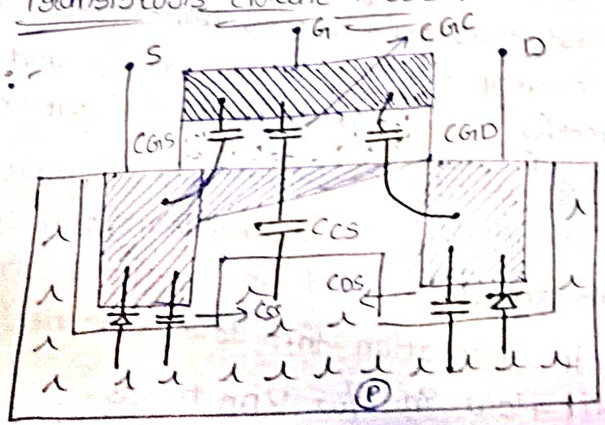
$$\boxed{I_{ds} = -I_{dsn}}$$

$$I_{dsp} = \frac{\beta}{2}\left[V_{in} - V_{DD} - V_t\right]^2$$

$$I_{dsn} = \frac{\beta}{2}\left[V_{in} - V_t\right]^2$$

MOS Transistors circuit Model:-

NMOS :-

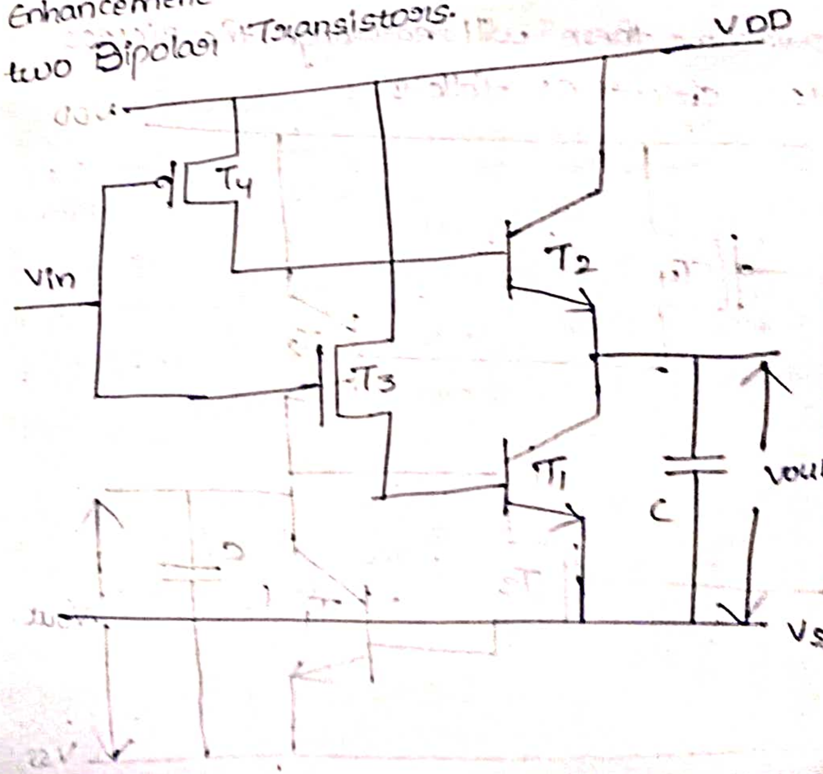


BiCMOS inverters:-

The NMOS, CMOS Transistor logics have their own advantages and we need a circuit with high o/p drive current & switching application. These two can be available in BiCMOS Transistor logic where BiCMOS is the combination of Bipolar Transistors (BJT) and CMOS Transistors.

→ The BJT Transistors are used to drive more o/p drive current and CMOS Transistor is used for switching applications.

→ let us consider BiCMOS inverter with use one Enhancement mode p-MOS, N-MOS Transistor and two Bipolar Transistors.



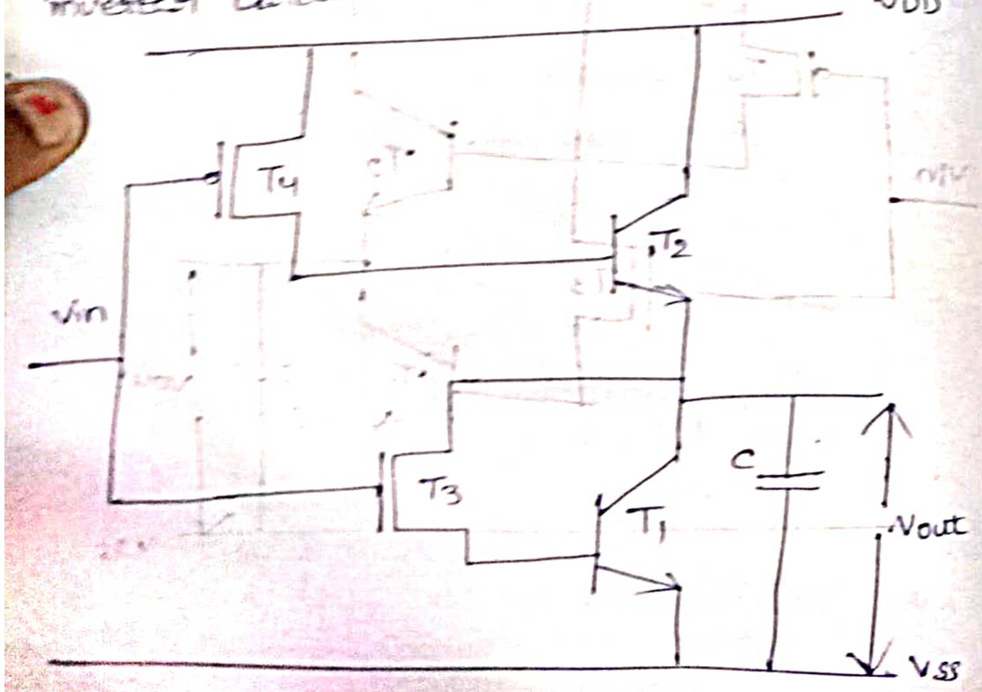
Operation:

1) If $V_{in} = 0$ then T_3 is off state, T_4 is ON state and if T_3 is off then T_1 is also off and if T_4 is ON then T_2 will ON then the o/p becomes VDD and the capacitor 'C' will charges to its max VDD.

2) If $V_{in} = 1$ then T_3 is ON state, T_4 is off state. If T_3 is ON then T_1 is also ON and if T_4 is off, then T_2 is also off then the o/p becomes zero and the capacitor 'C' will discharges.

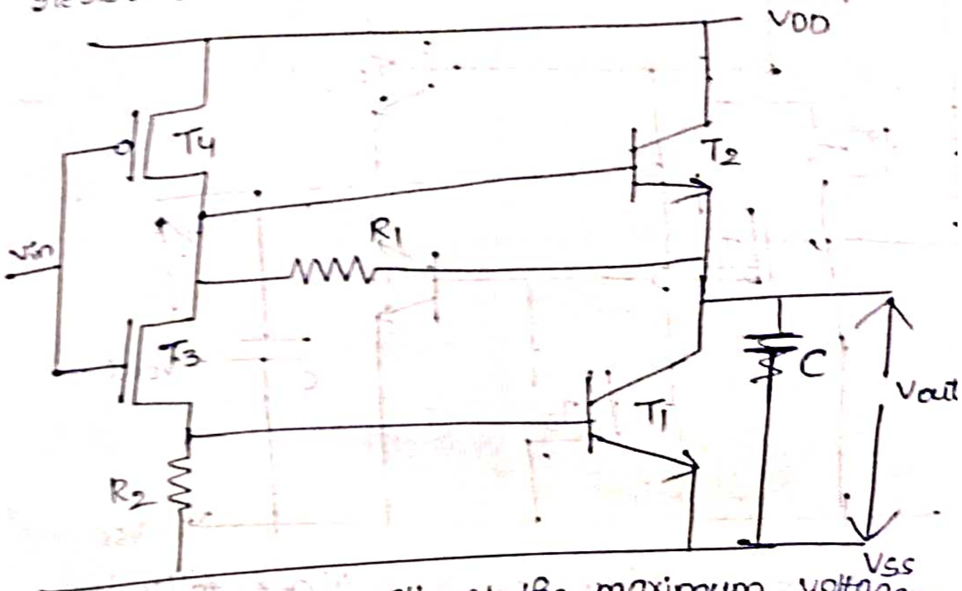
3) In this logic a direct dc path is establish b/w VDD to VSS when the i/p is at logic 1.

4) To overcome these will rearrange the BiCMOS inverter circuit as follow:



→ From these logic they perform BiCMOS inverter logic but the o/p voltage never reach the maximum voltage rails (VDD).

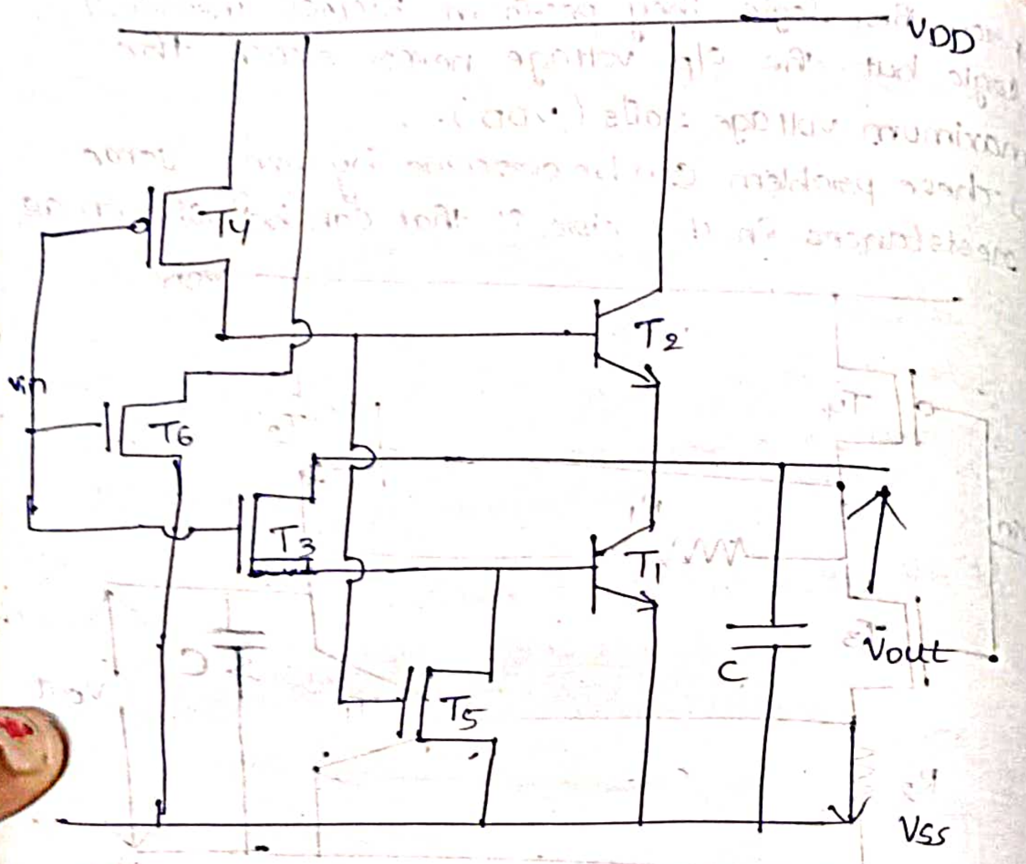
→ These problem can be overcome by using some resistances in the circuit that can be shown as



→ In this model we will get the maximum voltage rails but the circuit complexity will increases and very difficulty fabrication process.

→ And this can be avoid by replace the resistor with a enhancement mode N-MOS Transistors.

Blank



· From this BiCMOS inverter we will get following such as

① High i/p impedance
② High o/p impedance
③ High o/p drive current
④ High Noise margin

<u>Latch-up</u> in <u>MOS</u> <u>transistor</u> <u>circuit</u>:-

→ In N-MOS & P-MOS fabrication process No problem of latch-up. In N-well and P-well process more No. of junctions are form these junctions are establish some parasatic components like diodes, transistors, capacitors.

→ latch-up is the process of establish a low resistance path in b/w VDD to Vss. Because of these latch-up the power desipation is increases.

→ The latch-up is form because of improper fabrication and glitches in the voltage rails.

→ let us consider a p-well fabrication process

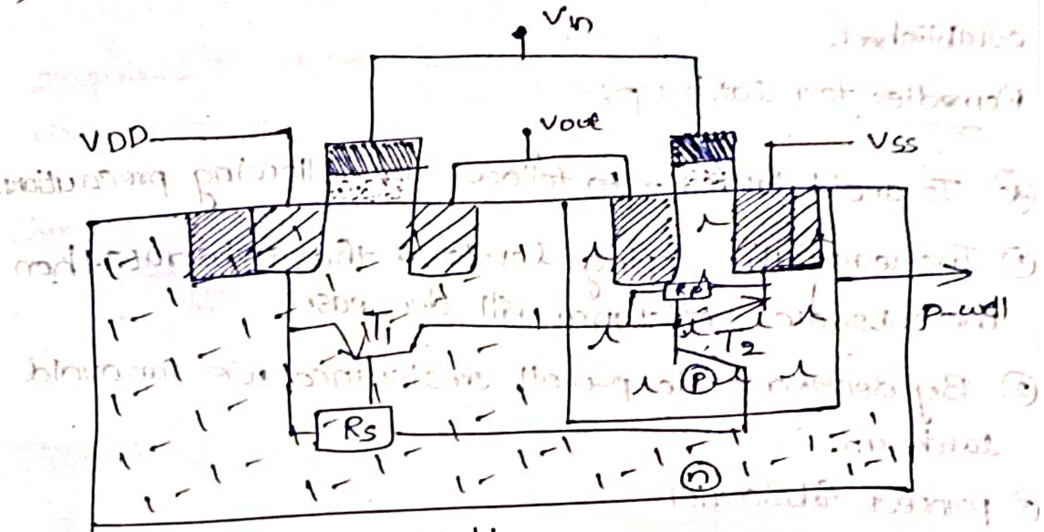


Fig:. P-Well

<u>Construction</u>:-

From the above p-well fabrication the latch-up arrangement can be form by using two Transistors T_1 and T_2 and 2 resistors Rs and Rp. where Rs is the N-substrate resistance and Rp is the p-well resistance. T_1 is the PNP Transistor and T_2 is NPN Transistor.

From the circuit substrate resistance Rs have a sufficient current to conduct, the Transistor T_1 then the transistor comes into saturation and the transistor T_2 is also draws some current from the p-well resistance R_p. Then the transistor T_2 is also ON. Then a low resistance path is establish in b/w VDD to Vss.

→ The latch is also form based on the gain of the transistor T_1 and T_2.

$\beta_1 \times \beta_2 > 1$ then the latch-up is established.

Remedies for latch-up:-

(4) To avoid latch-up to follow the following precautions.

(1) Increase the doping levels of the substrate then the substrate resistance will decrease.

(2) By decrease the p-well resistance we can avoid latch-up.

(3) proper fabrication.

latch-up-fig

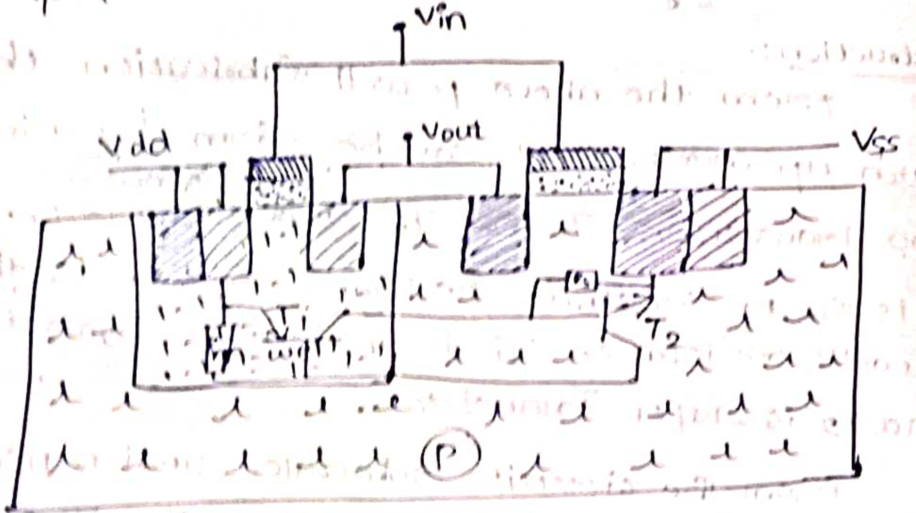


fig:- N-well using latch-up

latch-up Sustainability:-

→ In p-well and N-well process to avoid the latch-up by consider the following properties

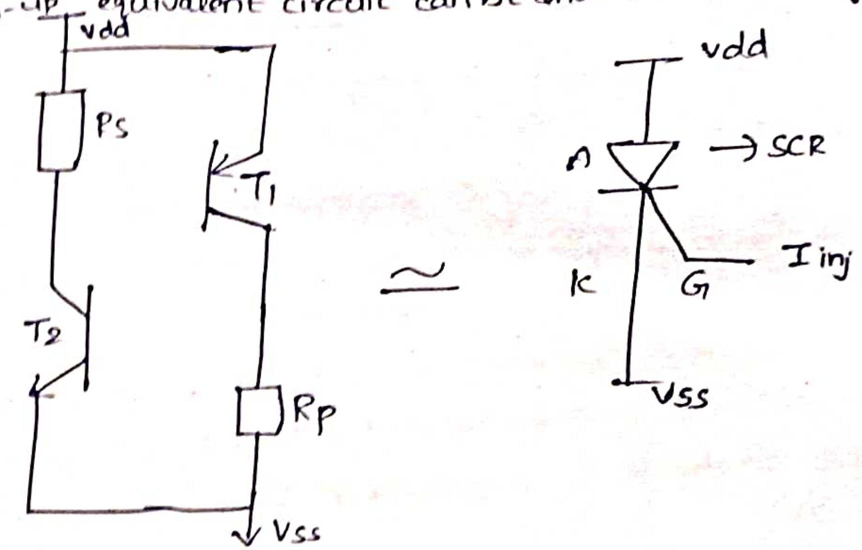
(1) Reduce the substrate resistance (Rs).

(2) By Reduce the well resistance (Rw).

(3) By reduction of Rs and Rw. The transistors requires a more amount of current to form the latch-up.

Simplified latch-up Diagram:-

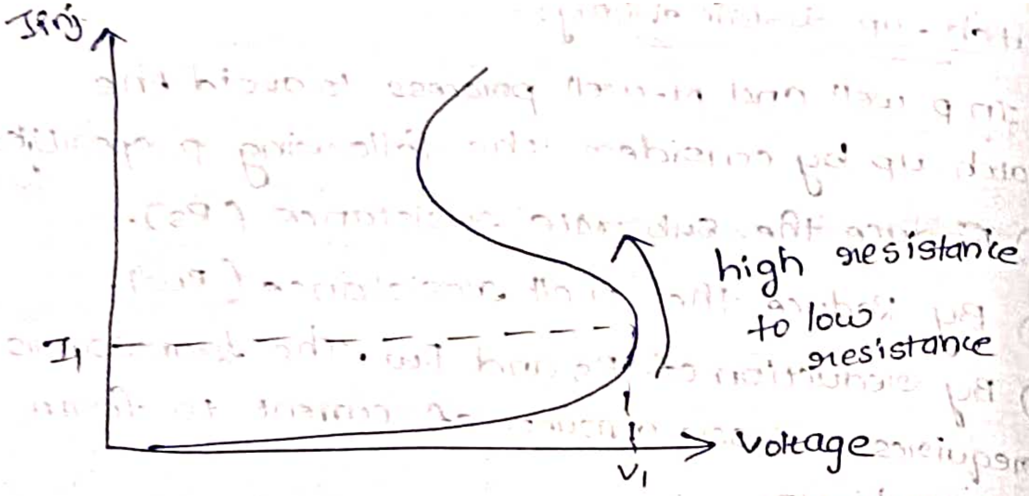
latch-up equivalent circuit can be shown as below fig.



→ If NO injected circuit then latch-up will not establish.

→ If the latch-up injected current is increases then, the latch-up will establish

→ The voltage to current response can be shown as below

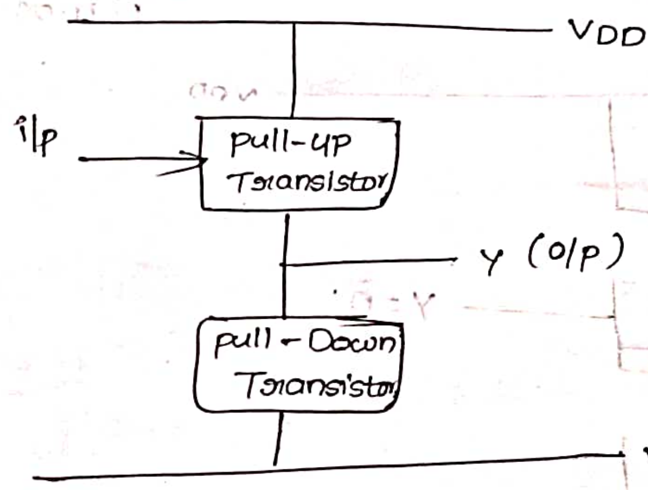


UNIT-3

MOS & BiCMOS circuit Design process

PMOS Transistor logic:-

① Block diagram:-



② All the pull-up transistors will be enhancement mode p-MOS Transistors Only.

→ All the pull-Down transistor will be depletion mode p-mode Transistors Only.

→ The Number of pull-up Transistors will be equal to the Number of i/p variables.

→ The Number of pull-Down Transistor will be only one.

→ The i/p variables can be applied to pull-up Transistors only.

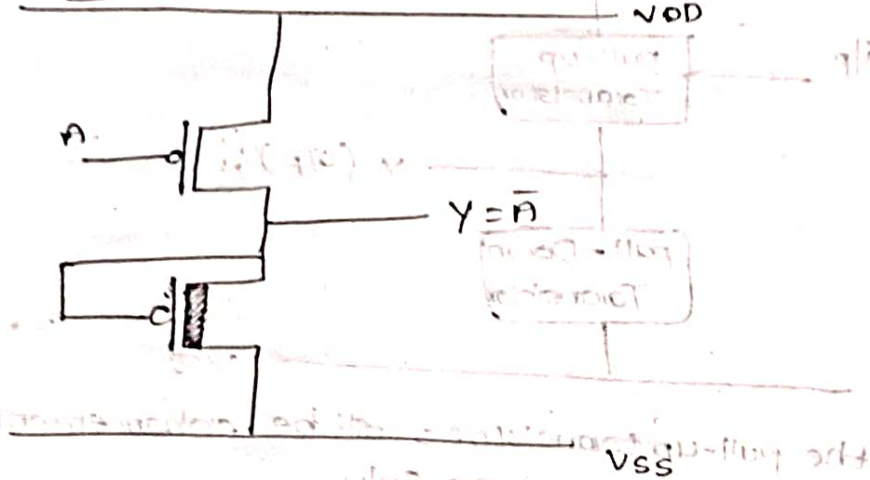
* Design an inverter using P-MOS Transistor logic.

(A) $\boxed{Y=\overline{A}}$

No.of P.U transistor = 1 (Enhancement mode P-mos)

No.of P.D transistor = 1 (Depletion mode P-mos)

Block Diagram :-



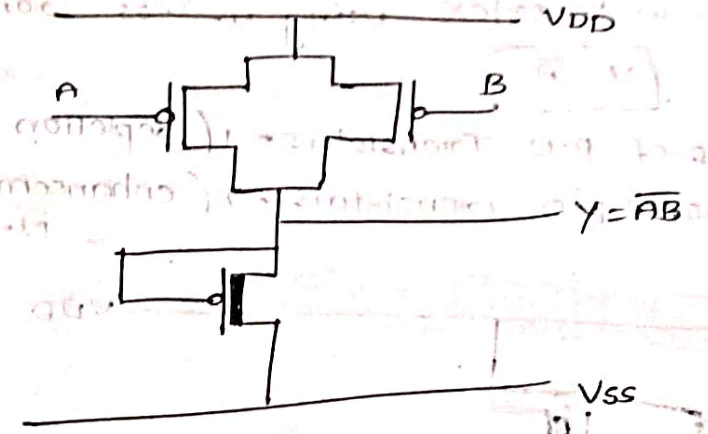
* Desing NAND Two i/p gate using P-MOS Transistor logic.

$\boxed{Y=\overline{AB}}$

No.of P.U Transistor = 2 (Enhancement P-MOS)

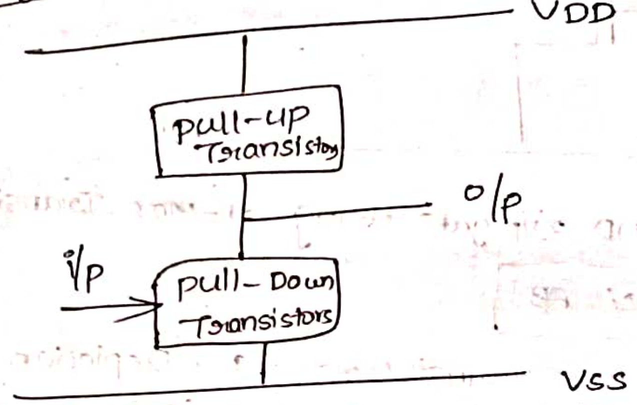
No.of P.D Transistor = 1 (Depletion P-MOS)

	PMOS	NMOS
AND	Parallel	Series
OR	Series	Parallel



N-MOS Transistor logic :-

(1) Block diagram :-



(2) All the pull-up Transistors will be depletion mode N-MOS Transistors only

→ All the pull-Down Transistors will be enhancement mode N-MOS only.

→ The pull-up Transistor will be only one.

→ The pull-down Transistors will be equal to the number of i/p variables.

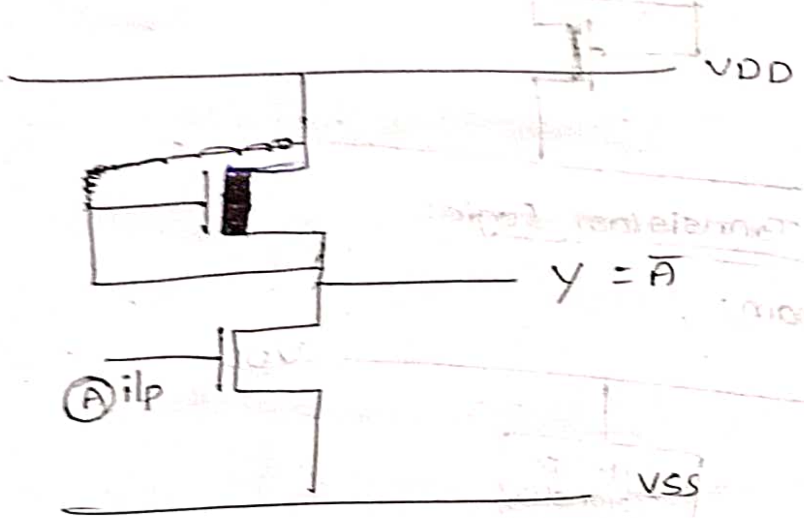
→ The i/p variables can be applied for pull-Down Transistor only.

① Design an inverter using N-MOS Transistor

$\boxed{Y = \bar{A}}$

No. of P.U Transistors = 1 (depletion N-MOS)

No. of P.D Transistors = 1 (enhancement N-MOS)

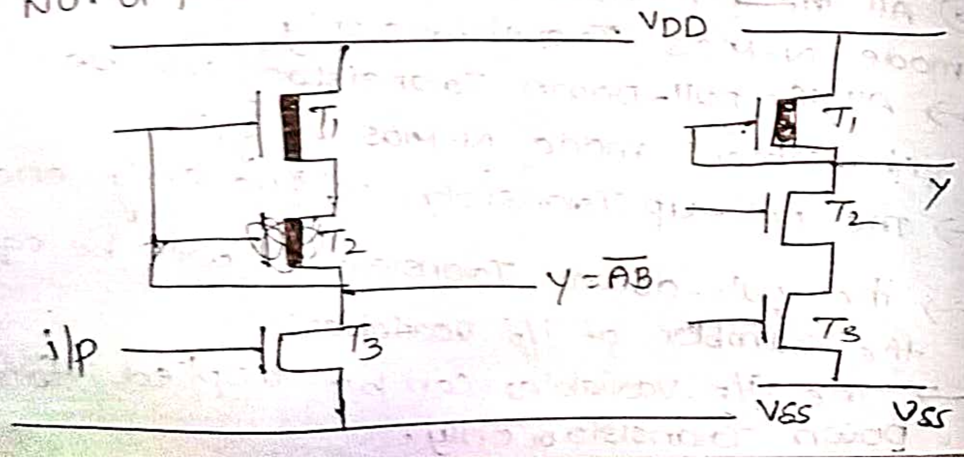


② Design NAND 2i/p gate using N-MOS Transistor

$\boxed{Y = \overline{AB}}$

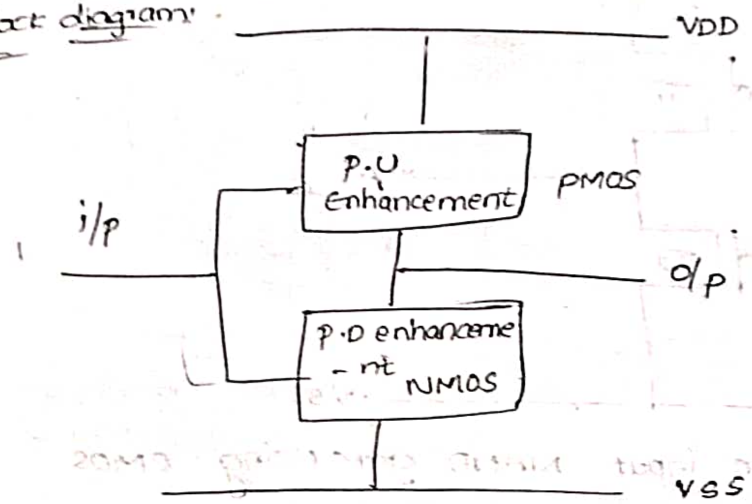
No. of P.U Transistors = 1 (Depletion N-MOS)

No. of P.D Transistors = 2 (Enhancement N-MOS)



CMOS Transistor logic :-

Block diagram:



② All the pull-up Transistor are enhancement mode p-MOS Transistors only.

→ All the P.D Transistors are enhancement mode N-MOS Transistors only.

→ The no. of P.U and P.D Transistors will be equal to Number of i/p variables/ literals.

→ In this MOS Transistor logic the o/p will be self complemented

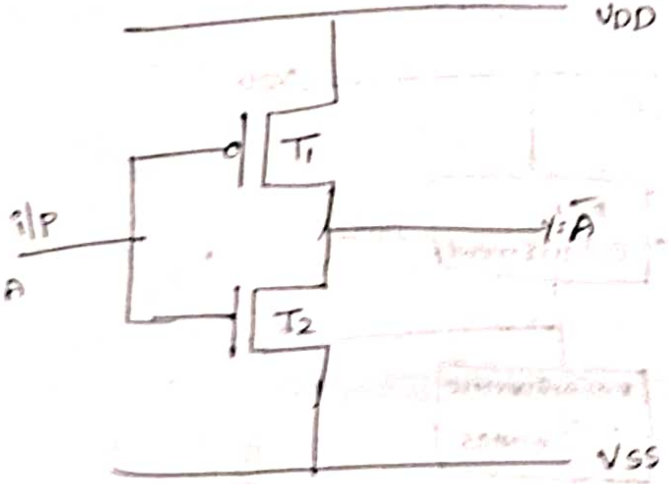
* Design an inverter using CMOS Transistor logic :-

Block diagram :-

$\boxed{Y = \bar{A}}$

No. of P.U Transistor = 1 [Enhancement p-MOS

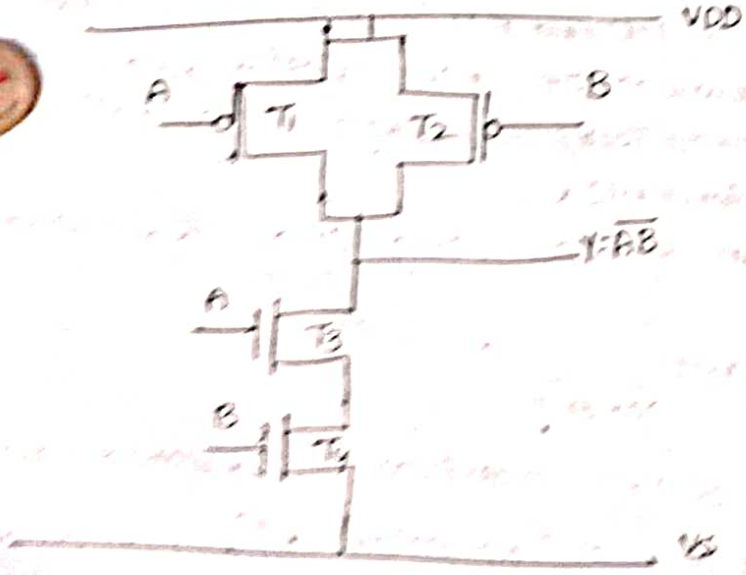
No. of P.D Transistor = 1 [Enhancement N-mos]



* Design a Two input NAND gate using CMOS Transistors.

$\boxed{Y = \overline{AB}}$

No. of P.U Transistors = 2 (Enhancement p-mos)
No. of P.D Transistors = 2 (Enhancement N-MOS)



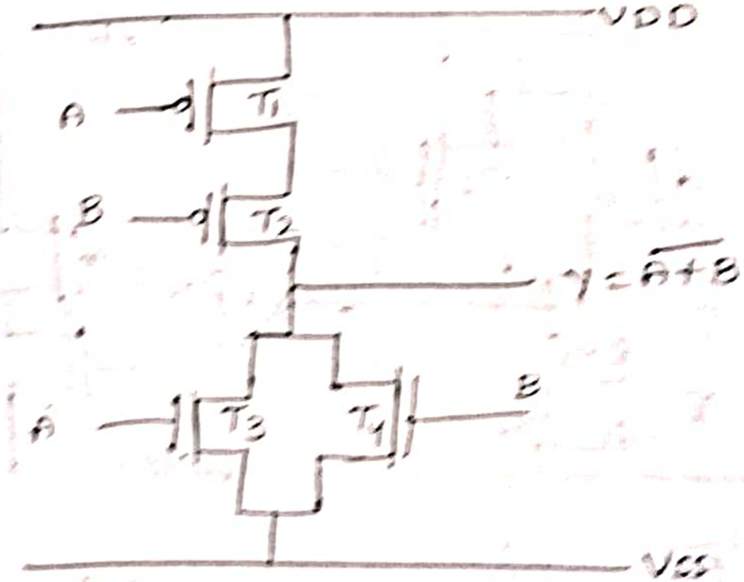
Truth Table:

A	B	$Y = \overline{AB}$
0	0	1
0	1	1
1	0	1
1	1	0

* Design a Two input NOR gate using CMOS Transistors

$\boxed{Y = \overline{A+B}}$

No. of P.U Transistors = 2 (Enhancement p-mos)
No. of P.D Transistor = 2 (Enhancement N-mos)

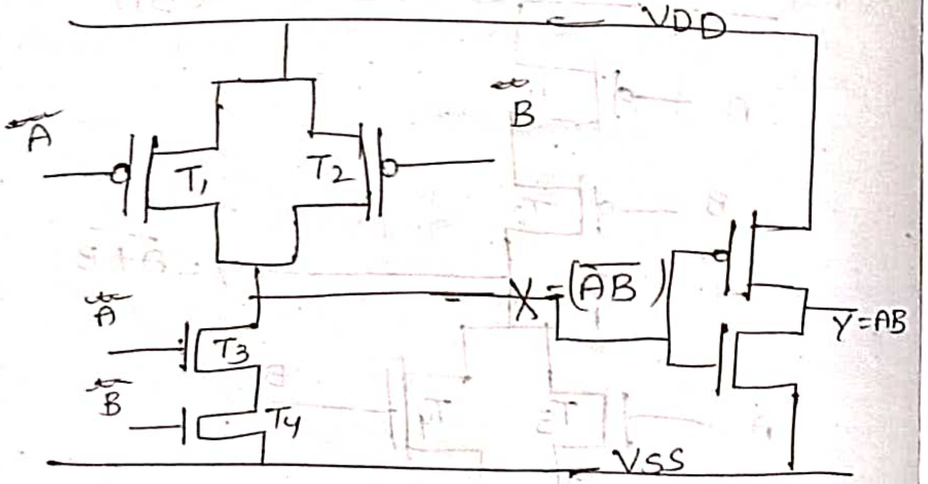


A	B	A+B'
0	0	1
0	1	0
1	0	0
1	1	0

* Design 2 i/p AND gate using CMOS Transistor logic

$\boxed{Y = A \cdot B}$

No. of P.U Transistor = 2 (Enhancement PMOS)
No. of P.D Transistor = 2 (Enhancement NMOS)

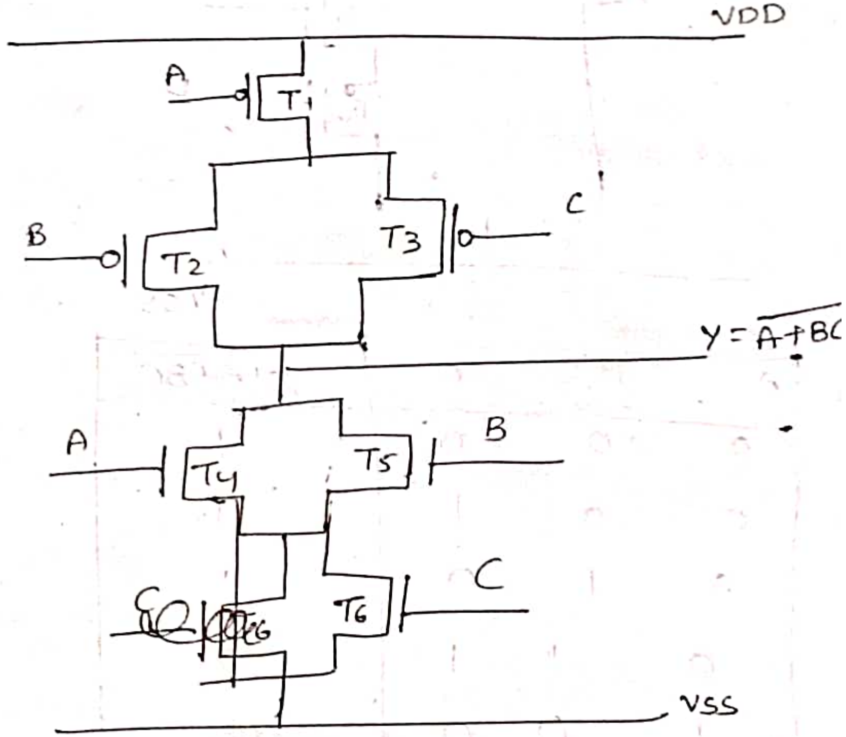


* $Y = \overline{A+BC}$
* $Z = \overline{AB+BCD}$

using C-MOS logic

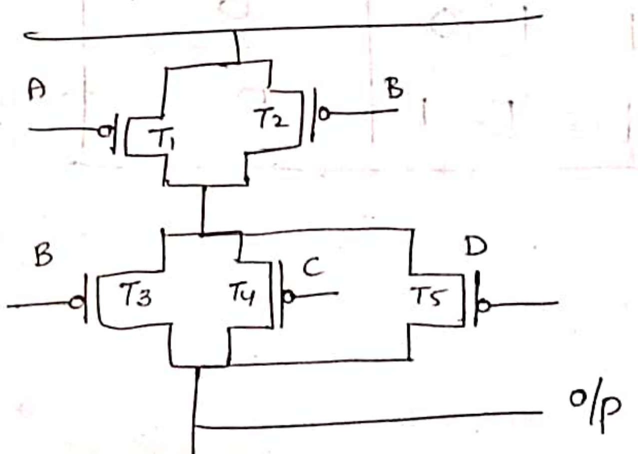
$Y = \overline{A+BC}$

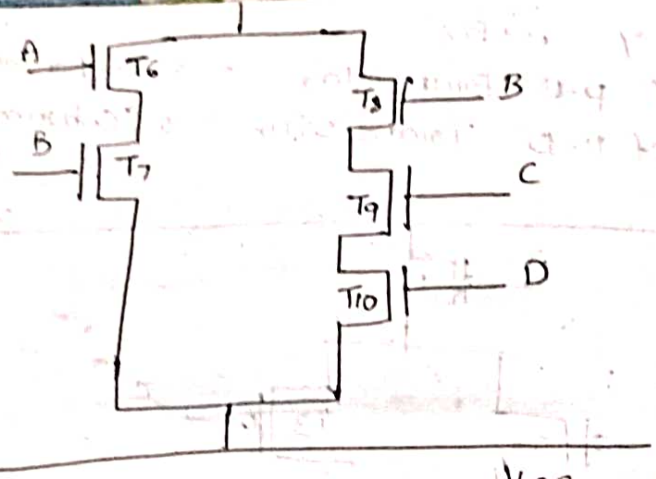
No. of P.U Transistor = 3 (Enhancement PMOS)
No. of P.D Transistor = 3 (Enhancement NMOS)



* $Z = \overline{AB+BCD}$

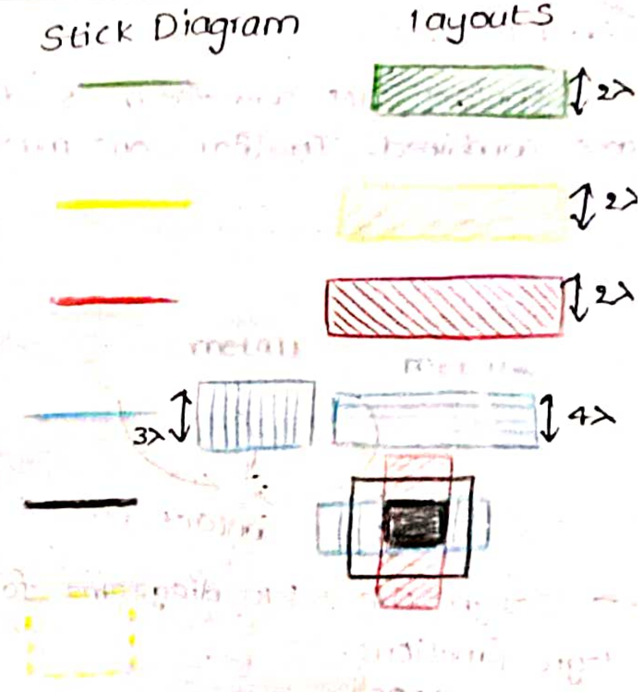
No. of P.U Transistors = 5 (Enhancement PMOS
No. of P.D Transistors = 5 (Enhancement N-MOS





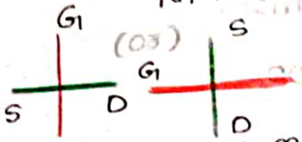
A	B	C	$Y = \overline{A+BC}$
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

	Stick Diagram	layouts
n- diffusion		2λ
p- diffusion		2λ
polysilicon		2λ
Metal	3λ	4λ
contact cut		
Implant		
Demarcation line		



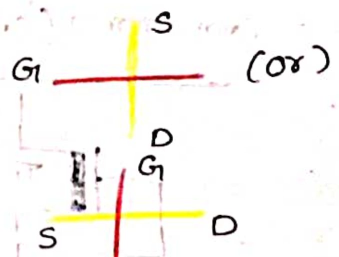
Formation of transistors:

NMOS



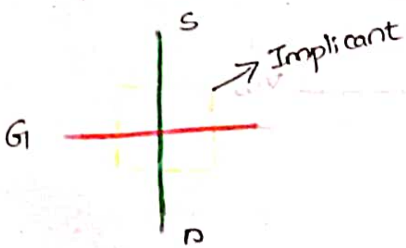
Enhancement mode
N-MOS

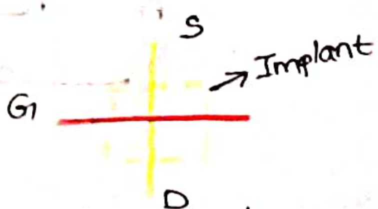
PMOS



Enhancement mode
P-MOS

Depletion mode
N-MOS:

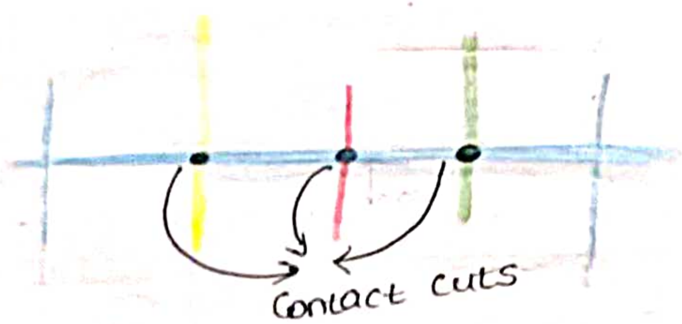




Depletion mode P-MOS

Contact cuts:-

Contact cuts are form 2 different materials are combined. In that one must be metal.



→ Design the stick diagrams for the following logic functions.

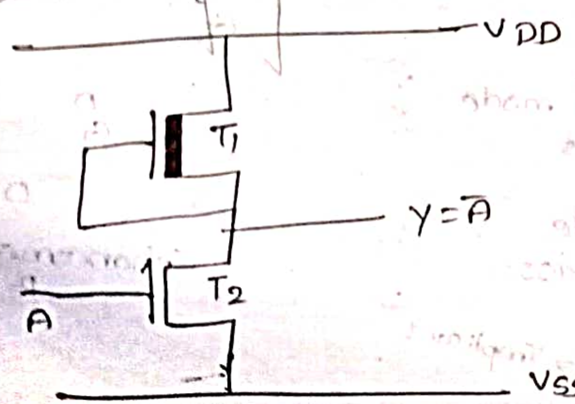
* Using NMOS:-

Inverter:

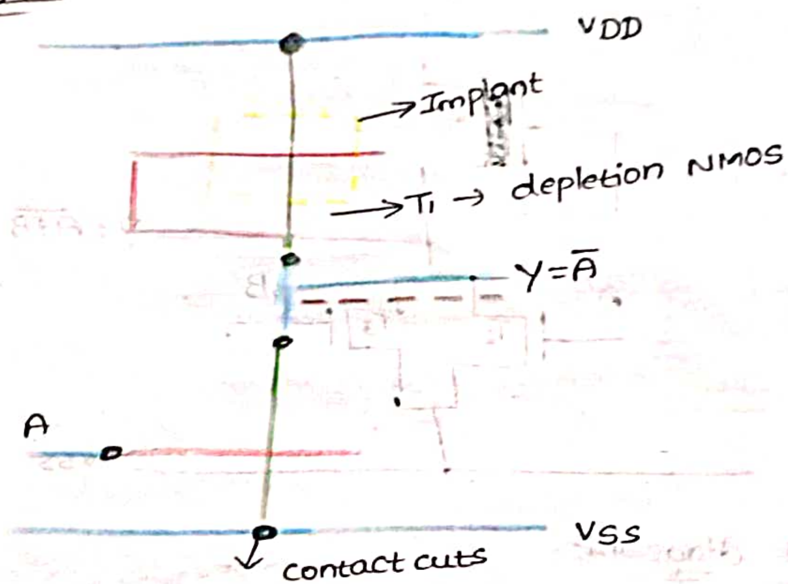
$\boxed{Y=\bar{A}}$

P.U Transistor = 1 (depletion N-mode)

P.U Transistor = 1 (Enhancement N-mos)



stick diagram:-



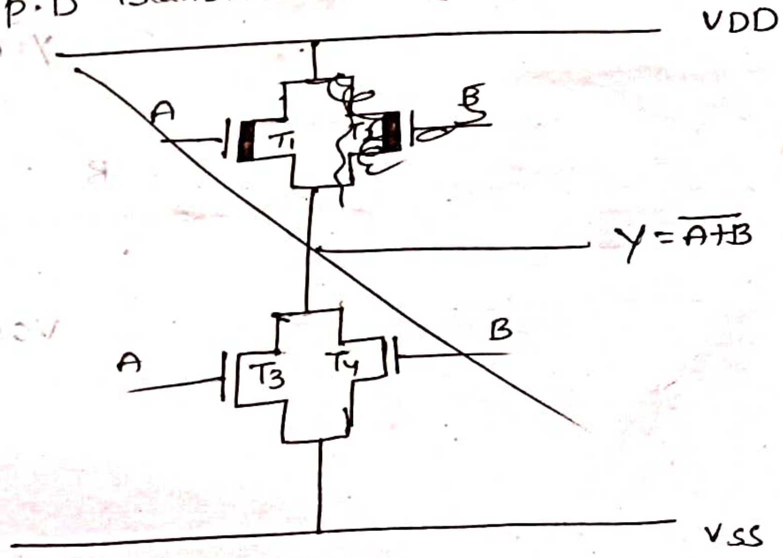
Ex-2:

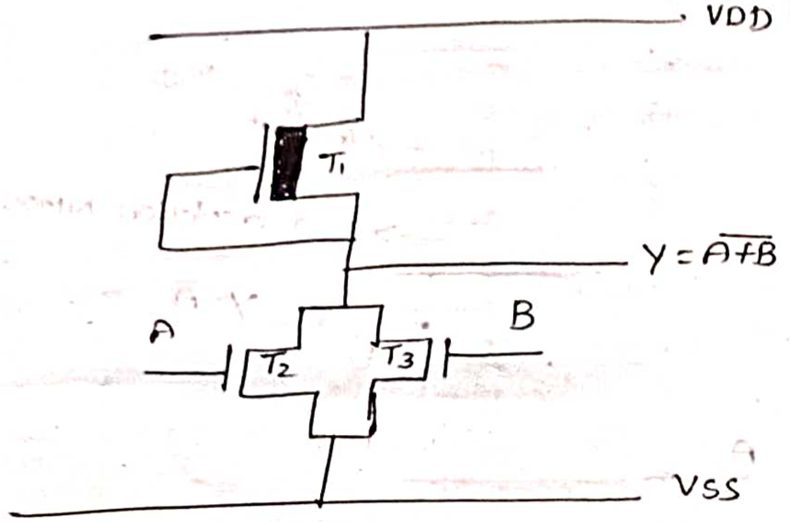
Design Two input NOR gate

$\boxed{Y=\overline{A+B}}$

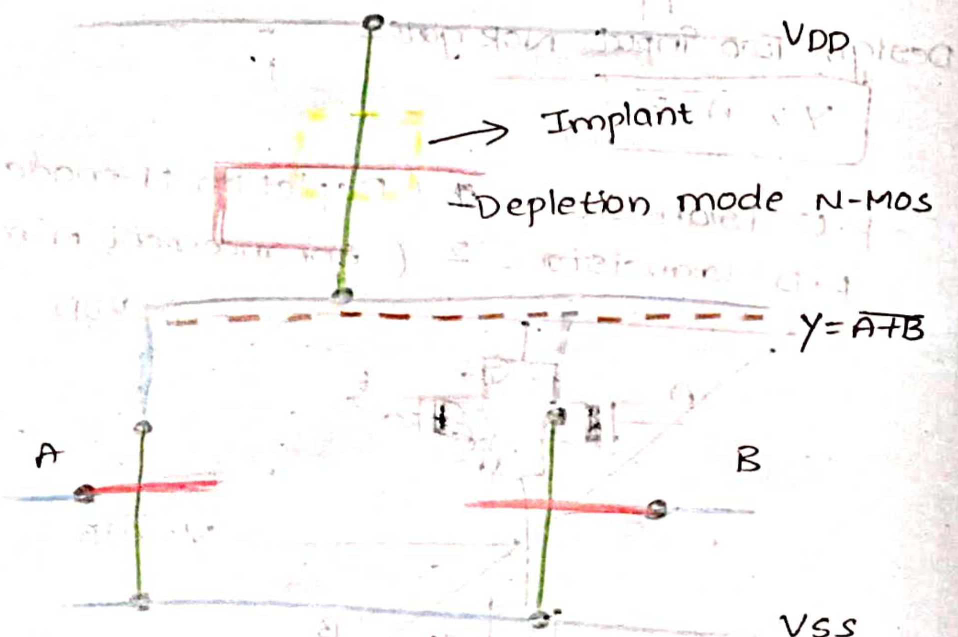
P.U Transistor = 2 (Depletion N-mode)

P.D Transistor = 2 (Enhancement N-mos)





stick diagram:-



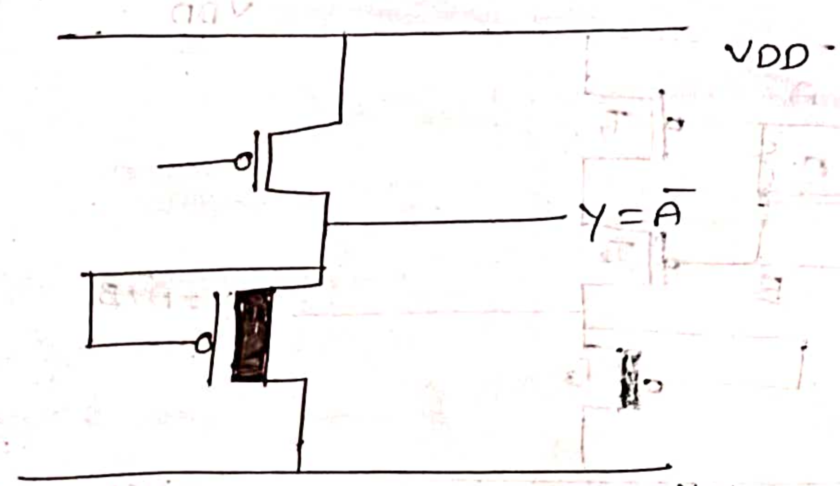
P-MOS

Inverter

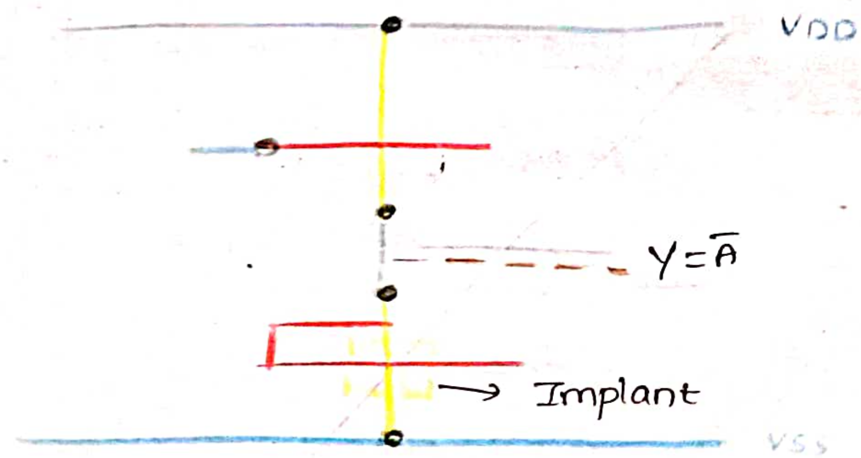
$\boxed{Y = \overline{A}}$

No. of P.U Transistors = 1 (Enhancement P-MOS)

No. of P.D Transistors = 1 (Depletion P-MOS)



stick diagram:-

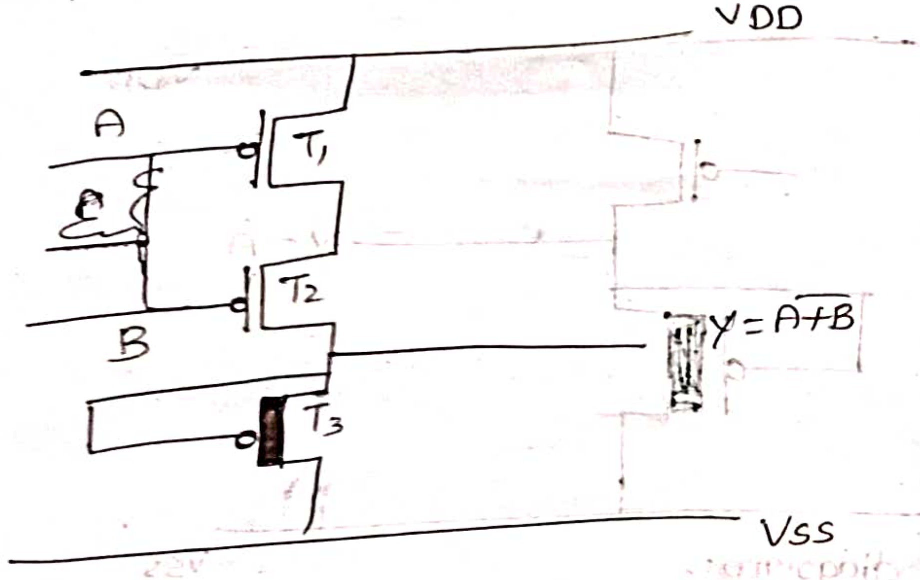


p-MOS

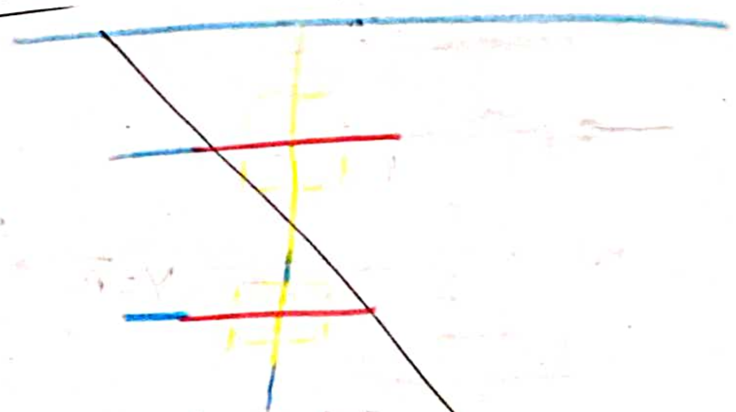
Two input NOR Gate

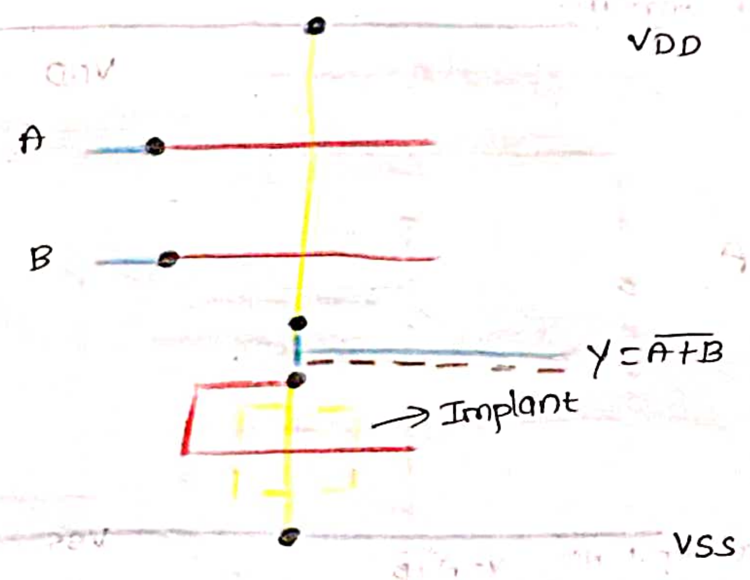
$\boxed{Y = \overline{A+B}}$

No. of P.U = 2 (Enhancement p-mos)
No. of P.D = 1 (depletion p-mos)



Stick diagram:





NAND (or) NOR conform

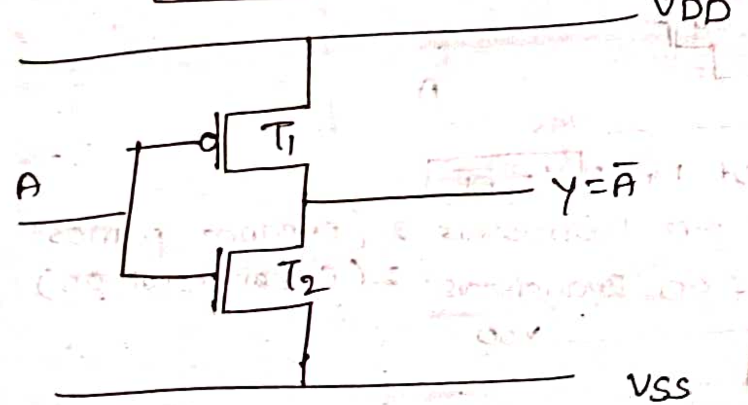
CMOS:

Design stick diagrams for the following Booleans functions using CMOS Transistors.

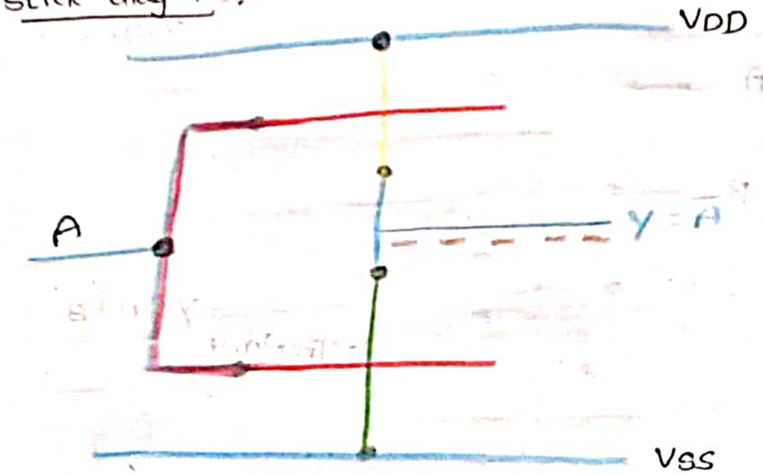
Inverter

$\boxed{Y = \overline{A}}$

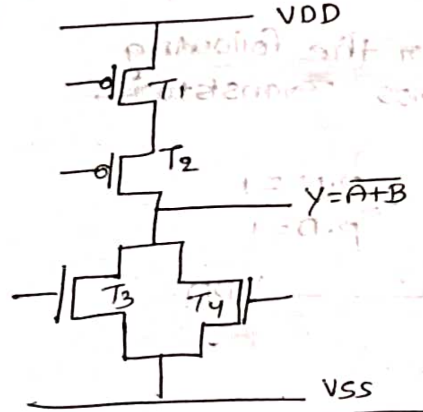
P.U = 1
P.D = 1

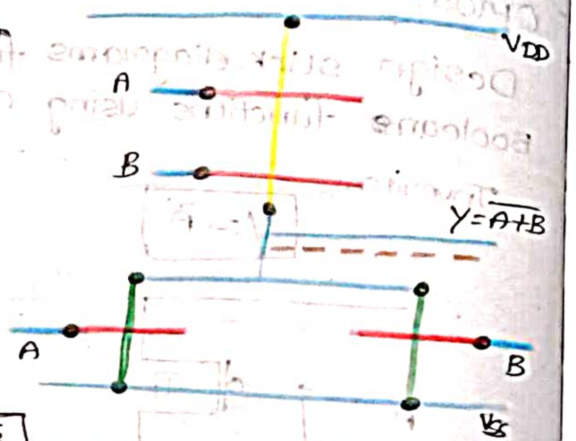


Stick diagram:-



② Two input NOR $Y = \overline{A+B}$

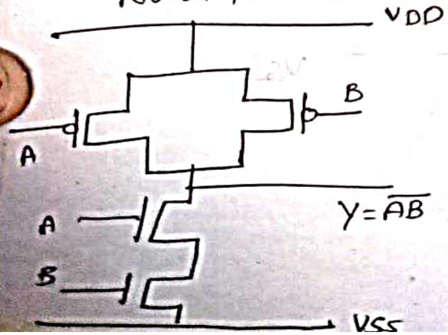


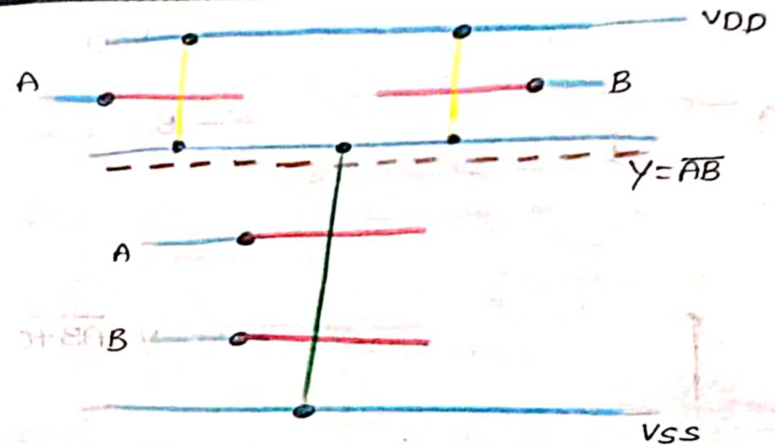


④ Two input NAND $\boxed{Y = \overline{AB}}$

NO.of P.U Transistors = 2 (enhance p-mos)

NO.of P.D Transistors = 2 (Enhance NMOS)



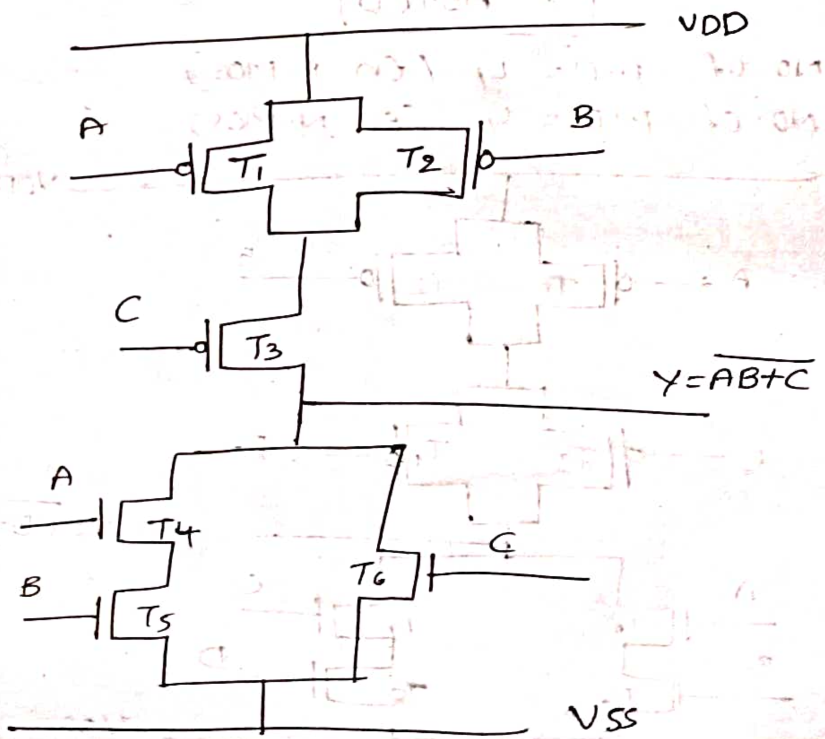


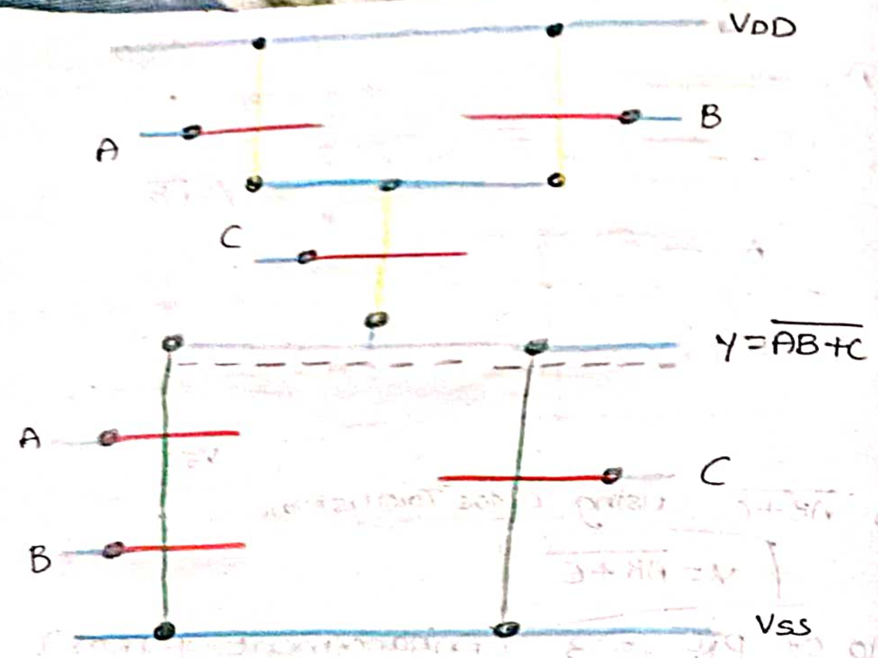
*Design $\overline{AB+C}$ using CMOS Transistors

$\boxed{Y = \overline{AB+C}}$

NO.Of P.U = 3 (Enhancement p-mos)

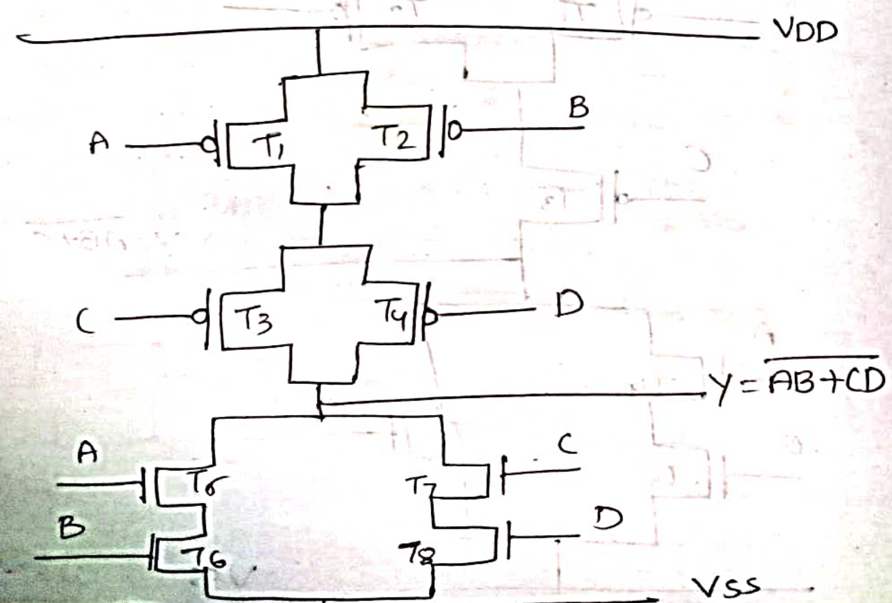
NO.Of P.D = 3 (Enhancement N-mos)

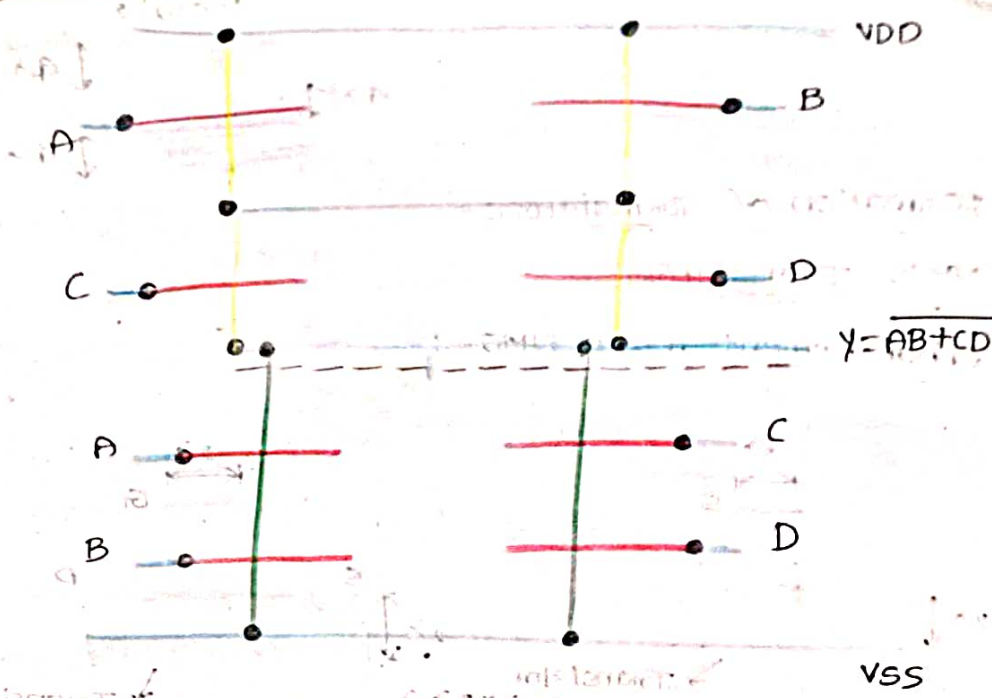




$\boxed{Y = \overline{AB+CD}}$

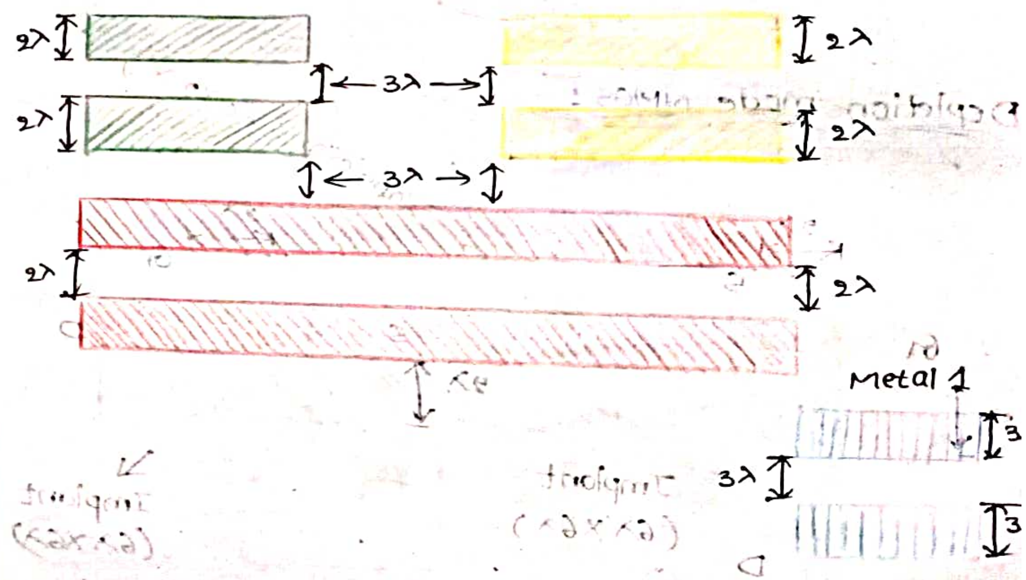
No. of p.U = 4 (En P-MOS)
No. of P.D = 4 (En N-MOS)

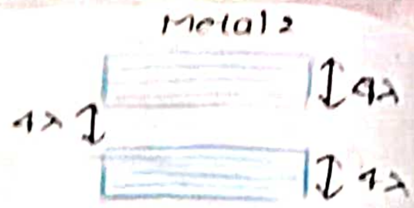




layout Design rules:-

λ based design rules

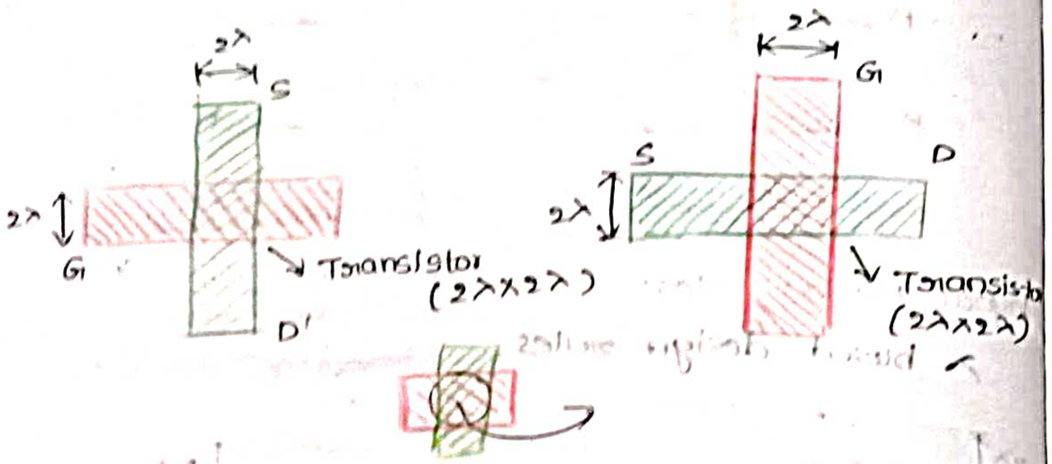




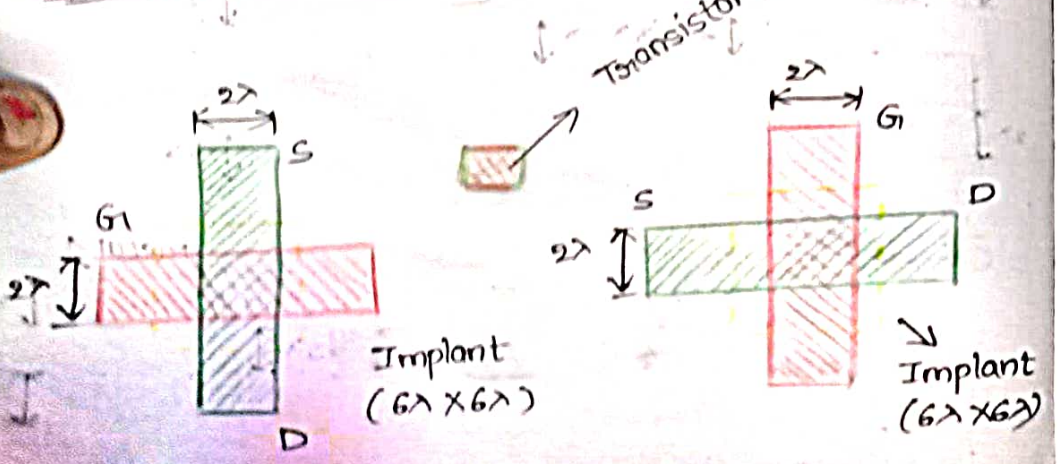
Formation of Transistor:-

NMOS Transistor:

Enhanchement mode NMOS :-

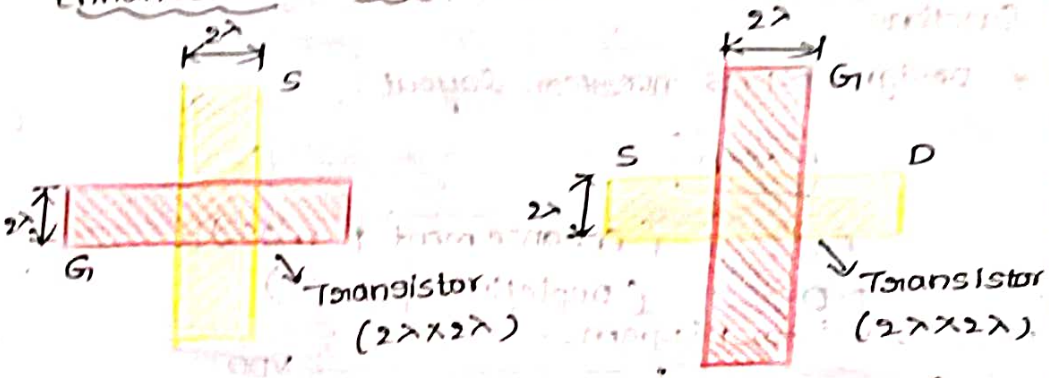


Depletion mode NMOS :-

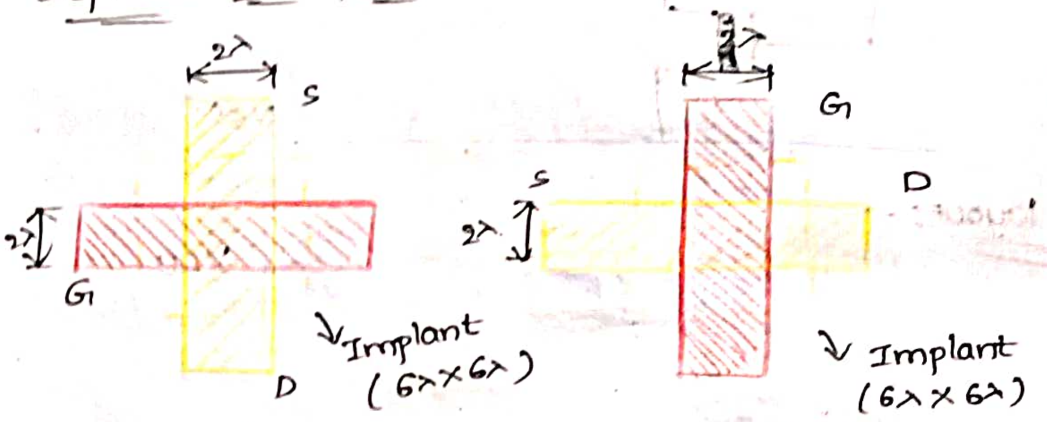


PMOS Transistor:-

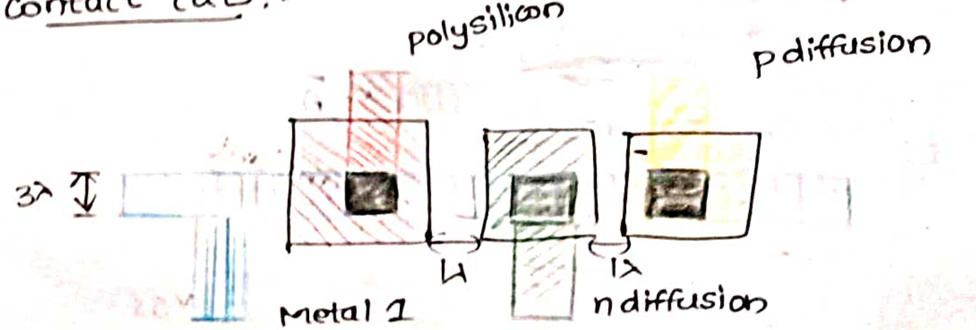
Enhancement mode PMOS transistor:-

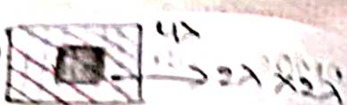


Depletion mode PMOS:-



Contact cuts:-



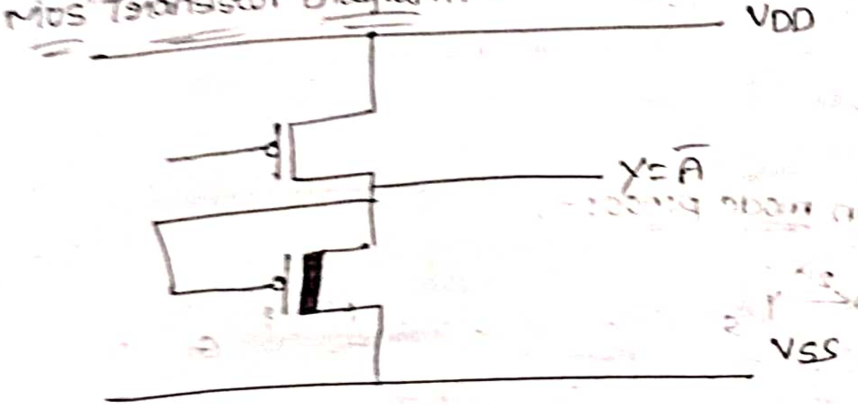
Contact cut (4λ) 

* Design the layouts for the following boolean functions.
* Design p-MOS inverter layout

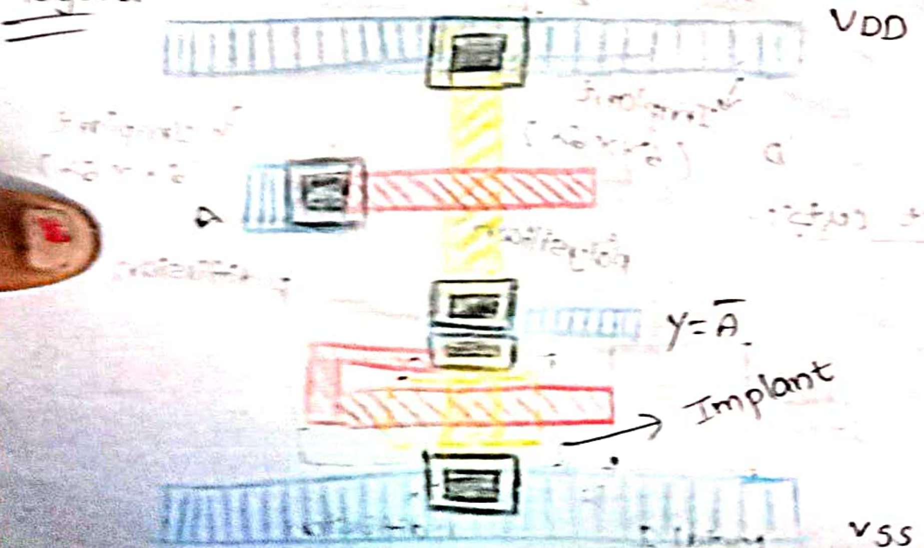
$\boxed{Y = \bar{A}}$

PU = 1 (enhancement p-mos)
PD = 1 (depletion p-mos)

MOS Transistor Diagram:-



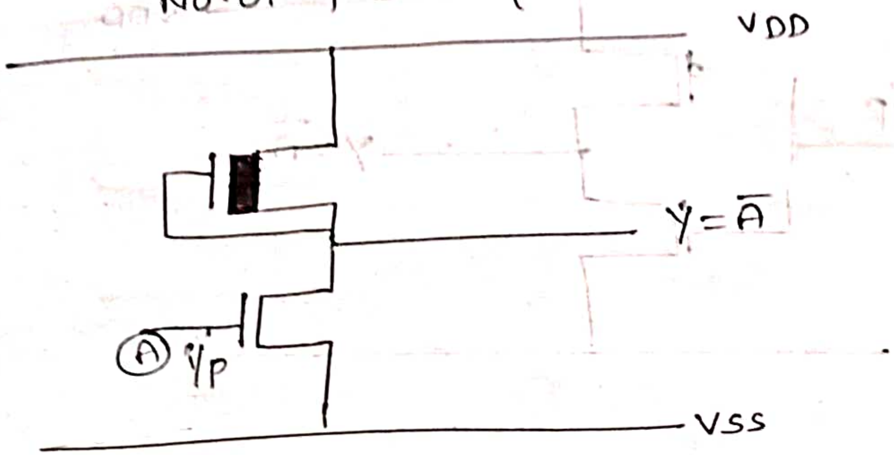
layout:-



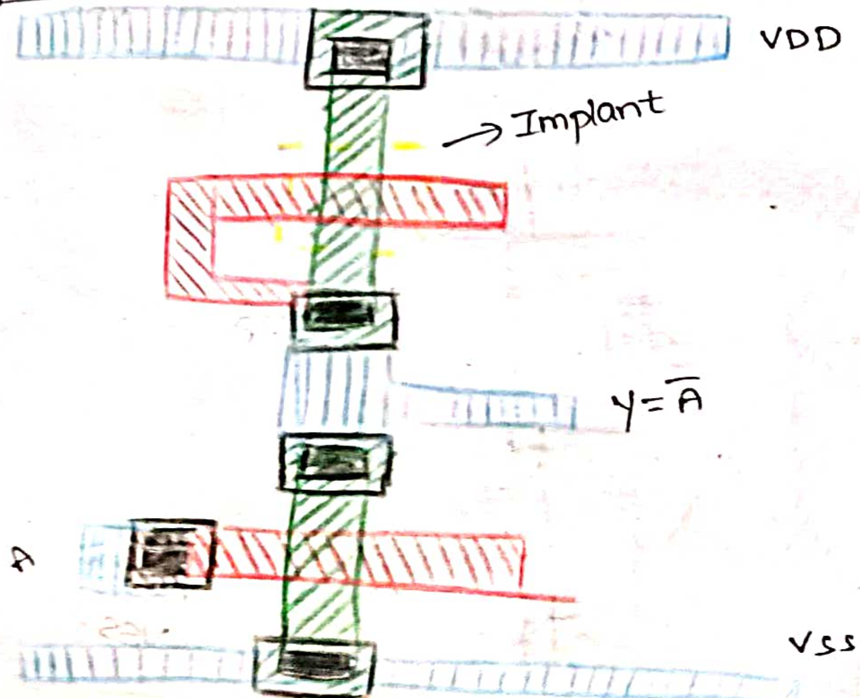
* Design the layout for the N-MOS inverter

$\boxed{Y = \bar{A}}$

No. of P.U = 1 (Depletion N-MOS)
No. of P.D = 1 (Enhancement N-MOS)



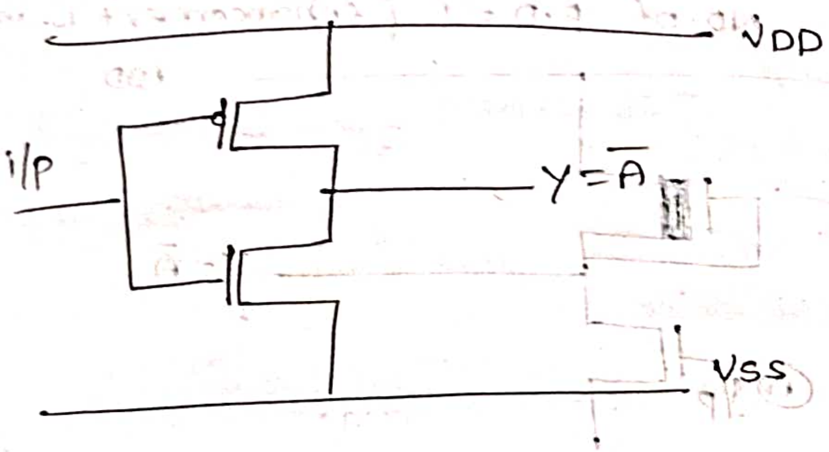
layout:-

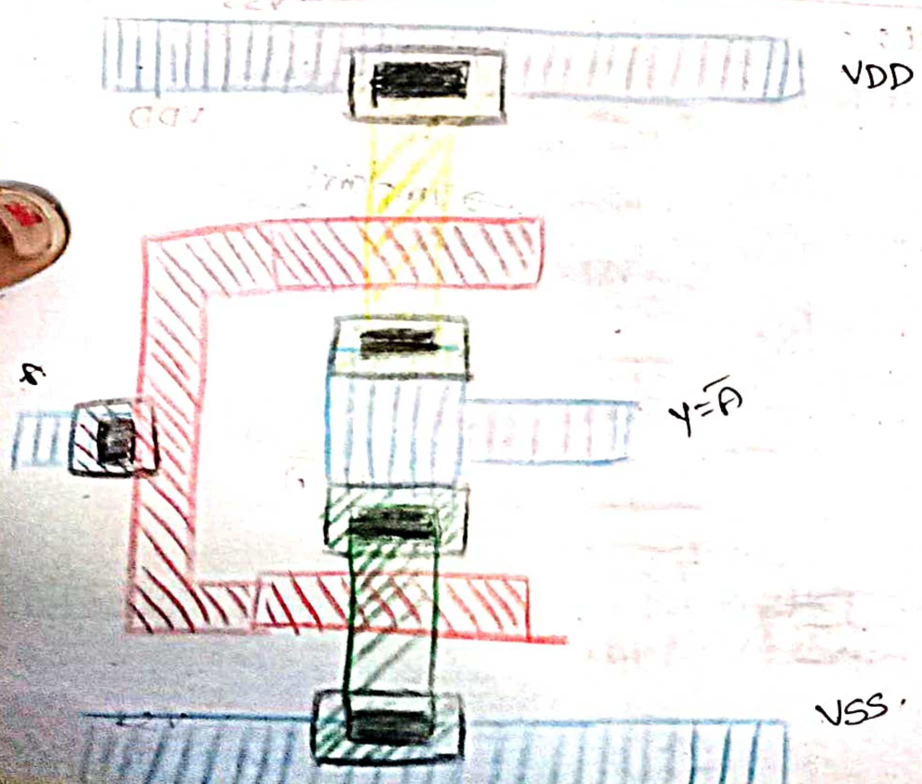


// contd
CMOS:-

$\boxed{Y=\overline{A}}$

No. of P.U = 1 (Enhancement P-MOS)
No. of P.D = 1 (Enhancement N-MOS)

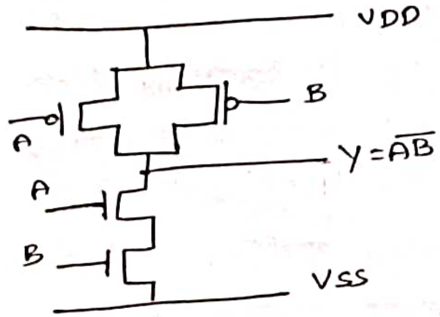




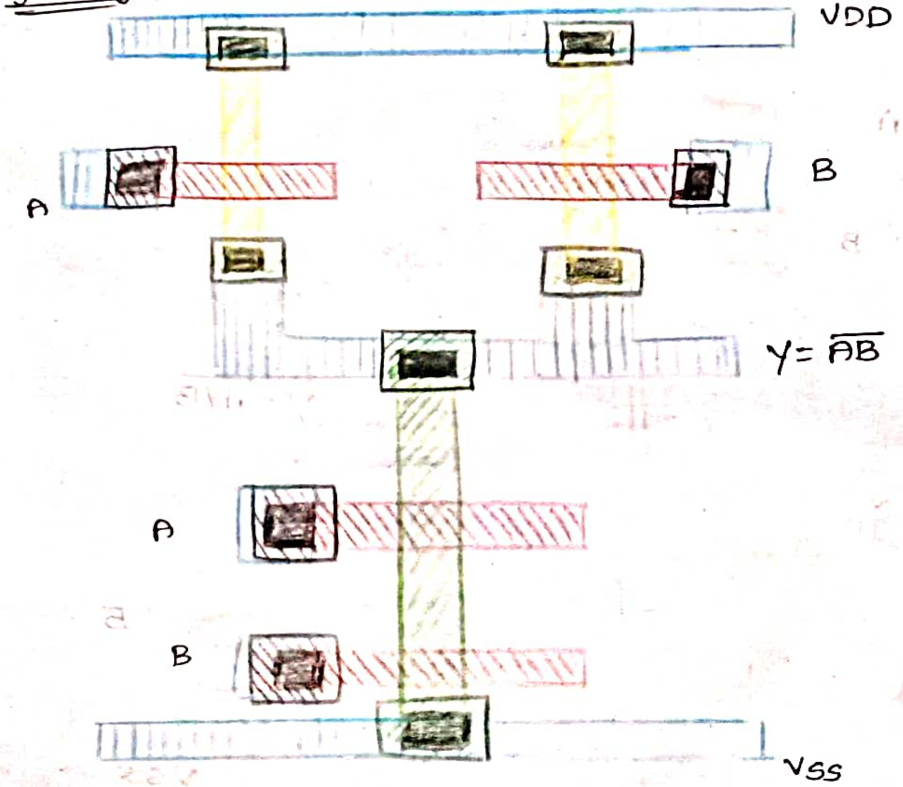
Two i/p NAND Gate by using CMOS:-

$\boxed{Y=\overline{AB}}$

No. of P.U = 2 (Enhancement P-mos)
No. of P.D = 2 (Enhancement N-mos)



Layout digram:-

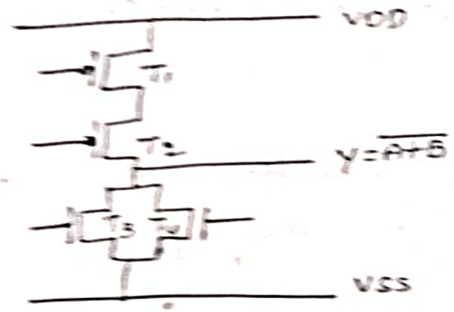


Two i/p NOR Gate using CMOS :-

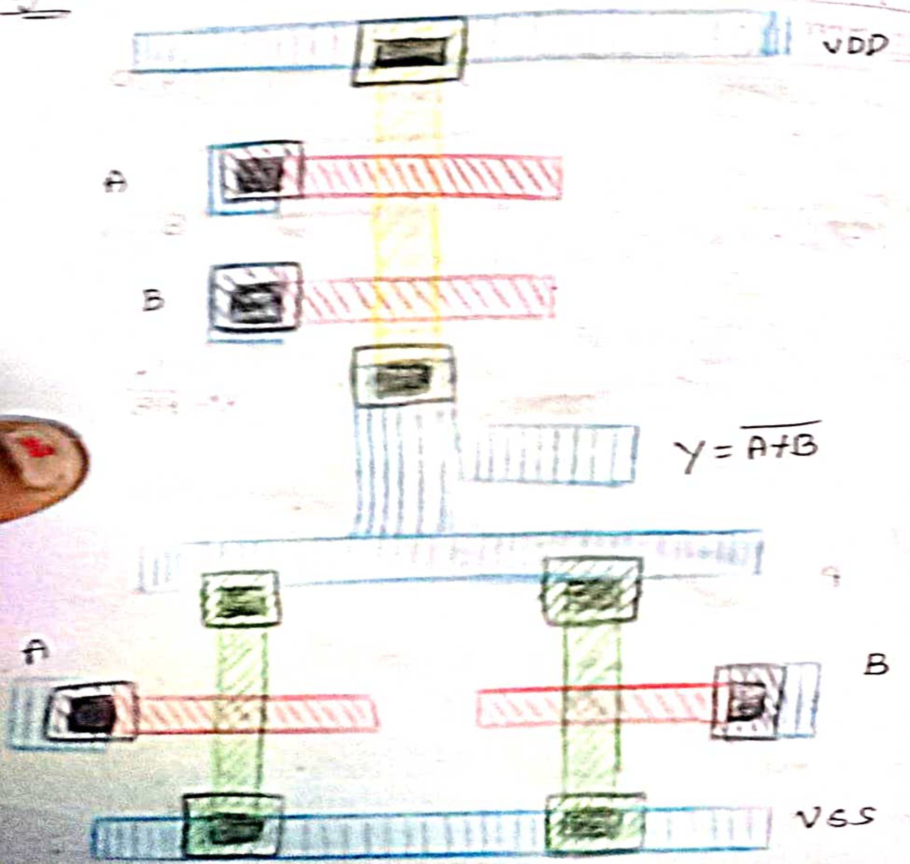
$Y = \overline{A+B}$

No. of P.U = 2 (Enhancement PMOS)

No. of P.D = 2 (Enhancement NMOS)



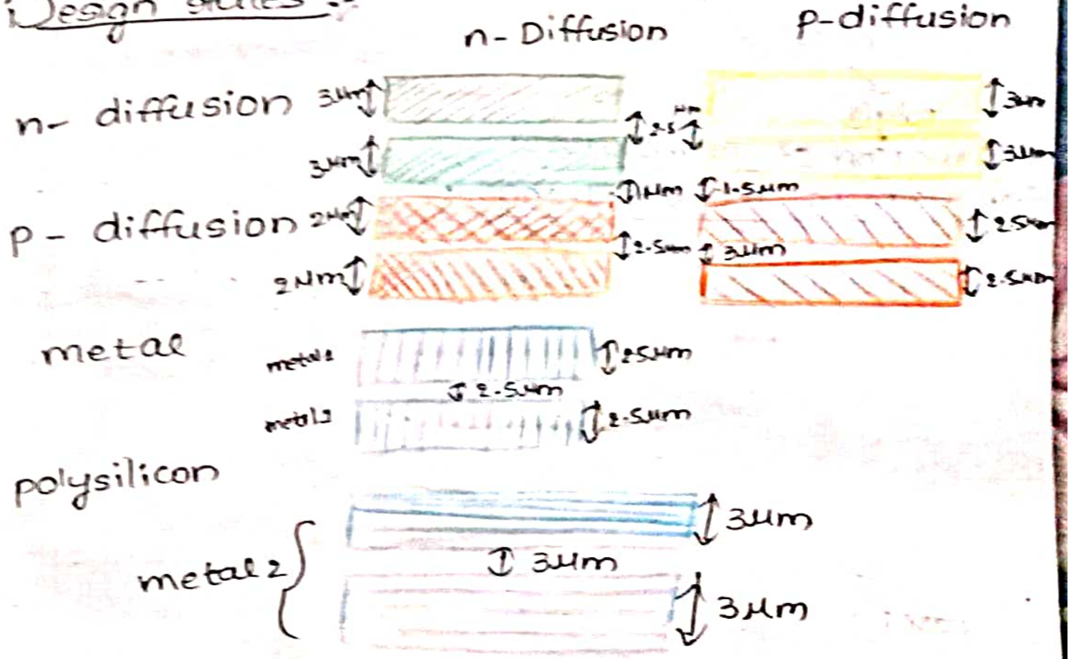
Layout :-



→ 2 micrometer Double metal Double poly CMOS bicmos Design rules:-

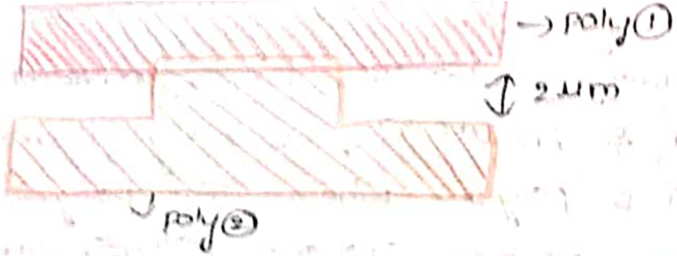
* In this design rules we have, 2 metals i.e., metal ① & metal ②, 2 polysilicons such as poly① & poly② where the poly① can be represented as red colour & poly② can be represented as Orange colour.
* Here all the dimensions represent with in terms of (µm) only.

Design rules :-

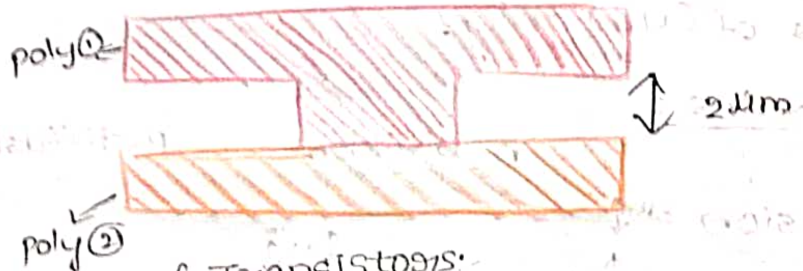


Overlapping of polysilicons :.

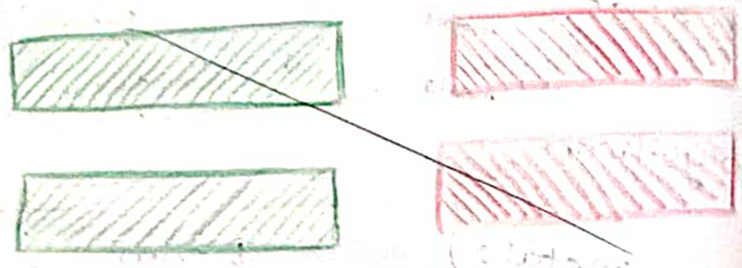
Polysilicon ② on polysilicon ①



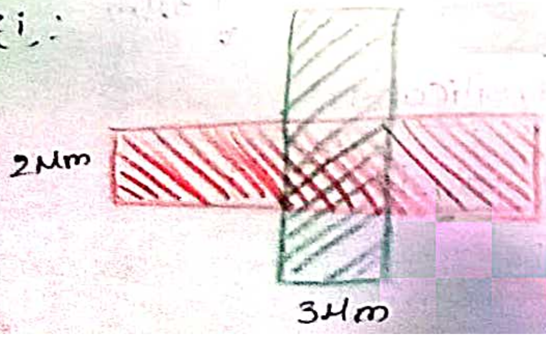
Poly① on poly② :-



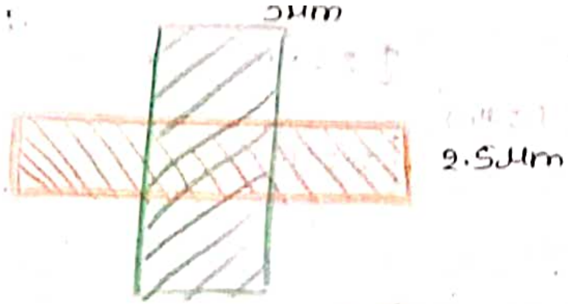
Formation of Transistors:
Enhancement Mode N-MOS :-



case i :

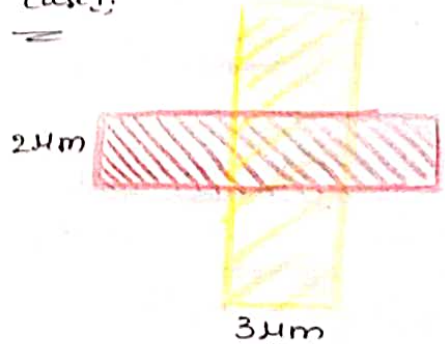


case ii :

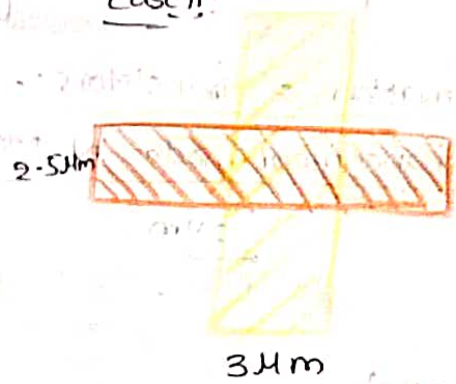


Enhancement mode p-MOS :-

case i;

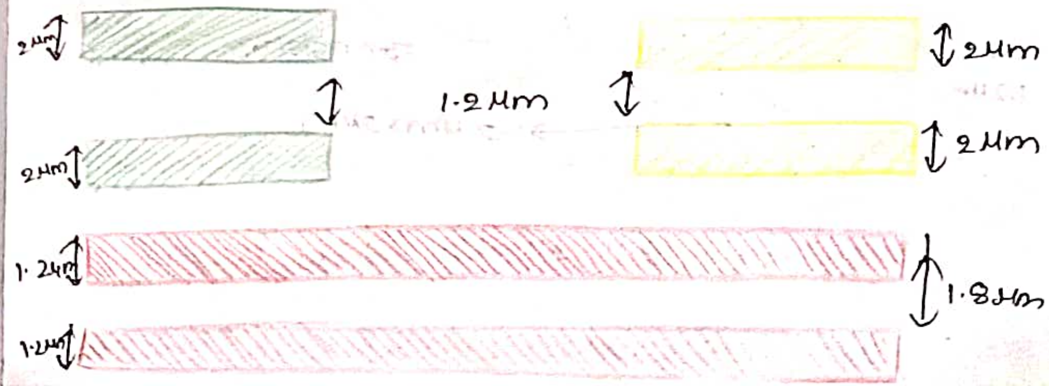


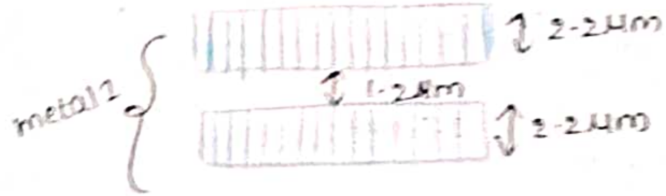
case ii;



1.2 µm Double metal, single poly CMOS/ bi-CMOS.

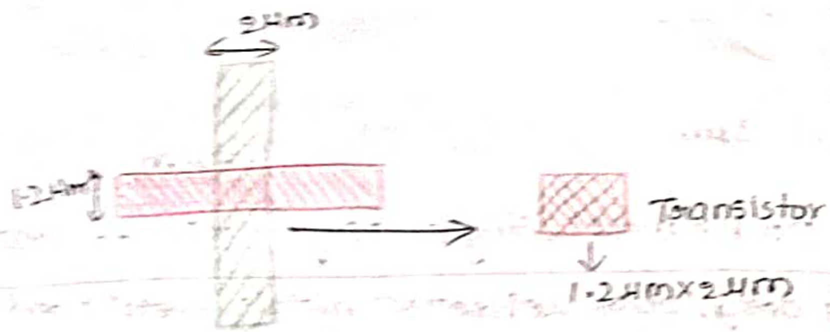
In this design rules we have two metals metal 1 and Two, n-diffusion & p-diffusion with single polysilicon.
→ Here all the dimensions will represent µm only.



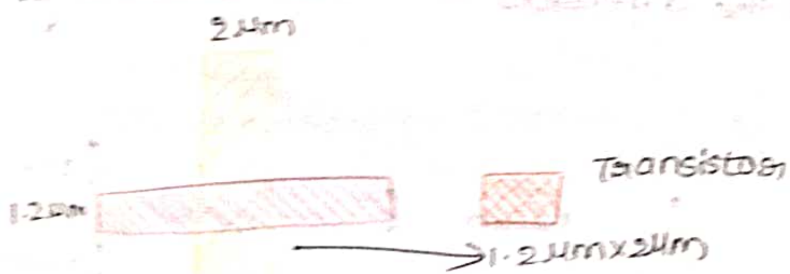


Formation of transistors:-

Enhancement mode N-MOS:-



Enhancement mode p-MOS:-



Vlsi Design flow:-

~~System speci~~

In the vlsi design flow will discuss above the design process of an IC. In this the initial step is the system specifications. In that the requirements of the product can be enclosed

System specification
↓
Architectural Design
↓
Functional Design
↓
logic Design
↓
Circuit Design
↓
Fabrication
↓
Testing
↓
packaging

→ The next step is block diagram of circuit. How many ~~units~~ inputs and outputs obtain and then

→ Functional design means boolean function obtain based on output. ~~packaging~~ ←

→ logic design is based on the boolean function by using logic gates obtained the diagram

→ Circuit design means Combination of P-MOS and N-MOS transistors. The logic gates converts the gates into Transistors.

→ Fabrication means the circuit design is comple

→ In the Architectural design we will design a block diagram with the requirements of system specifications

→ In this functional design we design the boolean functions for the outputs of the block diagram.

→ logic design we implement the boolean function by using logic gates only.

→ the circuit design, we convert the logic gates into transistor level.

→ In this fabrication step the fabricate the all transistors will adjust in the transistor level

→ Testing means verifying the final product with the system specifications

→ packaging means the testing process is qualified all the steps then it is package.

4. Basic circuit concepts

→ The active devices of MOS transistor technology having deal with some measures how it's consider the appropriate interconnections as circuits. The

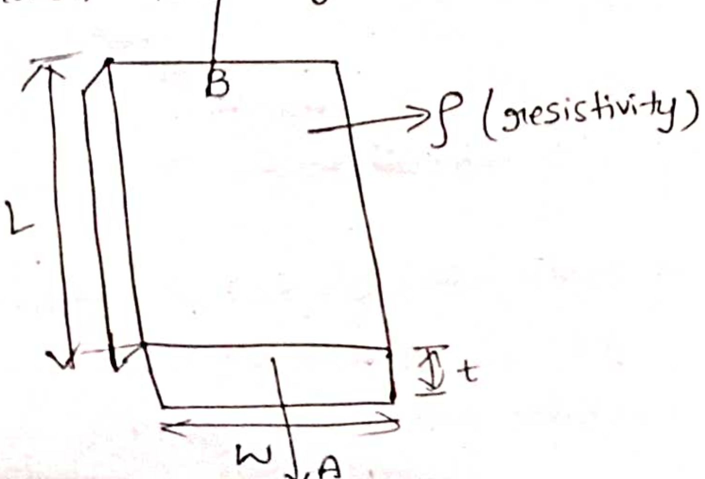
① The wiring of the circuit takes place in different conductive layers which are formed in the MOS fabrication.

The concept of sheet Resistance and standard unit of capacitance can be evaluate the wiring and i/p and o/p capacitances along with that will discuss delay's & inverter delay's.

→ The sheet resistance and capacitance values can be vary with different design Technology 2λ (5μm), 2μm, 1.2μm, λ(2.5μm).

Imp
Sheet. Resistance :- (Rs)

→ let us consider a uniform slab material with a resistivity of (ρ), (L) length and width (w) and Thickness (T) as it shown in the figure



From the ckt the resistance in b/w the two nodes A and B can be obtained as

$$R_{AB} = \frac{\rho L}{A} \text{ ohms}$$

Where

A → area of cross section

ρ → resistivity

L → length of the material

$$A = tW$$

$$\boxed{R_{AB} = \frac{\rho L}{tW} \text{ ohms}}$$

→ The sheet resistance can be obtained when the material is in the form of a square

i.e.; $\boxed{L=W}$ therefore

$$R_{AB} = \frac{\rho L}{tL} \text{ ohms}$$

$$\boxed{R_S = \frac{\rho}{t} \text{ ohms}}$$

→ The sheet resistance $\boxed{(R_S) = \frac{\rho}{t} \text{ ohms per square}}$

→ The units for sheet resistance is ohms per square.

→ The sheet resistance is independent of its length and width when thickness and resistivity is constant

→ Let us consider two materials with dimensions of 1cm and 10cm with same resistivity and same thickness then so both the material should have same sheet resistance.

→ The sheet resistance for different layers will be depends on resistivity and thickness of that layer. For metal and polysilicon the thickness can be easy to identify and for diffusion it is not easy to find the thickness because of it is not easy to find the depth of the diffusion.

→ The different sheet resistance values for different materials can be written as

Techologys:- 5 μm, 2 μm, 1.2 μm

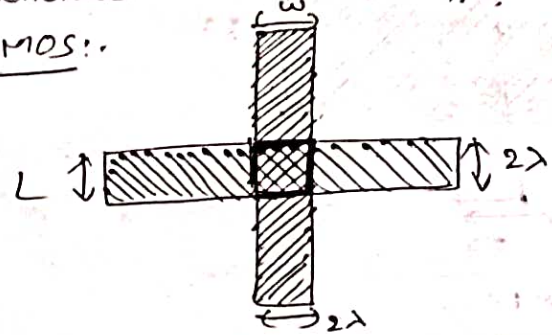
sheet resistance / materials	5 μm	2 μm	1.2 μm
Metal	0.03	0.04	0.04
n-diffusion	10 - 50	20-45	20-45
p-diffusion	25-125	50-112	50-112
silicide	2-4	—	—
polysilicon	15-100	15-30	15-30
n-transistor channel	10^4	2×10^4	2×10^4
p-Transistor channel	2.5×10^4	4.5×10^4	4.5×10^4

Where silicon is equivalent for a low resistance polysilicon and the p-diffusion is around 2.5 times of the n-diffusion sheet resistance.

Sheet resistance concept is applied to MOS Transistors and Inverters:-

Let us consider N-MOS and P-MOS transistors

For N-MOS:

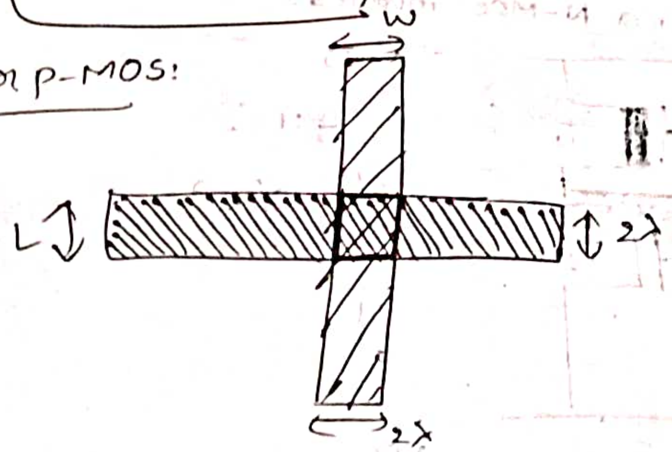


$R =$ square $\times$ RS

$=$ square $\times 10^4$ ohms per square

$$\boxed{R = 10k\Omega}$$

For P-MOS:



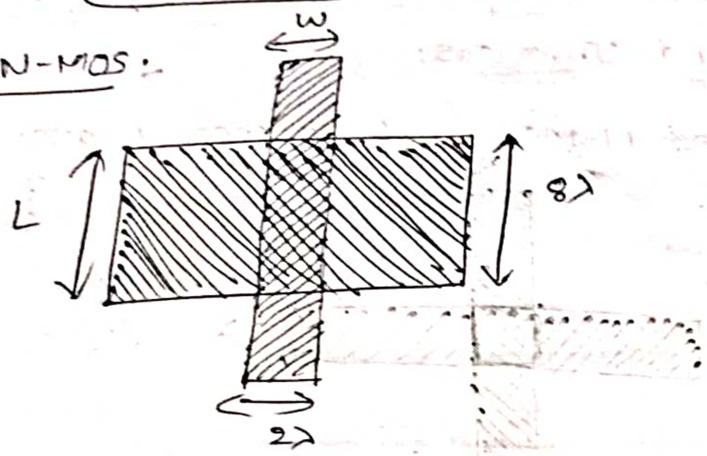
$R =$ square $\times$ RS

$R =$ square $\times\ 2.5\times10^4$ ohms per square

$R = 2.5 \times 10^4$ ohms

$\boxed{R = 25k\Omega}$

For N-MOS:



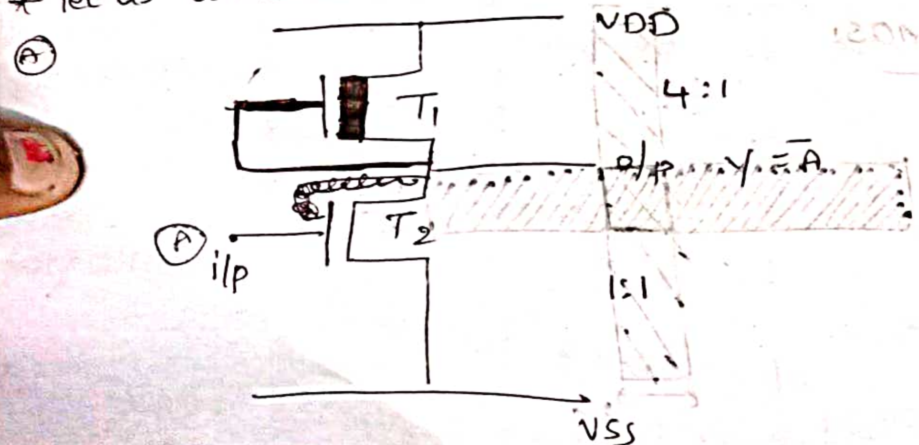
$R = (\text{Square}) \times R_S$

$= \frac{8\lambda}{2\lambda} \times R_S$ ohms per square

$= 4 \times 10^4$

$\boxed{R = 40k\Omega}$

* let us consider a N-MOS inverter.



For P.U:

$R_{P.U} = Z_{P.U} R_S$

$= \frac{4}{1} \times 10^4$

$= 40k\Omega$

For P.D:

$R_{P.D} = Z_{P.D} R_S$

$= \frac{1}{1} \times 10^4$ Ω/square

$= 10^4 \Omega$

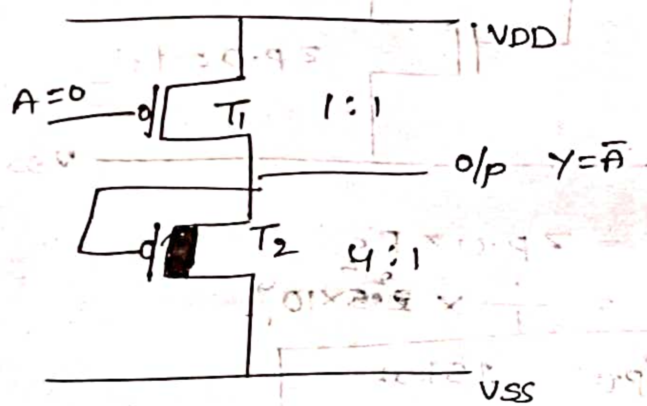
$= 10k\Omega$

Total resistance $= R_{P.U} + R_{P.D}$

$= 40k\Omega + 10k\Omega$

$\boxed{R_{ON} = 50k\Omega}$

P-MOS inverter:



$R_{P.U} = Z_{P.U} R_S$

$= \frac{1}{1} \times 2.5 \times 10^4$

$= 2.5 \times 10^4$

$\boxed{R_{PU} = 25k\Omega.}$

For P.D :

$R_{P.D} = Z_{P.D}\, R_s$

$= \frac{4}{1} \times 2.5 \times 10^4$

$\boxed{R_{P.D} = 100K\Omega}$

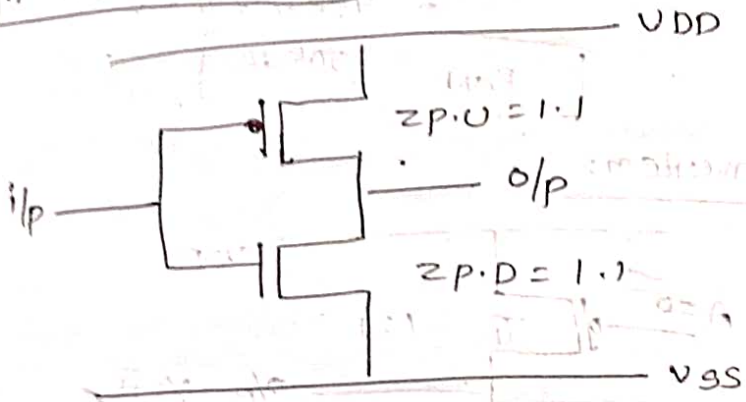
Total resistance $R_{ON} = R_{P.U} + R_{P.D}$

$= 25K\Omega + 100K\Omega$

$\boxed{R_{ON} = 125K\Omega}$

model imp

Design for C-MOS inverter:-



$R_{P.U} = Z_{P.U} \times R_s$

$= \frac{1}{1} \times \cancel{2.5}\,2.5 \times 10^4$

$\boxed{R_{P.U} = 25K\Omega}$

$R_{P.D} = Z_{P.D} \times R_s$

$= \frac{1}{1} \times 10^4$

$\boxed{R_{P.D} = 10K\Omega}$

Total resistance $R_{ON} = \cancel{R_{P.U} + R_{P.D}}$

$= \cancel{25K\Omega + 10K\Omega}$

$= \cancel{35K\Omega}$

RON resistance is not possible in CMOS transistor logic.

Silicides:-

The transistor width varies the resistance will increases then the delay ($\tau = RC$) is also increases. then the speed will be reduces. To improve the speed we want to reduce the resistance. where the resistance of the polysilicon will be (15-100) instead of these we use a low resistance around (2-4) which is equivalent properties of polysilicon. That is silicide.

→ The silicide can be form by the deposition of metal on to the polysilicon and then sintering where the sintering is the process of heating the material and apply some pressure.

→ ~~The~~ sintering can be done by using any one of the Techniques as follows

1. Sputtering / Evaporation
2. Co-sputtering
3. Co-evaporation

Area capacitance of layers :-

$$C = \frac{\varepsilon_0 \varepsilon_{ins} A}{D} \text{ Farads}$$

ε_0 = permittivity of free space SiO_2 4.0

ε_{ins} = permittivity of insulating material =

$\varepsilon_0 = 8.85 \times 10^{-14}$ F/cm

A = Area of two parallel plates

D = Thickness of the dielectric material

→ As in the process of fabrication we use different layers like substrate, polysilicon, metal, diffusion etc... The combination of any two layers a separated with oxidation layer i.e; (dielectric material) should form a capacitor. To find the value of a capacitor the formula when we know that value of thickness of the oxidation.

→ The different capacitance values in 3 different technologies such as 5µm, 2µm, 1.2µm. In this the units for area capacitance layers is

$pF \times 10^{-14} / \mu m^2$

Capacitance	5µm	2µm	1.2µm
Gate to channel	4(1.0)	8(1.0)	16(1.0)
Diffusion (n-type)	1 (0.25)	1.75(0.22)	3.75 (0.23)
Polysilicon to substrate	0.4 (0.1)	0.6 (0.075)	0.6 (0.038)
Metal 1 to Substrate	0.3 (0.075)	0.33 (0.04)	0.33 (0.02)
Metal 2 to Substrate	0.2 (0.05)	0.17 (0.02)	0.17 (0.01)
Metal 2 to Metal 1	0.4 (0.1)	0.5 (0.06)	0.5 (0.03)
Metal 2 to polysilicon	0.3 (0.075)	0.3 (0.038)	0.3 (0.018)

Standard unit of capacitance $\square C_g$:-

→ The standard unit of capacitance can be given a value appropriate value to the Technology & it is also used to calculate the absolute value where the standard unit of capacitance can be defined with $\square C_g$. The standard unit can be represented as L = W and it can be represented with feature. It is also known standard size (or) feature size i.e; $(2\lambda \times 2\lambda)$.

In 5µm :-

Area) standard square = 5µm × 5µm

= 25µm²

Capacitance = $4 \times pF \times 10^{-4} / \mu m^2$

Standard unit of capacitance = Area/standard square × capacitance

$= 25 \mu m^2 \times 4 \times pF \times 10^{-4} / \mu m^2$

$= 100 \times 10^{-4}$ pF

$= 10^2 \times 10^{-4}$ pF

$$\boxed{\square C_g = 0.01 pF}$$

In 2μm Technology:-

Area/standard square = $2\mu m \times 2\mu m = 4\mu m^2$

Capacitor = $8 pF \times 10^{-4}/\mu m^2$

Standard unit of capacitance =

$= 4\mu m^2 \times 8 \times pF \times 10^{-4}/\mu m^2$

$= 32 \times pF \times 10^{-4}$

$= 32 \times 10^{-4} \, pF$

$$\boxed{\square C_g = 0.0032 pF}$$

In 1.2μm Technology:-

Area/standard square = $1.2\mu m \times 1.2\mu m = 1.44\mu m^2$

capacitor = $16 \times pF \times 10^{-4}/\mu m^2$

Standard unit of capacitance =

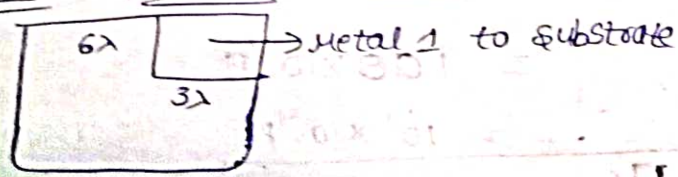
$= 1.44\mu m^2 \times 16 \times pF \times 10^{-4}/\mu m^2$

$= 23.04 \times 10^{-4} \times pF$

$$\boxed{\square C_g = 0.0023 \, PF}$$

Some area capacitance calculations:-

Eg:-



Relative Area = $\frac{\text{Selected Area}}{\text{Standard size Area}} = \frac{6\lambda \times 3\lambda}{2\lambda \times 2\lambda}$

$= \frac{18\lambda^2}{4\lambda^2}$

$= 4.5$

Relative Area x Relative Capacitance

$= 4.5 \times 0.075$

$= 0.3375 \, \square C_g$

→ In this all the dimensions represent in terms of λ based designed rules. To find the capacitance of a specific area.

Capacitance = Relative Area x Relative Capacitance

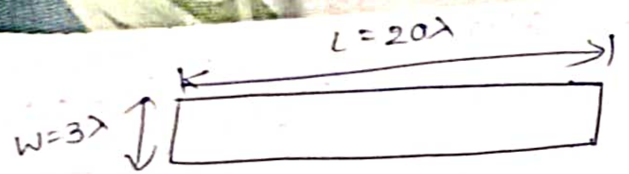
→ all these capacitances values represented in terms of standard units of capacitance($\square C_g$). In that

→ Where the relative capacitance can be obtained from the capacitance table and the relative area can be obtained as the ratio of selected Area to the standard size Area

$$\text{Relative Area} = \frac{\text{Selected Area}}{\text{Standard size Area}}$$

in that the standard size Area is fixed that is $(2\lambda \times 2\lambda)$.

let us consider a material with the following dimensions



Relative Area = $\frac{\text{Selected Area}}{\text{Standard size}}$

$= \frac{20\lambda \times 3\lambda}{2\lambda \times 2\lambda}$

$= \frac{60\lambda^2}{4\lambda^2}$

$\boxed{\text{Relative Area} = 15}$

(i) Relative Area × Relative capacitance = Capacitance

$= 15 \times 0.075 \square C_g$

let us consider the material will be metal 1

$\boxed{\text{capacitance} = 1.125 \square C_g}$

2) For polysilicon :-

Capacitance = Relative Area × Relative capacitance

$= 15 \times 0.1 \square C_g$

$= 1.5 \square C_g$

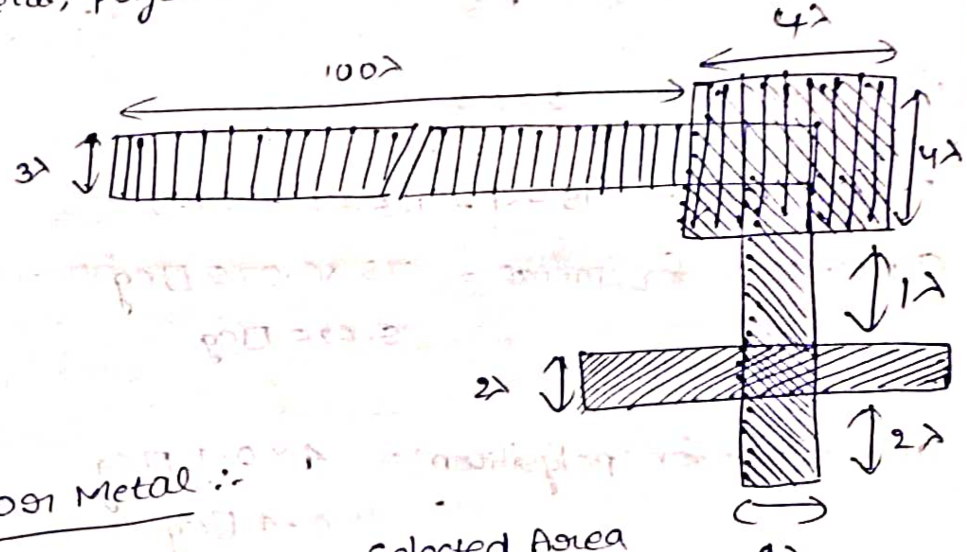
(3) For N-Diffusion :-

Capacitance $= 15 \times 0.25 \square C_g$

$= 3.75 \square C_g$

Delay unit

→ let us consider a circuit with a combination of metal, polysilicon and diffusion.



1) For Metal :-

Relative Area = $\frac{\text{Selected Area}}{\text{Standard size}}$

$= \frac{100\lambda \times 3\lambda}{2\lambda \times 2\lambda}$

$= \frac{300\lambda^2}{4\lambda^2} = 75$

2 Contact cut :-

Relative Area $= \frac{4\lambda \times 4\lambda}{2\lambda \times 2\lambda}$

$= \frac{16\lambda^2}{4\lambda^2} = 4$

③ Polysilicon:-

Relative Area = $\frac{3\lambda \times 2\lambda}{2\lambda \times 2\lambda}$

= 1.5

④ Gate = $\frac{2\lambda \times 2\lambda}{2\lambda \times 2\lambda}$ = 1

Total Relative Area = 75 + 4 + 1.5 + 1 = 81.5

Capacitance for metal = $75 \times 0.075 \square C_g$

= $5.625 \square C_g$

Capacitance for polysilicon = $4 \times 0.1 \square C_g$

= $0.4 \square C_g$

Capacitance for polysilicon = $1.5 \times 0.1 \square C_g$

= $0.15 \square C_g$

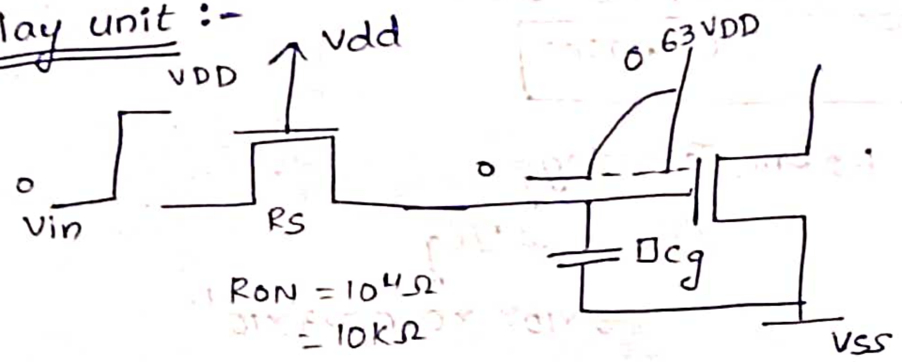
Gate Capacitance = $1 \times 1 \square C_g$

= $1 \square C_g$

Total capacitance = 5.625 + 0.4 + 0.15 + 1

= $7.175 \square C_g$

Delay unit :-



→ we have discuss above sheet resistance (Rs) and the standard unit of capacitance ($\square C_g$).

→ let us consider the standard unit of capacitor is charge through a N-channel transistor then the capacitor will charges 0.63VDD (or) 63% VDD

→ The time constant (τ) can be obtain by the product of resistance and capacitance

$\tau = R \times C$

In 5μm Technology the time constant

$\tau = R_S \times \square C_g$

$= 10^4 \times 0.01 \square C_g \times 10^{-12}$

$= 0.01 \times 10^{-8}$ sec

$= 0.1 \times 10^{-9}$ sec

$\boxed{\tau = 0.1 \text{nsec}}$

In 2μm Technology

$\tau = R_S \times \square C_g$

$= 2 \times 10^4 \times 0.0032 \times 10^{-12}$

$= 0.006 \times 10^{-8}$ sec

$= 0.064 \times 10$

$\boxed{\tau = 0.064 nsec}$

1.2 μm Technology

$\tau = R_s \times \square C_g$

$= 2 \times 10^4 \times 0.0023 \times 10^{-12}$

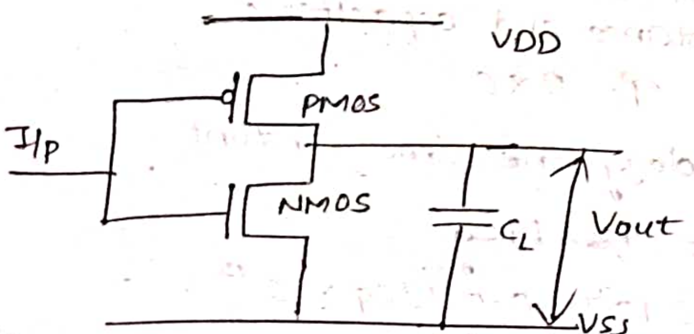
$= 0.005 \times 10^8$

$= 0.05 \times 10^{-9}$ sec

$\boxed{\tau = 0.05 nsec}$

Imp?

Estimation of raise time and fall time in a CMOS inverter:-



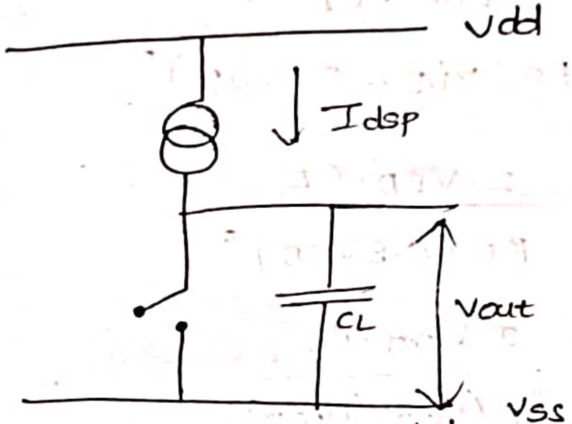
let us consider a CMOS inverter. In that the load capacitor C_L will charge or discharge.

→ The time taken by the capacitor to charge is known as Rise time (T_r).

→ The time taken by the capacitor to discharge is known as fall time (T_f).

→ The conditions for these two can be derived as follows.

Rise Time:-



The rise time can be establish when the i/p voltage is at 'zero'. Then the circuit becomes

From the ckt the p-Mos Transistor is acts as a current source (Idsp) and the N-MOS transistor is acts as open switch then the current $I_{dsp} = \frac{\beta_p (V_{gs} - |V_t|)^2}{2}$

→ the o/p voltage across the capacitor can be obtained as $V_{out} = \frac{I_{dsp} \cdot t}{C_L}$

i.e., $t = \frac{V_{out} \cdot C_L}{I_{dsp}}$

$t = \frac{2 V_{out} \cdot C_L}{\beta_p (V_{gs} - |V_t|)^2}$

let's assume some typical values

$t = T_r$

$V_{out} = V_{DD}$

$V_{gs} = V_{DD}$ & $V_t = 0.2 V_{DD}$

$$\tau_r = \frac{2\,V_{DD}\cdot C_L}{\beta_p (V_{DD} - 0.2V_{DD})^2}$$

$$= \frac{2\cdot V_{DD}\cdot C_L}{\beta_p (0.8V_{DD})^2}$$

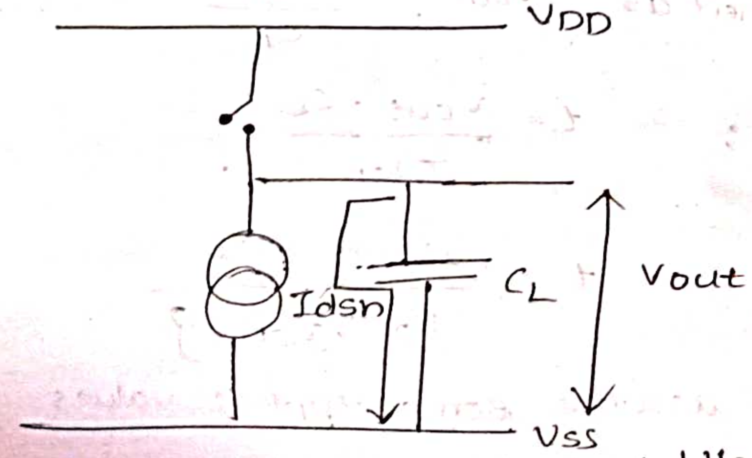
$$= \frac{2\cdot V_{DD}\cdot C_L}{\beta_p (0.64) V_{DD}^2}$$

$$= \frac{2}{0.64}\,\frac{C_L}{\beta_p\cdot V_{DD}}$$

$$\boxed{\tau_r = 3\,\frac{C_L}{\beta_p\cdot V_{DD}}}$$

Fall time:-

The fall time can be establish when the i/p voltage is at 'one'. Then the ckt becomes



Similarly the fall time can be establish as

$$\boxed{\tau_f = \frac{3\cdot C_L}{\beta_n\cdot V_{DD}}}$$

Relation b/w τ_r and τ_f:-

$$\frac{\tau_r}{\tau_f} = \frac{\frac{3\cdot C_L}{\beta_p\cdot V_{DD}}}{\frac{3\cdot C_L}{\beta_n\cdot V_{DD}}} = \frac{\beta_n}{\beta_p}$$

$$\boxed{\tau_r = \frac{\beta_n}{\beta_f}\,\tau_f}$$

Wiring capacitance:-

Generally the capacitances can be establish through different layers such as substrate, metal, polysilicon, diffusion etc.. beyond these some other capacitance may also be form through wiring. Such capacitances are known as wiring capacitances. These are 3 types of wiring capacitances such as

1. Fringing Fields
2. Interlayer capacitance
3. Pheriperal capacitance

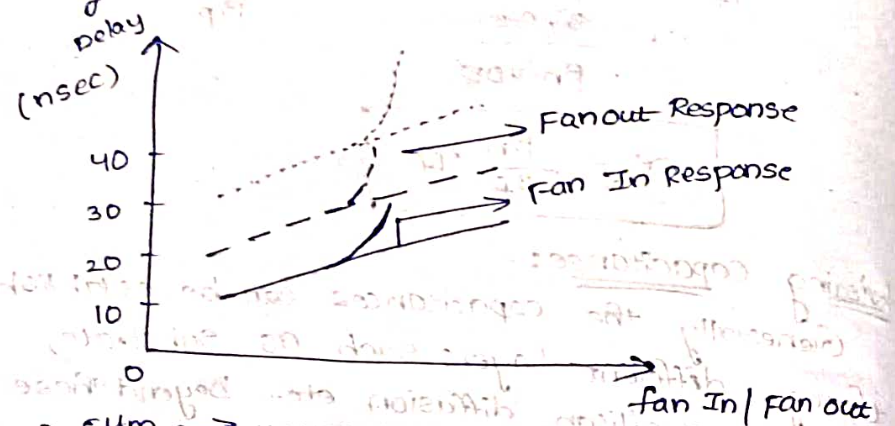
Fan-In, Fan-out characteristics:-

In the vlsi design we have, fan in and fan-out where, Fan-In in No. of inputs are given and Fan-out means No. of gates connected to the o/p. Where No. of i/p's are increases then the delay is linearly increases.

1

→ If fan-out will increase then the delay will be expotentially increases.

→ Fan-In and Fan-Out characteristic response can be given as



→ 5µm → ....

2µm → — — — — — —

1.2µm → ————————

part-II

Scaling of MOS Transistor ckt

In the vlsi design consider the fabrication is the process of combination of different layers with different dimensions. The technology will improve then try to reduce the length and width then No. of Transistor count will increase

→ The scaling down the feature size it improves the performance interms of speed.

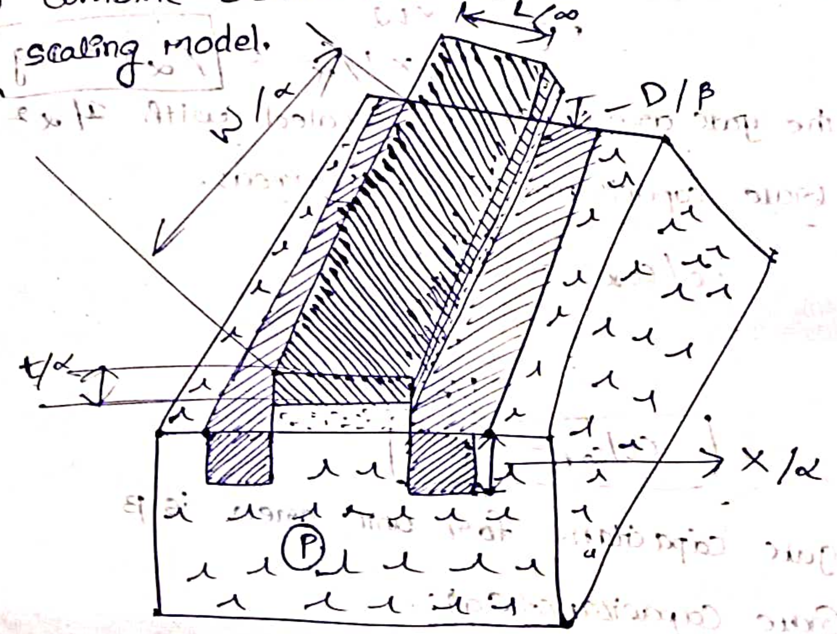
→ To improve the scaling we considered different indicators such as

1. Feature size
2. No. of transistors in a chip
3. power desipation
4. Maximum operating frequency
5. Die size
6. production cost.

Scaling models & scaling factors :-

→ In the scaling process we have 3 different scaling models such as

1. Constant electric field (E) scaling model
2. Constant voltage (V) scaling model
3. Combine Electric field (E) and dimensions (D) scaling model.



→ In order to accomidate these 3 scaling models we use 2 scaling factors such as $1/\alpha$, $1/\beta$ where $1/\beta$ can be scaling to voltage (power supply), Thickness of the oxidation (D) and remaining all can be scaling with $1/\alpha$.

→ In constant e^- field we will replace β with α ($\beta = \alpha$).

→ In constant voltage we consider $\boxed{\beta = 1}$

→ For any constant can be scaled with 1. constants (k, μ, ε_{ox}, ε_{ins}, 1, 2, 3 .... n).

Scaling factors for Device parameters :-

① Gate area (A_g) :-

$$A_g = L \times W$$
$$= 1/\alpha \times 1/\alpha = \boxed{1/\alpha^2 = A_g}$$

→ the gate area can be scaled with $1/\alpha^2$.

② Gate capacitance for unit area :-

$$C_0/C_{ox} = \frac{\varepsilon_{ox}}{D}$$
$$= \frac{1}{1/\beta}$$
$$\boxed{C_0/C_{ox} = \beta}$$

→ gate capacitance for unit area is β

③ Gate Capacitance (C_g) :-

$$C_g = C_0 W L$$
$$= \beta \times 1/\alpha \times 1/\alpha$$
$$C_g = \frac{\beta}{\alpha^2 \beta}$$

The Gate capacitance C_g is scaled with β/α^2

④ parasitic capacitance (C_x) :-

$$C_x = \frac{A_x}{d}$$
$$= \frac{1/\alpha^2}{1/\alpha}$$
$$= \cancel{\alpha}/\alpha^{\cancel{2}}$$
$$\boxed{C_x = 1/\alpha}$$

∴ The parasitic capacitance C_x can be scaled with $1/\alpha$.

⑤ carrier density :- (Q_{ON})

$$Q_{ON} = C_0 V_{gs}$$
$$= \beta \cdot 1/\beta$$
$$\boxed{Q_{ON} = 1}$$

∴ The carrier density can be scaled by 1.

⑥ channel resistance :- (R_{ON})

$$R_{ON} = \frac{L}{W} \cdot \frac{1}{Q_{ON}\,\mu}$$
$$= \frac{\frac{1}{\alpha}}{\frac{1}{\alpha}} \cdot \frac{1}{1} = 1$$

$\boxed{R_{ON} = 1}$

∴ The channel resistance R_{ON} is scaled with 1.

7) Gate delay :- (T_d)

$$T_d = R_{ON} \times C_g$$

$$= 1 \times \beta/\alpha^2$$

$$\boxed{T_d = \beta/\alpha^2}$$

∴ The gate delay T_d is scaled by β/α^2.

8) Maximum operating frequency (f_0) :-

$$f_0 = \frac{W}{L} \frac{C_0 \cdot \mu \cdot V_{DD}}{C_g}$$

$$= \frac{1/\alpha}{1/\alpha} \frac{\beta \cdot 1 \cdot 1/\beta}{\beta/\alpha^2}$$

$$\boxed{f_0 = \alpha^2/\beta}$$

∴ The maximum operating frequency is scaled by α^2/β.

9) Saturation current (I_{dss}) :-

$$I_{dss} = \frac{C_0 \cdot \mu}{2} \times \frac{W}{L} \times (V_{gs} - V_t)^2$$

$$= \frac{\beta \cdot 1}{1} \times \frac{1/\alpha}{1/\alpha} \times (1/\beta - 1/\beta)^2$$

$$\boxed{I_{dss} = 1/\beta}$$

∴ The saturation current I_{dss} is scaled by $1/\beta$.

10) current density :- (J)

$$J = \frac{I_{dss}}{A}$$

$$= \frac{1/\beta}{1/\alpha^2}$$

$$\boxed{J = \alpha^2/\beta}$$

∴ The current density J is scaled by α^2/β

11) Switching energy for Gate [E_g] :-

$$E_g = \frac{C_g}{2} \times (V_{dd})^2$$

$$= \frac{\beta/\alpha^2}{} \times (1/\beta)^2$$

$$\boxed{E_g = \frac{1}{\alpha^2 \beta}}$$

∴ The switching energy for gate E_g is scaled by $1/\alpha^2\beta$.

ii) Power dissipation for gate (P_g) :-

$$P_g = P_{gs} + P_{gd}$$

where

P_{gs} is → gate to source power $= \frac{(V_{DD})^2}{R_{ON}}$

P_{gd} → gate to drain power $= E_g f_0$

$$P_g = \frac{(V_{DD})^2}{R_{ON}} \times E_g f_0$$

$$= \frac{1/\beta^2}{1} \times \frac{1}{\alpha^2 \beta} \times \frac{\alpha^2}{\beta}$$

$$\frac{1}{\beta^2} + \frac{1}{\beta^2} = \frac{2}{\beta^2} = \frac{1}{\beta^2}$$

$$\boxed{P_g = 1/\beta^2}$$

∴ the power dissipation for gate is scaled by $1/\beta^2$.

iii) Power dissipation for unit area :- (P_a)

$$P_a = \frac{P_g}{A_g} = \frac{1/\beta^2}{1/\alpha^2} = \alpha^2/\beta^2$$

$$\boxed{P_a = \frac{\alpha^2}{\beta^2}}$$

∴ Power dissipation for unit area P_a is scaled by α^2/β^2

iv) Power, speed - product :- (P_T)

$$P_T = P_g \cdot T_d$$

$$= 1/\beta^2 \times \beta/\alpha^2$$

$$\boxed{P_T = \frac{1}{\alpha^2 \beta}}$$

∴ The power - speed, product is scaled by $1/\alpha^2\beta$;

Parameter	Combined E & D	Constant E ($\beta = \alpha$)	Constant Voltage ($\beta = 1$)
Length of channel (L)	$\frac{1}{\alpha}$	$\frac{1}{\alpha}$	$1/\alpha$
Width of channel (W)	$1/\alpha$	$1/\alpha$	$1/\alpha$
Oxidation thickness (D)	$1/\beta$	$1/\alpha$	1
Supply Voltage (V_{DD})	$1/\beta$	$1/\alpha$	1
Depth of the Diffusion (X)	$1/\alpha$	$1/\alpha$	$1/\alpha$

* Grate area (Ag)	$1/\alpha^2$	$1/\alpha^2$	$1/\alpha^2$
+ Grate capacitance for unit area (Co/Cox)	β	α	1
+ gate capacitance (Cg)	β/α^2	$\frac{\alpha}{\alpha^2} = \frac{1}{\alpha}$	$1/\alpha^2$
+ parasitic capacitance (Cx)	$1/\alpha$	$1/\alpha$	$1/\alpha$
+ Carrier density (QON)	1	1	1
+ channel resistance (RON)	1	1	1
+ Grate delay (Td)	β/α^2	$1/\alpha$	$1/\alpha^2$
* Maximum operating frequency (fo)	α^2/β	α	α^2
* saturation current ($Idss$)	$1/\beta$	$1/\alpha$	1

+ current density (J)	α^2/β	α	α^2
+ switching energy for gate (Eg)	$1/\alpha^2\beta$	$1/\alpha^3$	$1/\alpha^2$
+ power desipation for gate (Pg)	$1/\beta^2$	$1/\alpha^2$	1
* power desipation for unit area (Pa)	α^2/β^2	1	α^2
+ power, speed -product (PT)	$1/\alpha^2\beta$	$1/\alpha^3$	$1/\alpha^2$

* limitations of Scaling :-

→ limitations due to sub threshold current

* one of the major ~~consist~~ concern in scaling devices is the affect of sub threshold current

i.e., $I_{sub} = e^{(V_{gs} - V_t) q/KT}$

→ where a transistor is in off state then the $V_{gs}-V_t$ value will be negative. Then the current value will be reduced. As voltage are scaled down then the ratio of $V_{gs}-V_t$ and K will be reduces. Then the current value will be increases.

limitations due to current density :-

① High purity aluminum is mostly used in the interconnects of IC. the scaling down of

② dimensions are also increases the current density in interconnects by the same factor.

② when the current density in the aluminium is approximately ~~10^6 amp/cm²~~ reaches to 10^6 Amp/cm² (or) 10 mA/µm². The allowable current density in the IC's will be 1 to 2 mA/µm² will be used nowadays.

Unit - V

FPGA Architecture

CPLD (complex programmable logic device) :-

→ CPLD are characterized by array of PAL blocks (or) PLD's with combinational logic implementation for multiple inputs sop functions. In this the CPLD's have predictable timming and cross bar interconnections and these are using low & ~~middle~~ medium density applications.

→ FPGA's have more complex architecture of functional unit and flexible channel based interconnections and these are used in medium & high density applications.

→ CPLD can be reprogrammable for a limited no of times but FPGA's have no limit for programmability.

→ The basic difference for FPGA as compared to CPLD as basically two types

① There performings depends on the routing

② The functionality of FPGA's can be implement using LUT (look up tables).

FPGA Technologies:-

In FPGA's can be designed we using different Technologies such as

1. Antifuse/PROM
2. EPROM/EEPROM
3. SRAM

***Antifuse/PROM:-**

In the Antifuse Technology we use fuses to store the information. Once the data was store the fuse will burn by passing some high current 20mA-30mA. Once the fuse will burn we can't replace it. So this memory element is known as permanent storage.

*** EPROM/EEPROM:-**

By using these concept we can erase the data as well as we can rewrite the data for a limited no. of times.

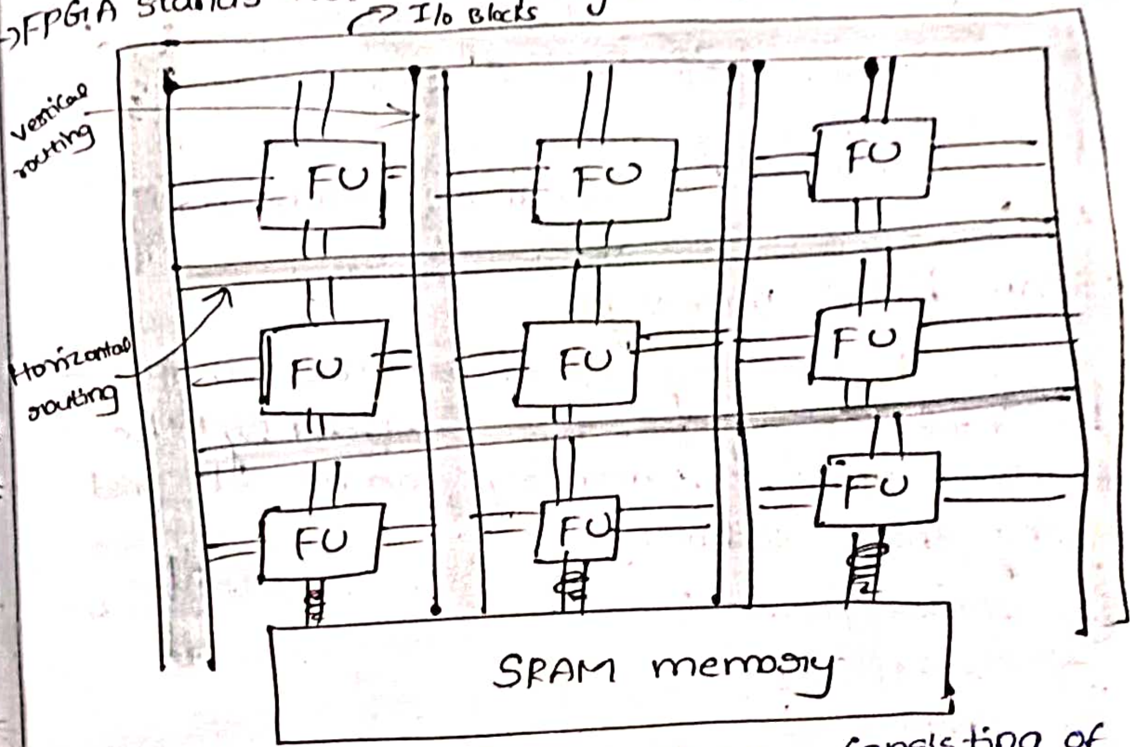
*** SRAM:-**

By using SRAM Technology we can reprogra-mmable or restore of the infinite no. of times.

The basic FPGA Architecture:-

imp

→FPGA stands for field programmable Gate array.



→ The basic architecture of FPGA consisting of following blocks such as

1. Functional unit (FU)
2. SRAM Memory
3. Input/output Blocks
4. Interconnection/Routing

*** Function unit:-**

It can be used to perform various functions such as adder, sub, multiplexer etc.. This will be major functional block in any

FPGA. The functional unit can be designed by using more no. of transistors/ logic gates and it widely used in medium and high density applications.

→ In some FPGA's the functional units have an inbuilt memory also.

→ The functional units can be made with LUT's.

*SRAM based memory:-

The memory unit can be used to store the information which is produced by the functional unit. The memory can be designed by using these technologies.

1. Antifuse/ PROM
2. EPROM/ EEPROM
3. SRAM

* Antifuse/ PROM :-

In the antifuse Technology we use fuses to store the information. Once the data was store the fuse will burn by passing some high current 20mA - 30mA. once the fuse will burn we can't replace it so this memory element is known as permanent storage

EPROM/ EEPROM :- By using these concept we can erase the data as well as we can rewrite the data for a limited no. of times.

*SRAM :-

By using SRAM Technology we can reprogrammable or restore of the infinite no. of times.

I/O Blocks :-

These blocks can be used to both the input and output applications where the i/p's blocks can be used to provide the inputs to the FPGA and the output blocks access the o/p from the FPGA and these blocks can be interchangable i.e; both blocks can be used as either inputs or outputs and these are also used to share the information from one FPGA to another FPGA.

* Interconnections (or) Routing:-

this can be used to share the information from one logic gate to another or from one functional unit to another with in the FPGA.

→ there are two types of routings such as

1. Horizontal routing
2. vertical routing

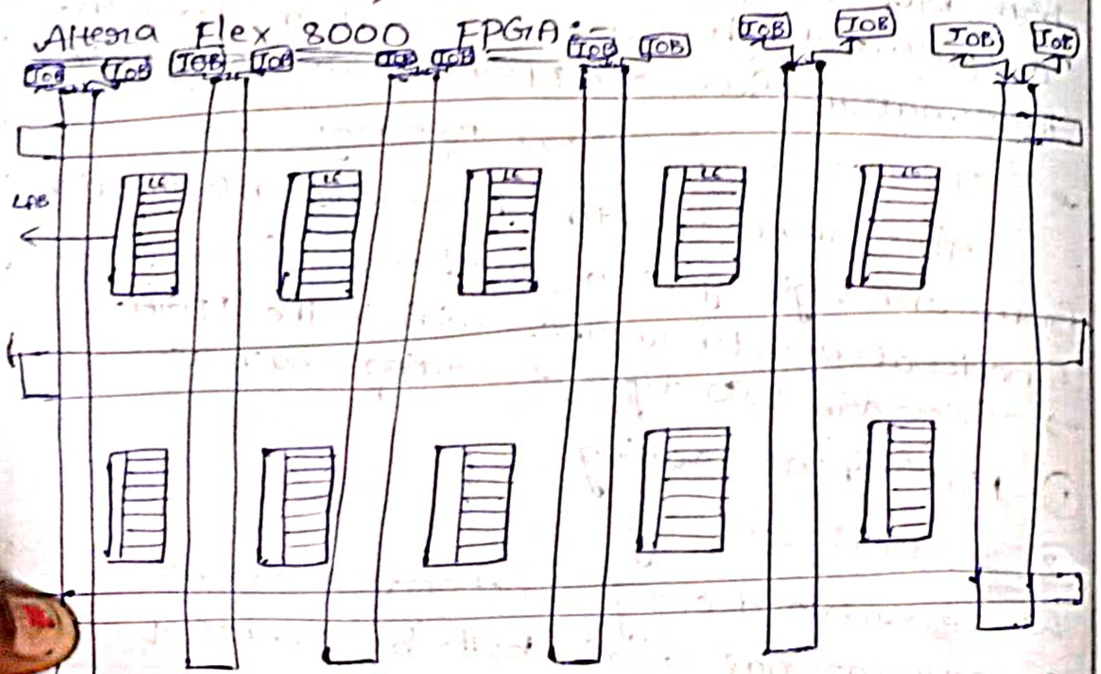
where each and every functional unit has a communicational with both horizontal & vertical routing.

★ Different venders:-
there are different vendors to produce FPGA such as
① Altera, ② xilinx ③ LG ④ Atmel ⑤ Lucent

① Altera → Altera Flex 8000 FPGA
→ Altera Flex 10 FPGA
→ Altera Apex FPGA

② xilinx → xilinx XC 4000 series
→ xilinx spartan II
→ xilinx spartan III
→ xilinx vertex FPGA

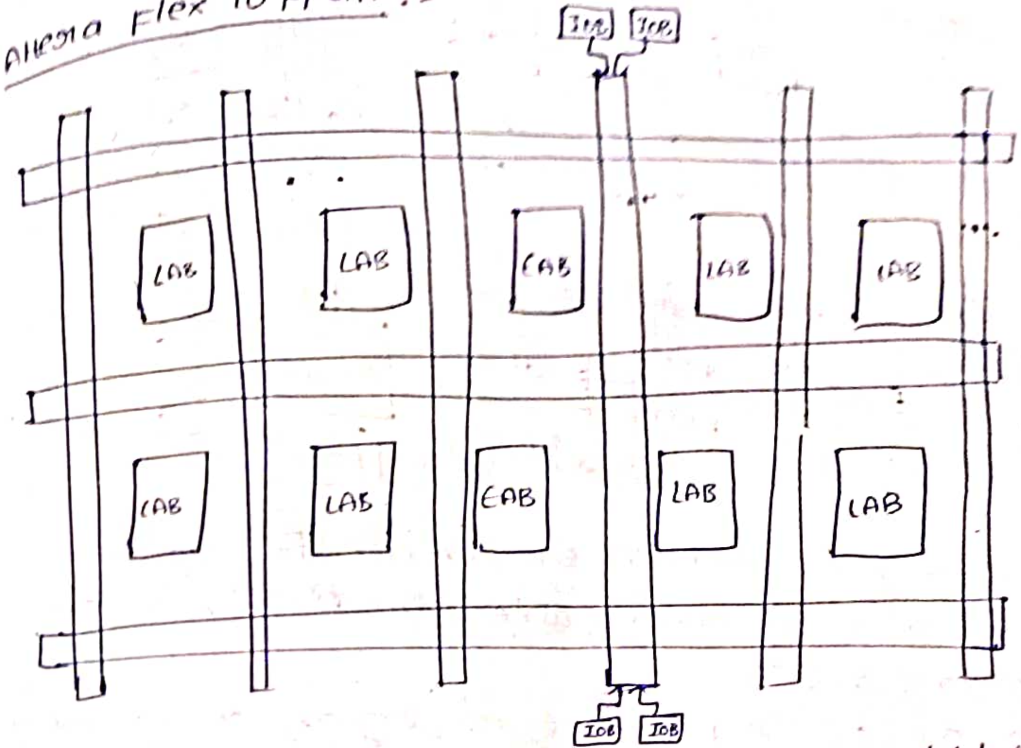
Altera Flex 8000 FPGA:-



→ In this FPGA architecture we use logical array block (LAB) instead of function unit. where each LAB have 8 logical elements and the register type memory. where logical element consist of 4 input LUT.

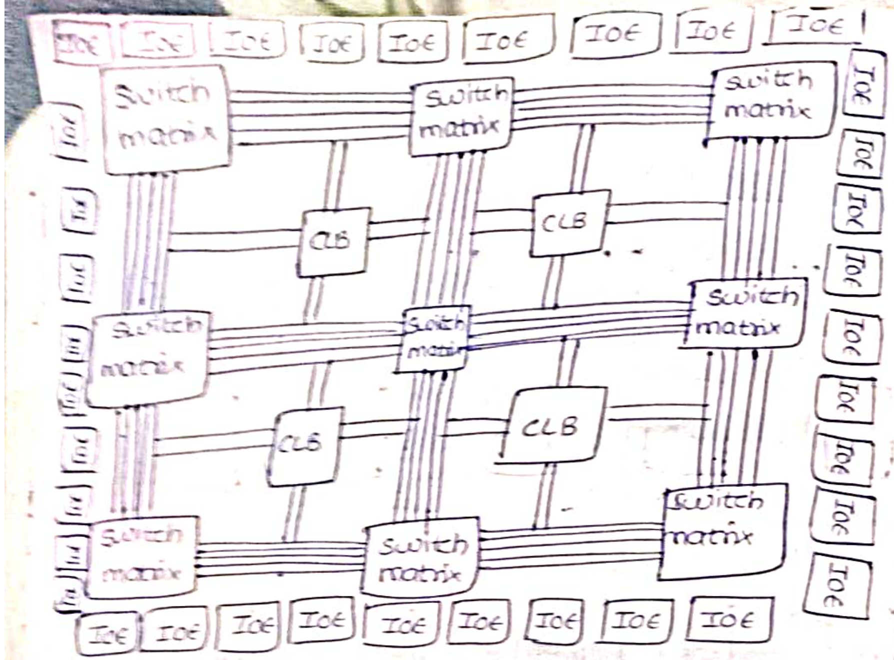
→ In FPGA architecture multiple No. of LAB's ....

Altera Flex 10 FPGA :-



→ EAB stands for embedded array block which is provides the common memory to all the logical array blocks.

xilinx XC 4000 series & xilinx spartan II / xilinx spartan III / xilinx vertex FPGA :-



→ xilinx FPGA architecture we use the major block of CLB (configurable logic block), switch matrix & IOE

→ The CLB can performs the all logical operations and it also consists SRAM Based memory.

→ Switch matrix where the switch matrix can be used to improve data speed as compared to horizontal & vertical routing in altera FPGA's switch matrix is more speed.

→ IOE (Input / output element) it can be used to share the information or to provide the i/p's or excess the o/p from the FPGA. So these IOE's

Synthesis :-

→ It is the process of converting one form of structure into another form.

→ Types of Synthesis :-

there are 3 types of synthesis

1. logic synthesis
2. RTL synthesis
3. High level synthesis

logic synthesis :-

In this synthesis process we convert the HDL code to RTL schematic circuit.

RTL synthesis :-

In this synthesis we convert RTL schematic to transistor level.

High level synthesis :-

→ In high level synthesis we can convert transistor level to boolean expression.

Synthesis :-

→ It is a process of converting one form of structure into another form.

→ Synthesis is process of converting RTL (synthesizable verilog code) to technology specific gate level netlist (includes nets, sequential and combinational cells and their connectivity).

Synthesis = Translation + optimization + mapping

→ Synthesis creates a sequence of transformations between views of a circuit.

Types of synthesis :-

① logic synthesis :-

→ In this synthesis process we can convert the HDL code to RTL schematic circuit

→ logic synthesis generates a structural view from a logic level view of a ckt.

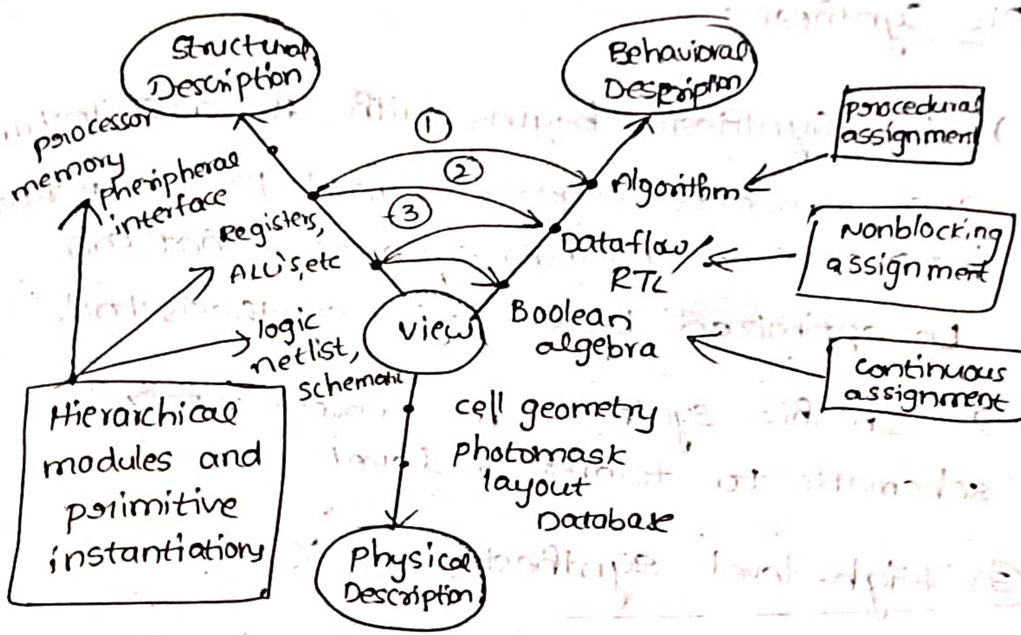


Fig :- Y-chart representation of verilog constructs synthesis

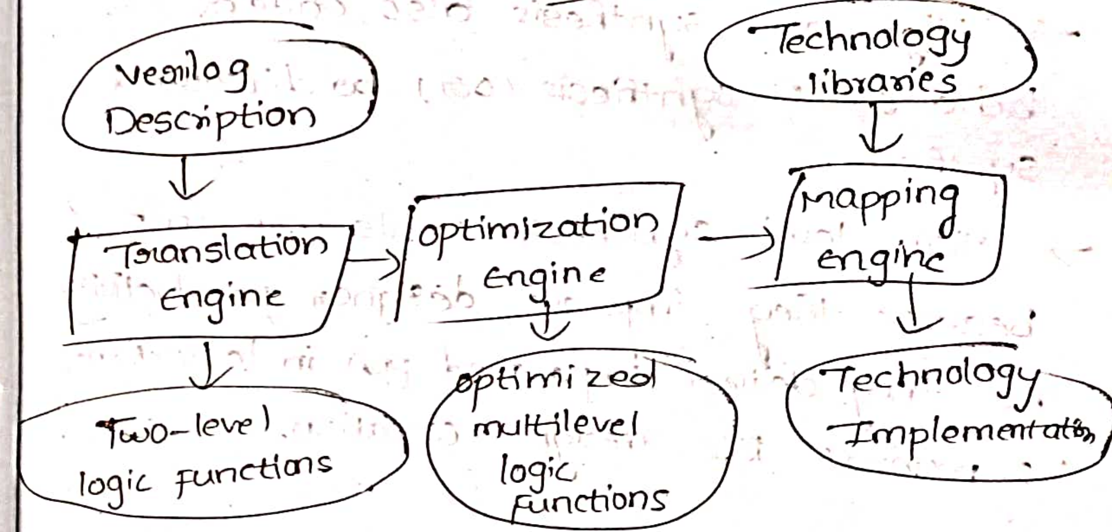


Fig: Synthesis-tool Organization

② RTL Synthesis :-

→ RTL synthesis begins with an architecture and converts language-based RTL statements into a set of Boolean equations that can be optimized by a logic synthesis tool.

→ In this synthesis we convert RTL schematic to transistor level.

③ High-level synthesis :-

→ In this level synthesis we convert transistor level to boolean expression

→ High-level synthesis also called behavioral synthesis (or) architectural synthesis.

→ High-level synthesis tools are imp because they improve designer productivity and they deliver improved PPA in less time vs. manual RTL design creation.